Prepp

Your Personal Exams Guide

NDA	CDS	SSC CGL	CBSE UGC NET
IAS	SSC CHSL	CTET	MPSC
AFCAT	CSIR UDC NET	IBPS PO	UP POLICE
SSC MTS	SBI PO	BPSC	UPTET
IBPS RRB	IBPS CLERK	IES	UPSC CAPF
SSC Stenogr..	RRB NTPC	SSC GD	RBI GRADE B
RBI Assistant	DSSSB		

NEET UG 2026 Question Paper (03-May-2026)

Total Time: 3 Hour **Total Marks:** 720

Instructions

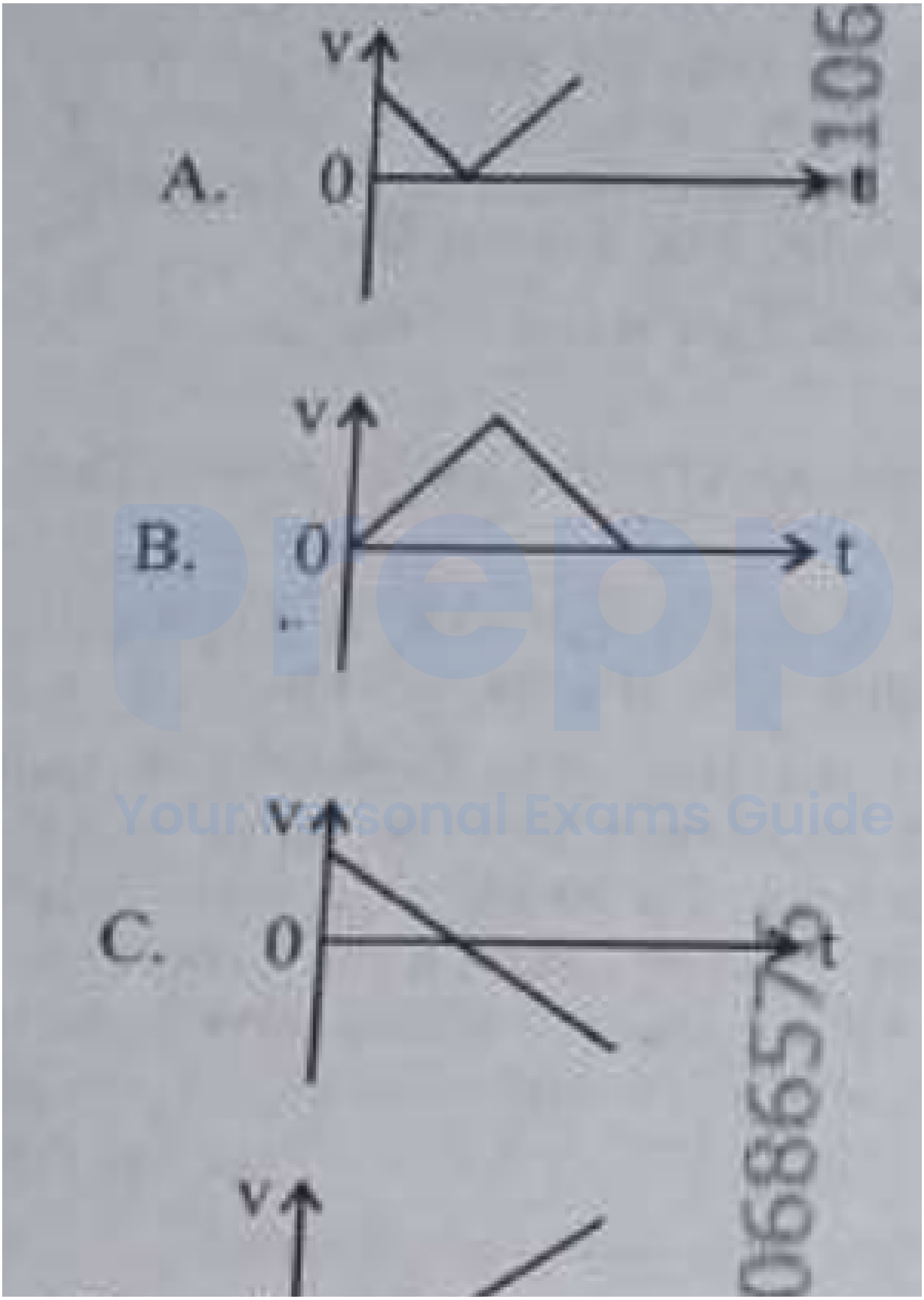
1. Test will auto submit when the Time is up.
2. The Test comprises of multiple choice questions (MCQ) with one correct answers.
3. The clock in the top right corner will display the remaining time available for you to complete the examination.

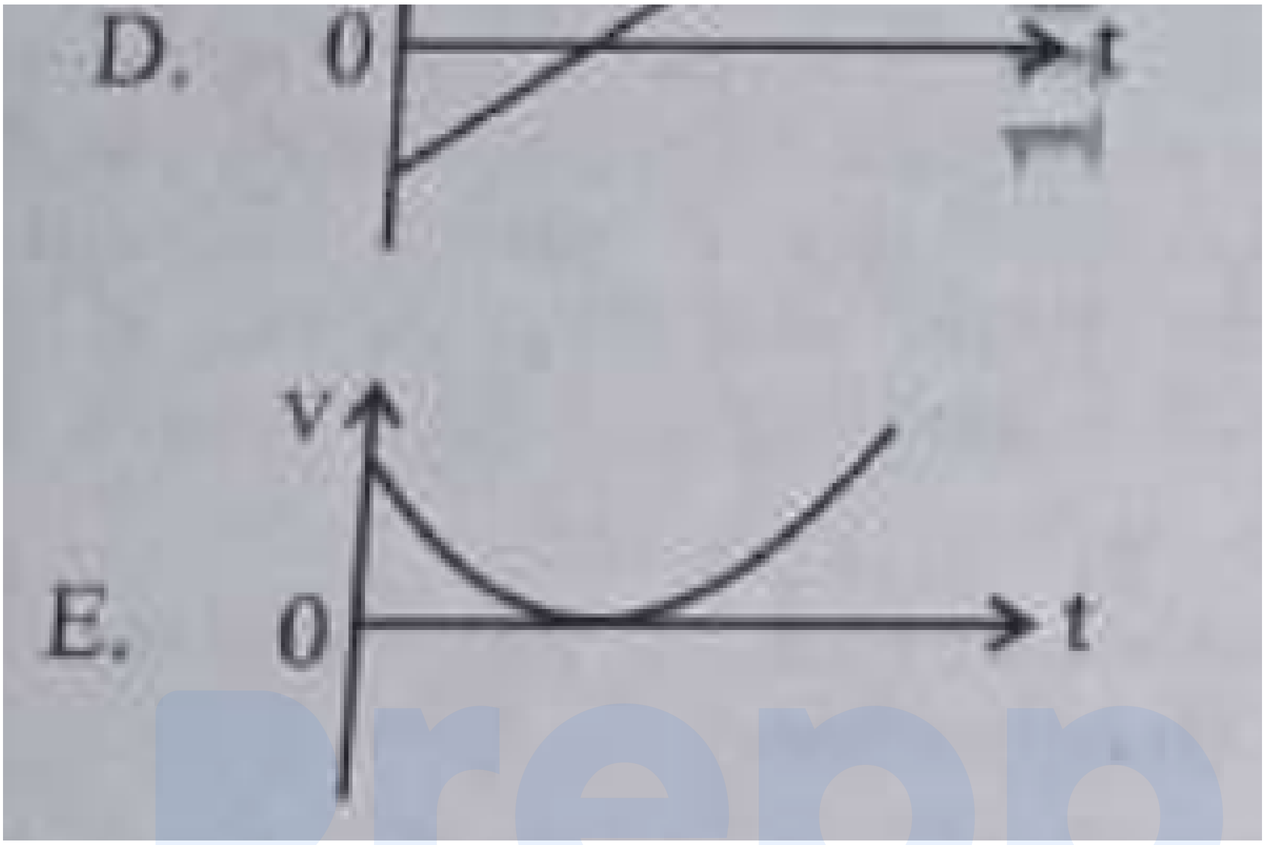
Navigating & Answering a Question

1. The answer will be saved automatically upon clicking on an option amongst the given choices of answer.
2. To deselect your chosen answer, click on the clear response button.
3. The marking scheme will be displayed for each question on the top right corner of the test window.

Physics

1. The following plots show variation of velocity (v) with time (t) of a ball thrown vertically upward, and falling back. Which of the following plots is/are correct? **(+4, -1)**


A.
v
0
B.
v
0
t
C.
v
0
t
v



a. B only

b. A and E only

c. D only

d. C only

2. For a metal of work function $6.6\ \mathrm{eV}$, which of the following wavelengths of incident radiation does not give rise to the photoelectric effect? **(+4, -1)**
(Take Planck's constant as $6.6 \times 10^{-34}\ \mathrm{J\ s}$)

a. $50\ \mathrm{nm}$

b. $100\ \mathrm{nm}$

c. $150\ \mathrm{nm}$

d. $200\ \mathrm{nm}$

3. The power of a crane, which lifts a mass of $1000\ \mathrm{kg}$ to a height of $20\ \mathrm{m}$ in $10\ \mathrm{s}$ is : $(g = 9.8\ \mathrm{m/s^2})$ **(+4, -1)**

a. $39.2\ \mathrm{kW}$

b. $39.2\ \mathrm{W}$

c. $19.6\ \mathrm{kW}$

d. $19.6\ \mathrm{W}$

---

4. Match List I with List II : **(+4, -1)**

List I	List II
A. $E = h\nu$	I. de Broglie wavelength
B. Interference	II. Particle nature of light
C. $\lambda = h/p$	III. Wave nature of light
D. Compton effect	IV. Energy of photon

Choose the correct answer from the options given below :

a. A-IV, B-I, C-II, D-III

b. A-I, B-IV, C-III, D-II

c. A-IV, B-III, C-II, D-I

d. A-IV, B-II, C-I, D-III

---

5. The magnitude and direction of the acceleration produced in a body of mass $5\ \mathrm{kg}$ when two mutually perpendicular forces $8\ \mathrm{N}$ and $6\ \mathrm{N}$ act on it, are respectively : **(+4, -1)**

a. $2\ \mathrm{m\,s^{-2}}, \tan^{-1}(4/3)$ with $8\ \mathrm{N}$ force

b. $2\ \mathrm{m\,s^{-2}}, \tan^{-1}(3/4)$ with $8\ \mathrm{N}$ force

c. $2\ \mathrm{m\,s^{-2}}, \tan^{-1}(3/4)$ with $6\ \mathrm{N}$ force

d. $20\ \mathrm{m\,s^{-2}}, \tan^{-1}(4/3)$ with $8\ \mathrm{N}$ force

---

6. The sum of kinetic energy and potential energy of a simple pendulum bob is $0.02\ \mathrm{joule}$. The speed of the simple pendulum bob at equilibrium position is approximately :
(Consider mass of the bob = $20\ \mathrm{g}$) **(+4, -1)**

a. $14.1\ \mathrm{m/s}$

b. $1.41\ \mathrm{m/s}$

c. $2.0\ \mathrm{m/s}$

d. $0.2\ \mathrm{m/s}$

---

7. A box of mass $15\ \mathrm{kg}$ is kept on the floor of a stationary trolley. The coefficient of static friction between the box and the trolley is 0.12. Keeping the box in stationary state over the trolley, the maximum acceleration with which the trolley can be moved horizontally is : $(g = 10\ \mathrm{m/s^2})$ **(+4, -1)**

a. $1.2\ \mathrm{m\,s^{-2}}$

b. $1.8\ \mathrm{m\,s^{-2}}$

c. $1.5\ \mathrm{m\,s^{-2}}$

d. $2.1\ \mathrm{m\,s^{-2}}$

---

8. The speed of light in vacuum is taken as unity. If light takes $6\ \mathrm{min}\ 40\ \mathrm{s}$ to reach the Earth from the Sun, the distance between the Sun and the Earth in new unit is : **(+4, -1)**

a. 500

b. 3×10^8

c. 400

d. 3×10^{10}

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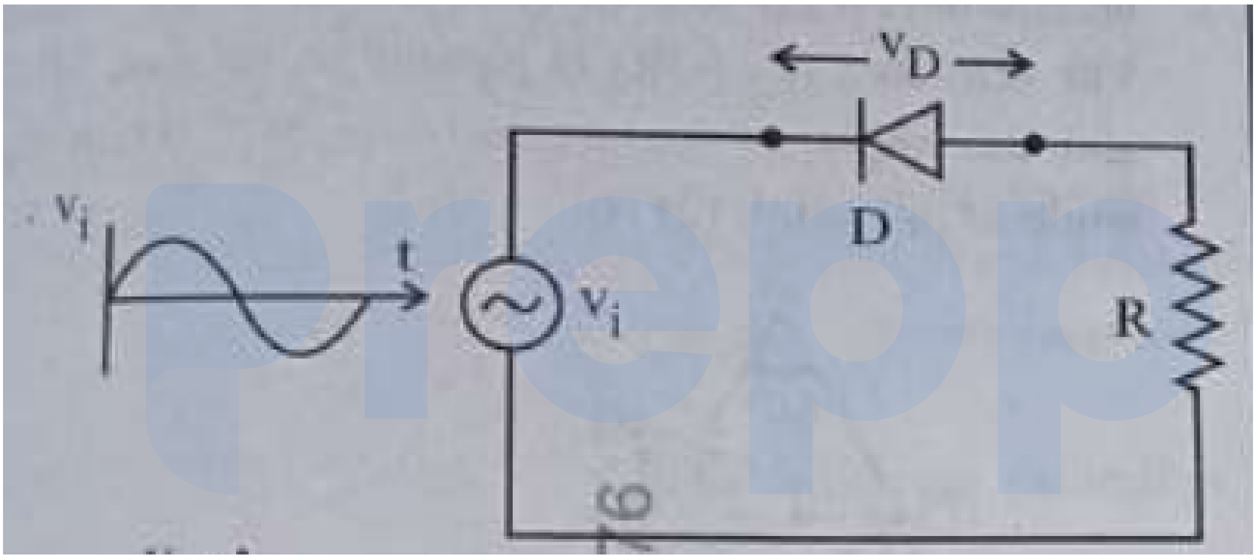
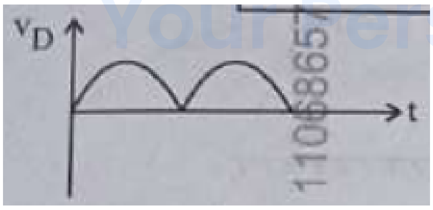
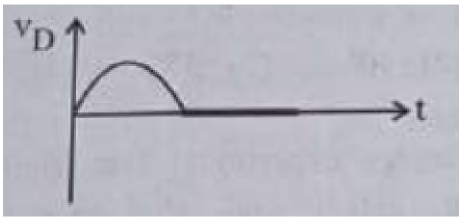
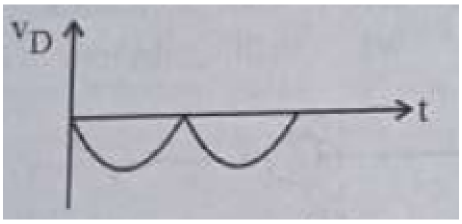
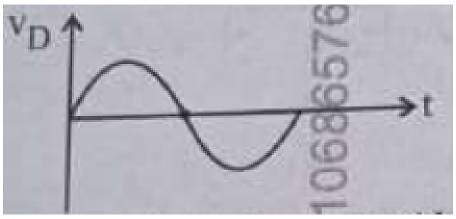
9. In the circuit shown below, the voltage appearing across the diode D will be of the form : (+4, -1)



a. 

b. 

c. 

d. 

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10. A submarine is designed to withstand an absolute pressure of 100 atm. How deep can it go below the water surface? **(+4, -1)**
(Consider the density of water = 1000 kg m^{-3}, $1 \text{ atm} = 1 \times 10^5 \text{ Pa}$ and gravitational acceleration $g = 10 \text{ m/s}^2$)

a. 990 m

b. 9000 m

c. 99 m

d. 9900 m

---

11. An electric heater supplies heat to a system at a rate of 100 W. If the system performs work at a rate of 75 J/s, then the rate at which internal energy increases will be : **(+4, -1)**

a. 125 W

b. 75 W

c. 100 W

d. 25 W

---

12. A 100-turn closely wound circular coil of radius 10 cm has a magnetic field of 3.14×10^{-3} T at its centre. The current passing through the coil, and the magnitude of the magnetic moment of this coil are, respectively : (+4, -1)
(Take $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$)

a. $2 \text{ A}, 4 \text{ A m}^2$

b. $2.5 \text{ A}, 20 \text{ A m}^2$

c. $2.5 \text{ A}, 2 \text{ A m}^2$

d. $2 \text{ A}, 10 \text{ A m}^2$

---

13. In Young's double slit experiment, using monochromatic light of wavelength λ, the intensity of light at a point on the screen where the path difference is λ, is K units. The intensity of light at a point where the path difference is $\lambda/3$ will be : (+4, -1)

a. $K/2$

b. $2K$

c. $K/4$

d. K

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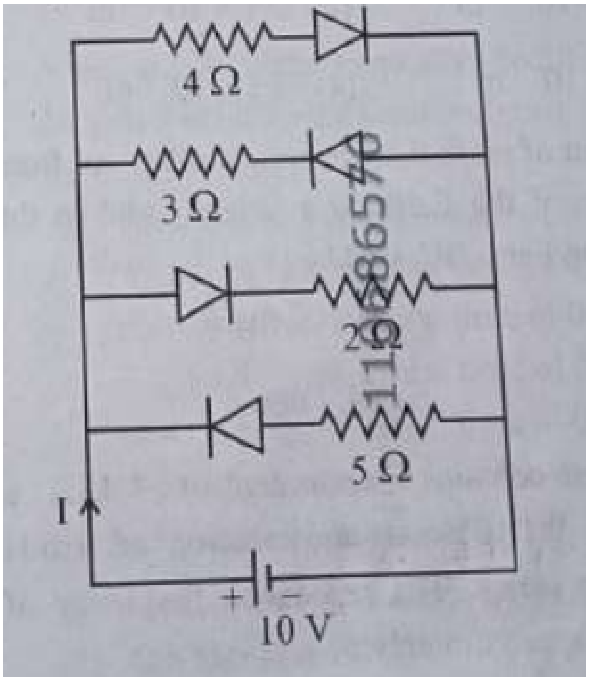
14. The current I in the circuit shown below is : (+4, -1)
(All diodes are ideal and identical.



a. $\frac{5}{3}$ A

b. $\frac{5}{9}$ A

c. $\frac{15}{2}$ A

d. $\frac{1}{3}$ A

---

15. In a concave lens, a ray of light emanating from the object parallel to the principal axis of the lens, after refraction : **(+4, -1)**

a. passes through the second principal focus.

b. appears to diverge from the first principal focus.

c. emerges parallel to the principal axis.

d. passes through 2F which is the radius of curvature of the lens.

---

16. A galvanometer of resistance $100\ \Omega$ gives full scale deflection for a current of $1\ \mathrm{mA}$. It is converted into an ammeter of range $0 - 10\ \mathrm{A}$. The shunt required is : **(+4, -1)**

a. $0.10\ \Omega$

b. $0.001\ \Omega$

c. $1.0\ \Omega$

d. $0.01\ \Omega$

---

17. In the first excited state of hydrogen atom, the energy of its electron is $-3.4\ \mathrm{eV}$. The radial distance of the electron from the hydrogen nucleus in this case is approximately : **(+4, -1)**
(Take $1\ \mathrm{eV} = 1.6 \times 10^{-19}\ \mathrm{J},\ \mathrm{e} = 1.6 \times 10^{-19}\ \mathrm{C}$ and $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ \mathrm{N\,m^2/C^2}$)

a. $2.1 \times 10^{-11}\ \mathrm{m}$

b. $2.1 \times 10^{-10}\ \mathrm{m}$

c. $2.1 \times 10^{-9}\ \mathrm{m}$

d. $2.1 \times 10^{-8}\ \mathrm{m}$

---

18. The amount of work done to raise a mass 'm' from the surface of the Earth to a height equal to the radius of the Earth 'R', will be : **(+4, -1)**

a. mgR

b. $2mgR$

c. $\frac{mgR}{4}$

d. $\frac{mgR}{2}$

---

19. An ac circuit contains a resistance of $1\ \mathrm{k\Omega}$, a capacitor of $0.1\ \mu\mathrm{F}$ and an inductor of $1\ \mathrm{mH}$ connected in series. The resonance frequency of the circuit is approximately : **(+4, -1)**

a. $13.5\ \mathrm{kHz}$

b. $15.9 \mathrm{~kHz}$

c. $10.1 \mathrm{~kHz}$

d. $20.7 \mathrm{~kHz}$

---

20. Consider two uncharged capacitors of equal capacitance $200 \mathrm{~pF}$. One of them is charged by a $100 \mathrm{~V}$ supply and disconnected. Now this capacitor is connected to the uncharged capacitor. The amount of electrostatic energy lost in the process is : **(+4, -1)**

a. $1.0 \mathrm{~J}$

b. $0.5 \mathrm{~J}$

c. $1.0 \times 10^{-6} \mathrm{~J}$

d. $0.5 \times 10^{-6} \mathrm{~J}$

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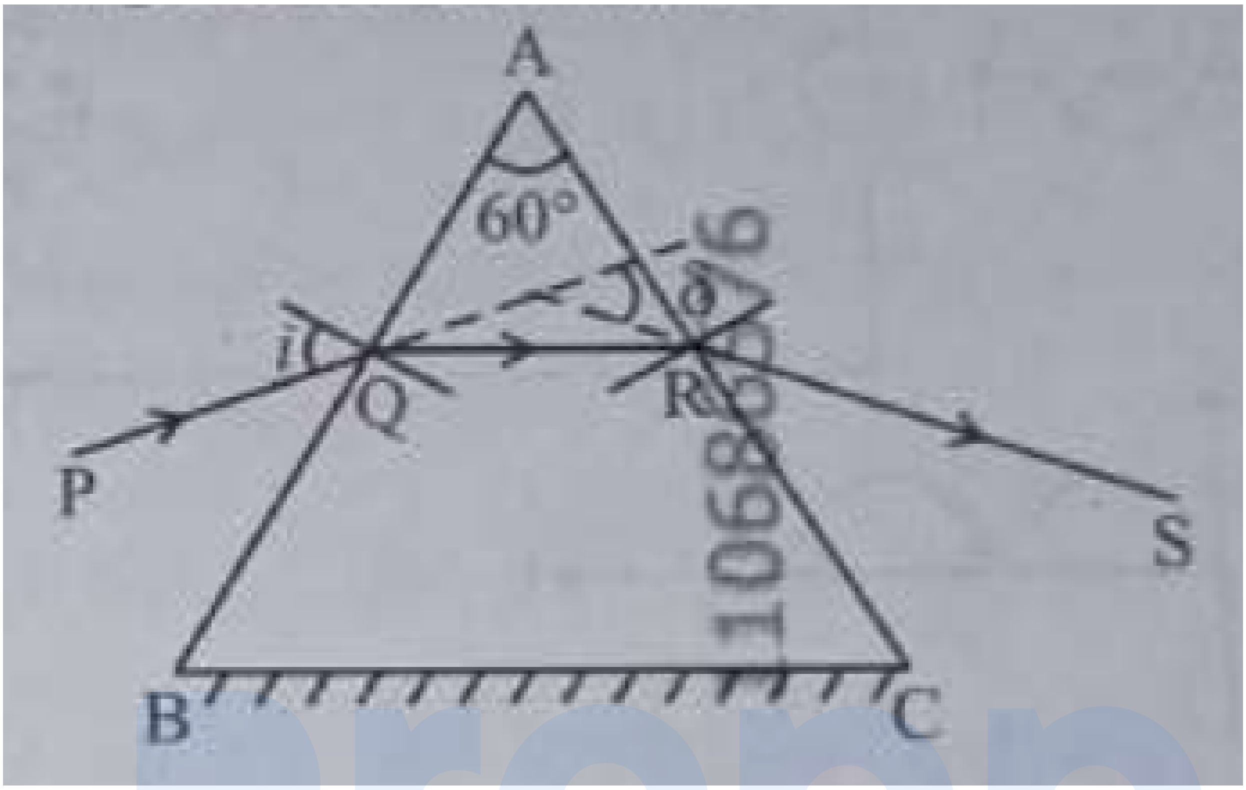
21. A ray of monochromatic light is passing through an equilateral prism (ABC) as shown in the figure. The refracted ray (QR) is parallel to the base (BC) and the angle of incidence (i) is 50°. Then the angle of deviation (δ) is : **(+4, -1)**



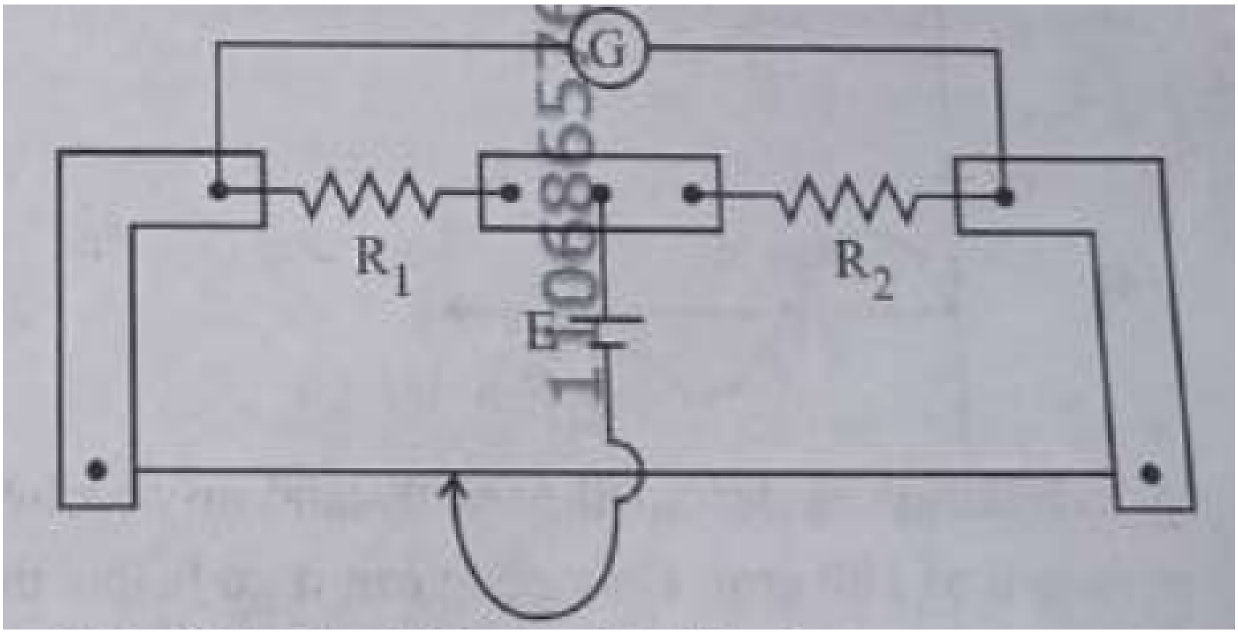
a. 45°

b. 40°

c. 35°

d. 55°

22. In a metre bridge experiment, the positions of the cell, E, and galvanometer, G, are interchanged. We shall observe in the galvanometer : **(+4, -1)**



a. Only the left-sided deflection

b. Both right-sided and left-sided deflection and at balance point no deflection

c. Only the right-sided deflection

d. There will be no deflection irrespective of the position of the jockey

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23. A flask contains argon and chlorine in the ratio of $2:1$ by mass. The temperature of the mixture is $27°\mathrm{C}$. The ratio of root mean square speed of the molecules of the two gases $\left(\frac{V_{rms}^{Ar}}{V_{rms}^{Cl}}\right)$ is :
(Atomic mass of argon = $40\ \mathrm{u}$ and molecular mass of chlorine = $70\ \mathrm{u}$) **(+4, -1)**

a. $\frac{7}{4}$

b. $\frac{\sqrt{7}}{2}$

c. $\frac{2}{\sqrt{7}}$

d. $\frac{7}{2}$

---

24. Two statements are given below : **(+4, -1)**

A. When the forward bias voltage across a p-n junction diode increases above a certain threshold voltage, the diode current increases significantly.

B. This current is called reverse saturation current.

Choose the correct answer from the options given below :

a. Statement A is true but Statement B is false

b. Both Statements A and B are true

c. Both Statements A and B are false

d. Statement A is false, but Statement B is true

---

25. For a travelling harmonic wave $y(x,t) = 2.0 \cos 2\pi(10t - 0.0080x + 0.35)$, where x and y are in cm and t in s. The phase difference between oscillatory motion of two points separated by a distance of 0.5 m is : **(+4, -1)**

a. $0.8\pi \text{ rad}$

b. $8\pi \text{ rad}$

c. $0.008\pi \text{ rad}$

d. $0.08\pi \text{ rad}$

---

26. A rectangular wire loop of sides 8 cm and 3 cm with a small cut, is moving out of a region of uniform magnetic field of magnitude 0.3 T directed normal to the plane of the loop. The emf developed across the cut, if the velocity of the loop is 2 cm s^{-1} in a direction normal to the shorter side of the loop, will be : **(+4, -1)**

a. 1.8×10^{-4} volt

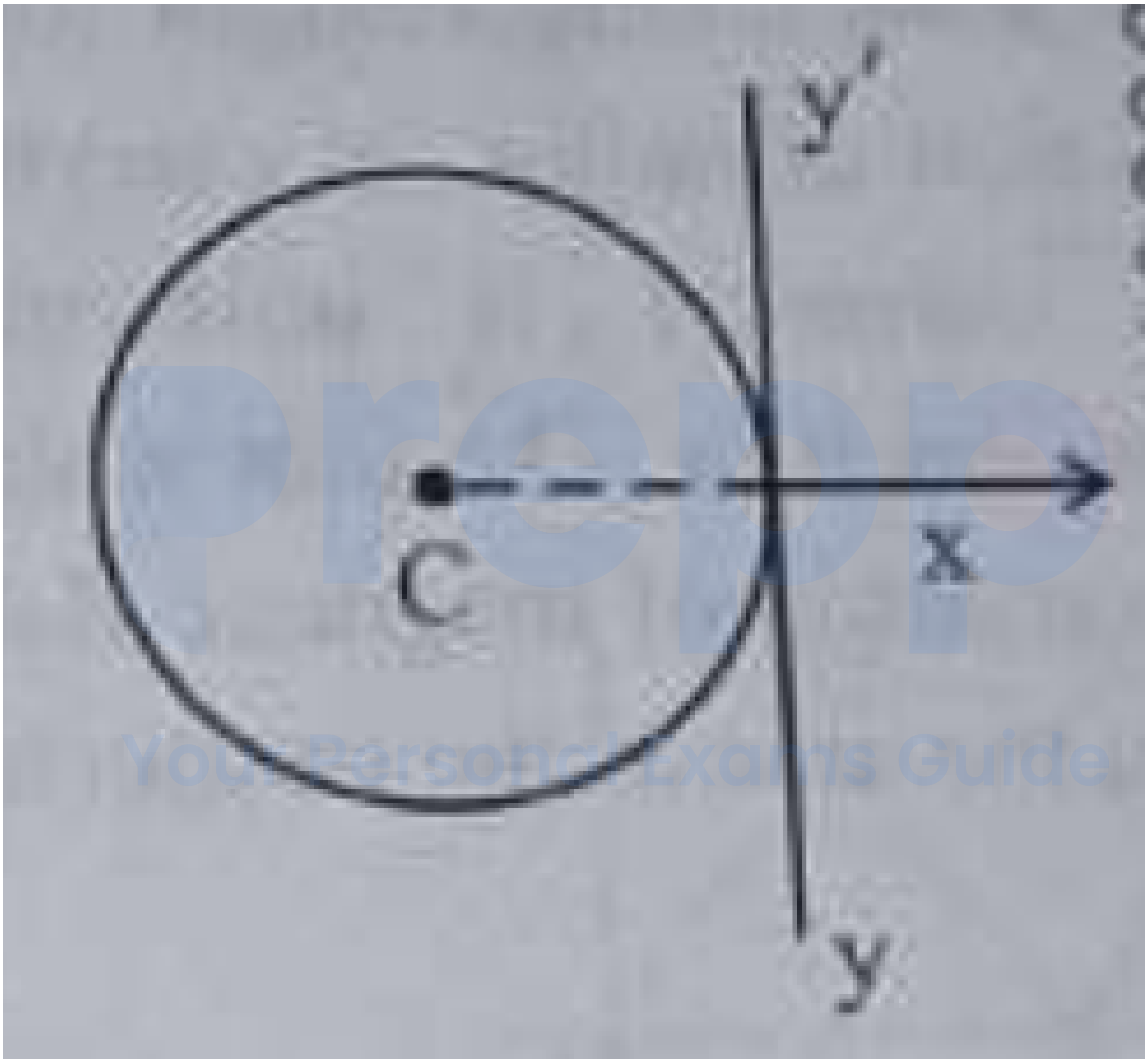
b. 1.3×10^{-4} volt

c. 1.2×10^{-4} volt

d. 4.8×10^{-4} volt

---

27. A thin wire of length 'L' and linear mass density 'm' is bent into a circular ring (in x-y plane) with centre 'C' as shown in figure. The moment of inertia of the ring about an axis yy' (tangent in the plane) will be : (+4, -1)



a. $\frac{3mL^2}{8\pi}$

b. $\frac{3mL^2}{8\pi^2}$

c. $\frac{3mL^3}{8\pi}$

d. $\frac{3mL^3}{8\pi^2}$

---

28. A resistor is connected to a battery of $12\ \mathrm{V}$ emf and internal resistance $2\ \Omega$. If the current in the circuit is $0.6\ \mathrm{A}$, the terminal voltage of the battery is : **(+4, -1)**

a. $10.8\ \mathrm{V}$

b. $1.2\ \mathrm{V}$

c. $12\ \mathrm{V}$

d. $10\ \mathrm{V}$

---

29. When a ruler falls vertically, 5 different persons catch it with different reaction times. $\left(g = 9.8\ \mathrm{m\,s^{-2}}\right)$ **(+4, -1)**

A. Person A has reaction time of $0.20\ \mathrm{s}$.
B. Person B has reaction time of $0.22\ \mathrm{s}$.
C. Person C has reaction time of $0.18\ \mathrm{s}$.
D. Person D has reaction time of $0.19\ \mathrm{s}$.
E. Person E has reaction time of $0.21\ \mathrm{s}$.
What is the correct order of the distance travelled by the ruler for each person ?

a. C > D > A > E > B

b. C > D > A > B > E

c. B > E > A > D > C

d. B > E > A > C > D

---

30. The angular speed of a flywheel is increased from $600\ \mathrm{rpm}$ to $1200\ \mathrm{rpm}$ in $10\ \mathrm{s}$. The number of revolutions completed by the flywheel during this time is : **(+4, -1)**

a. 300

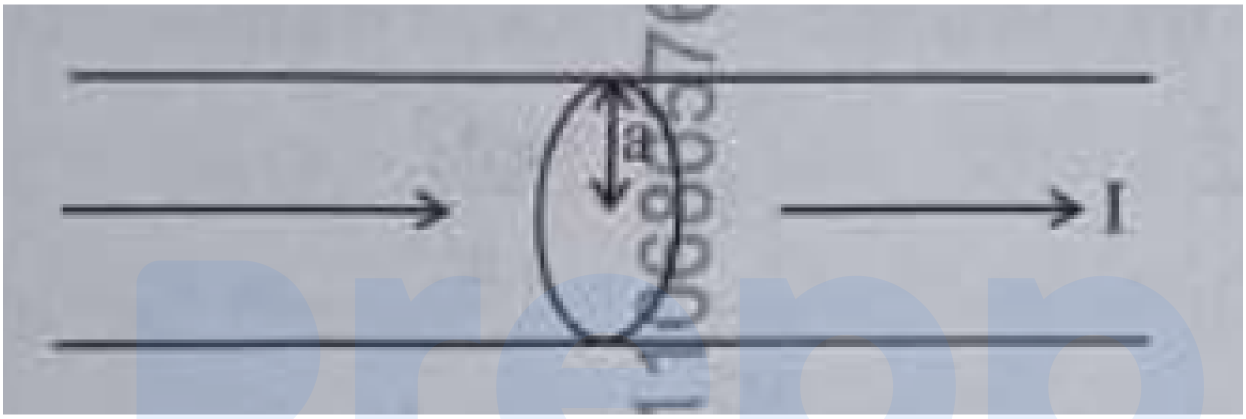
b. 150

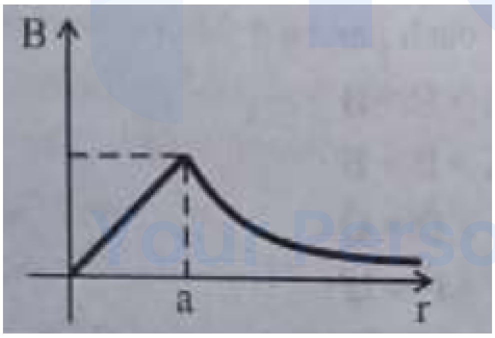
c. 900

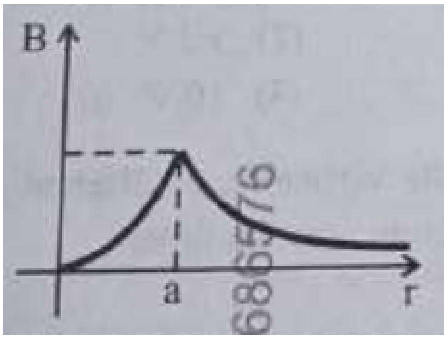
d. 600

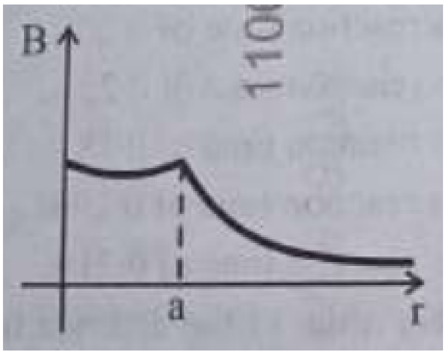
---

31. The figure given below shows a long straight solid wire of circular cross-section of radius 'a' carrying steady current I. The current I is uniformly distributed across its cross-section. The plot which correctly represents the variation of magnetic field (B) with distance (r) from the axis of the conductor in the region is : **(+4, -1)**

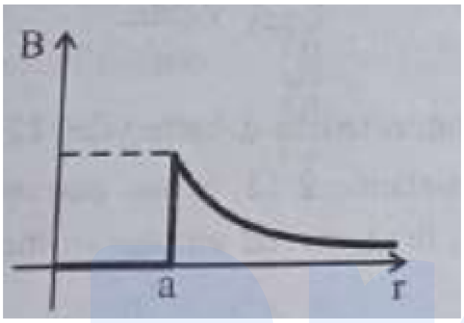


a. 

b. 

c. 

d.



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32. Four statements are given (A is mass number) : **(+4, -1)**

A. The volume of a nucleus is proportional to $A^{1/3}$.

B. The volume of a nucleus is proportional to A.

C. The difference in mass of an atom and its nucleus is called the mass defect.

D. The difference in mass of a nucleus and its constituent nucleons is called the mass defect.

Choose the correct answer from the options given below :

a. A and D are true, but B and C are false

b. B and D are true, but A and C are false

c. B and C are true, but A and D are false

d. A and C are true, but B and D are false

---

33. Savitha, a XI standard student, while conducting an experiment to determine the effective length of a simple pendulum L, notes down the data of time taken to complete 30 oscillations as $60\ \mathrm{s}$ and hence calculates the length of the **(+4, -1)**

simple pendulum as :
(Take $\pi^2 = 9.8$, and $g = 9.8 \text{ m/s}^2$)

a. 2 m

b. 1 m

c. 0.75 m

d. 1.5 m

---

34. In a vernier callipers, 20 VSD coincide with 16 MSD (each division of length 1 mm). The least count of the vernier callipers is : **(+4, -1)**

a. 0.1 cm

b. 0.02 cm

c. 0.01 cm

d. 0.2 cm

---

35. Each side of a metallic cube of mass 5.580 kg is measured to be 9.0 cm. Keeping the significant figures in view, the density of the material of the cube can be best expressed as $X \times 10^3 \text{ kg m}^{-3}$, where the value of X is : **(+4, -1)**

a. 7.654

b. 7.6

c. 7.65

d. 7.7

---

36. In interference and diffraction, the light energy is redistributed. If it reduces in one region, producing a dark fringe, it increases in another region, producing a bright fringe. **(+4, -1)**

A. As there is no gain or loss of energy, these phenomena are consistent with the principle of conservation of energy.
B. Diffraction and interference are characteristics exhibited only by light waves.
Choose the correct answer from the options given below :

a. A is false, but B is true

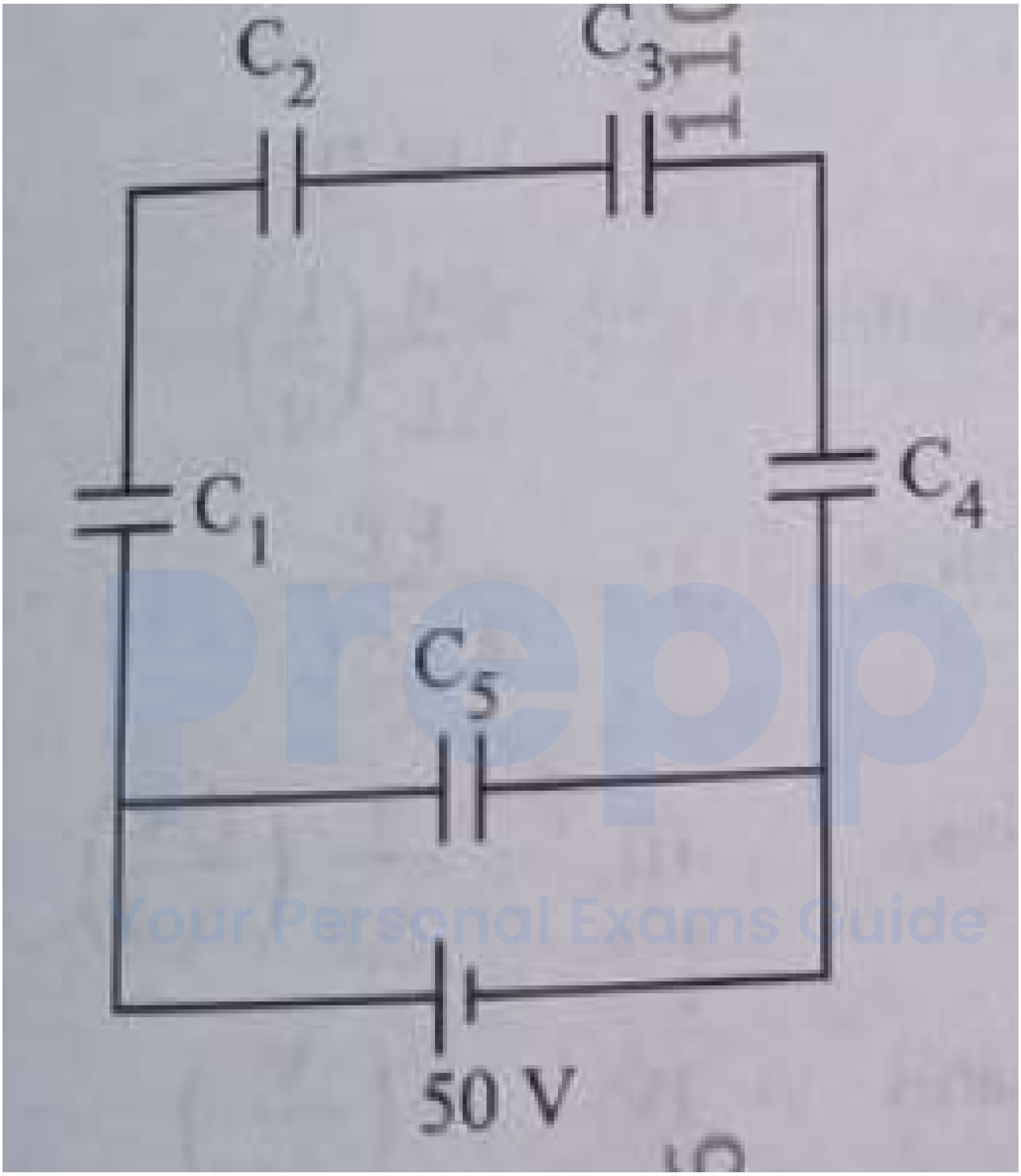
b. A is true and B is also true

c. A is true, but B is false

d. Both A and B are false

---

37. Five capacitors of capacitances $C_1 = C_2 = C_3 = C_4 = 10\ \mu\text{F}$ and $C_5 = 2.5\ \mu\text{F}$ are connected as shown along with a battery of $50\ \text{V}$. **(+4, -1)**



The equivalent capacitance and the charges on each capacitor respectively are :

a. $5\ \mu\text{F}, 125\ \mu\text{C}$ on C_1 to C_4 and $25\ \mu\text{C}$ on C_5

b. $4\ \mu\text{F}, 250\ \mu\text{C}$ on C_1 to C_4 and $125\ \mu\text{C}$ on C_5

c. $5\ \mu F, 250\ \mu C$ on all capacitors

d. $5\ \mu F, 125\ \mu C$ on all capacitors

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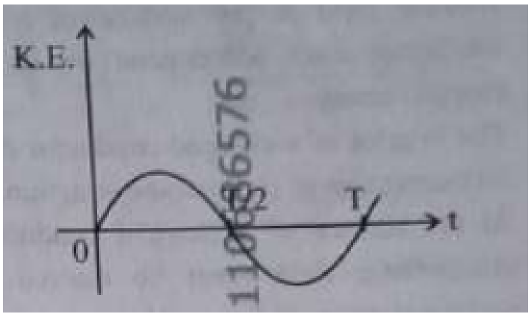
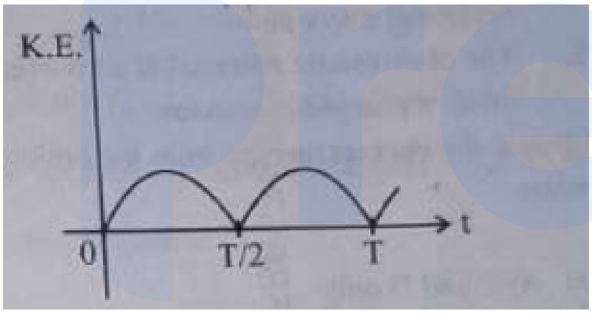
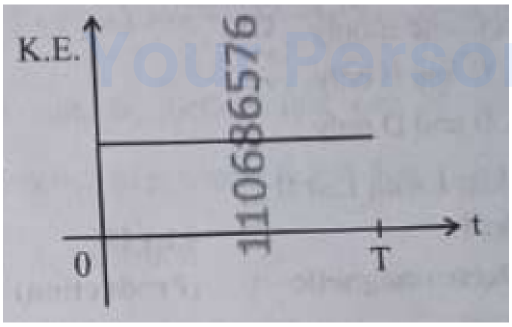
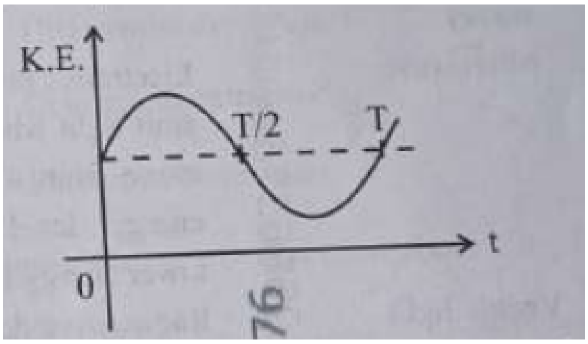
38. For a simple pendulum, having time period T, the variation of kinetic energy (K.E.) with time (t) is represented by : **(+4, -1)**

a. 

b. 

c. 



d.

39. A room heater is rated $400 \mathrm{~W}, 220 \mathrm{~V}$. If the supply voltage drops to $200 \mathrm{~V}$, what will be the power consumed (approximately) ? (+4, -1)

a. $121 \mathrm{~W}$

b. $200 \mathrm{~W}$

c. $400 \mathrm{~W}$

d. $331 \mathrm{~W}$

40. The peak value of an alternating current is $5 \mathrm{~A}$ and frequency is $60 \mathrm{~Hz}$. How long will the current, starting from zero, take to reach the peak value ? (+4, -1)

a. $1/60 \mathrm{~s}$

b. $1/240 \mathrm{~s}$

c. $1/30 \mathrm{~s}$

d. $1/120 \mathrm{~s}$

41. Which of the following statements are correct ? (+4, -1)
A. Inside a conductor the electrostatic field is zero.
B. Electric field at the surface of a charged conductor does not depend on its surface charge density.
C. The interior of a charged conductor can have no excess charge in the static situation.
D. At the surface of a charged conductor, the electrostatic field must be normal to the surface at every point.
E. The electrostatic potential is zero everywhere inside the charged conductor.
Choose the correct answer from the options given below :

a. A, C and D only

b. A, C and E only

c. C, D and E only

d. A, B and D only

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42. Match List I with List II : **(+4, -1)**

List I (Electromagnetic wave)	**List II (Production)**
A. Microwave	I. Electrons in atoms emit light when they move from a higher energy level to a lower energy level
B. Visible light	II. Radioactive decay of nucleus
C. Gamma rays	III. Vibration of atoms and molecules
D. Infra-red ray	IV. Klystron valve or magnetron valve

Choose the correct answer from the options given below :

a. A-III, B-I, C-II, D-IV

b. A-III, B-IV, C-I, D-II

c. A-IV, B-I, C-II, D-III

d. A-IV, B-III, C-II, D-I

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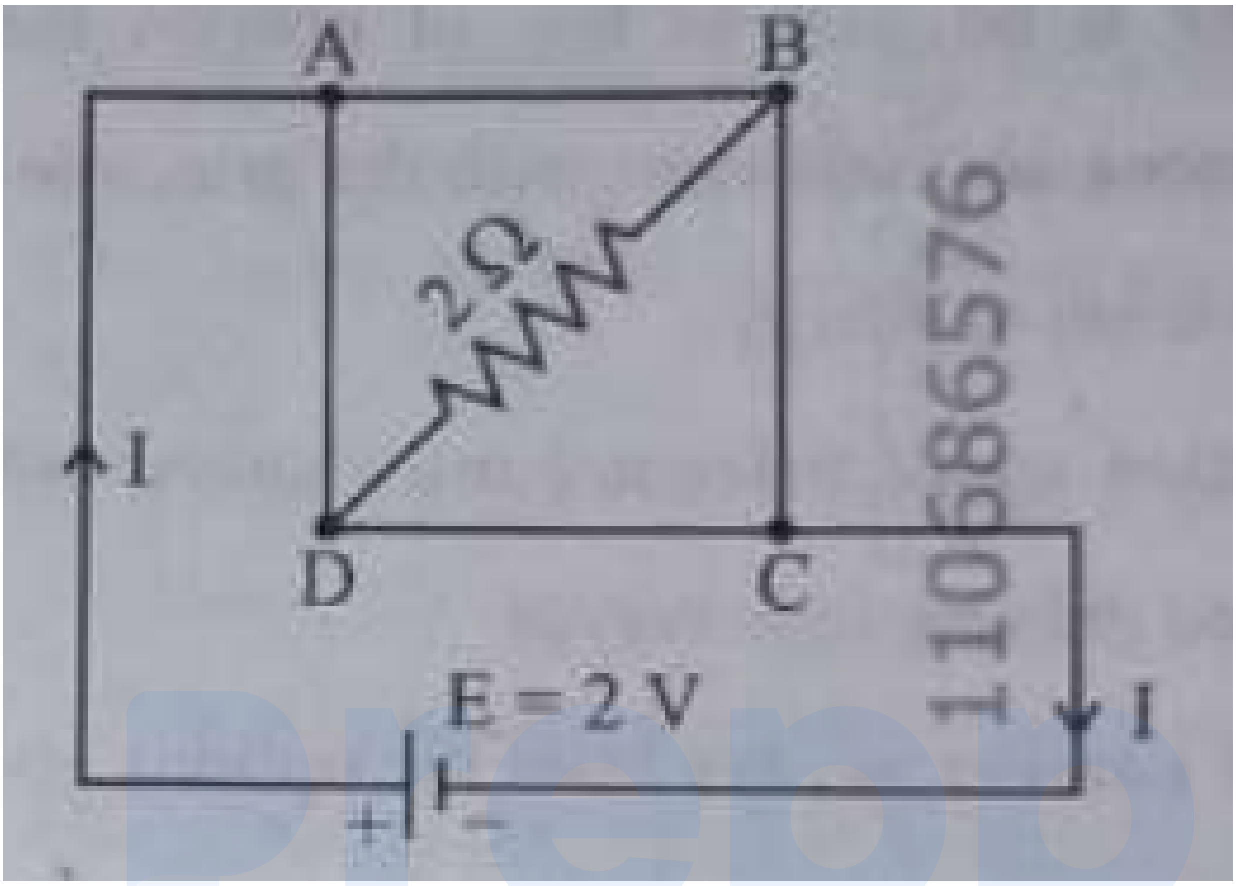
43. A uniform metallic wire having resistance $4\,\Omega$ is bent to form a square loop ABCD. A resistance of $2\,\Omega$ is connected between points B and D and a battery of 2 V is connected across points A and C as shown in the figure. Now the value of current (I) is : **(+4, -1)**



a. $4\ \mathrm{A}$

b. $8\ \mathrm{A}$

c. $4.5\ \mathrm{A}$

d. $2\ \mathrm{A}$

---

44. An unknown nucleus has a nuclear density of $2.29 \times 10^{17}\ \mathrm{kg/m^3}$ and mass of $19.926 \times 10^{-27}\ \mathrm{kg}$. Its mass number A is approximately : **(+4, -1)**
(Take $R_0 = 1.2 \times 10^{-15}\ \mathrm{m},\ 4\pi = 12.56$)

a. 12

b. 16

c. 19

d. 20

---

45. Match List I with List II : **(+4, -1)**

List I	List II
A. Young's Modulus	I. $\frac{\Delta d}{\Delta L}\left(\frac{L}{d}\right)$
B. Compressibility	II. $\frac{FL}{A(\Delta L)}$
C. Bulk Modulus	III. $-\frac{1}{\Delta P}\left(\frac{\Delta V}{V}\right)$
D. Poisson's Ratio	IV. $-P\left(\frac{V}{\Delta V}\right)$

Choose the correct answer from the options given below :

a. A-II, B-III, C-IV, D-I

b. A-III, B-II, C-I, D-IV

c. A-I, B-IV, C-III, D-II

d. A-IV, B-I, C-II, D-III

Chemistry

46. The correct statement with regard to the secondary structure of DNA/RNA is : **(+4, -1)**

a. DNA possesses a single strand helix structure and contains uracil as one of the four bases.

b. DNA possesses a double strand helix structure and contains thymine as one of the four bases.

c. RNA possesses a double strand helix structure and contains uracil as one of the four bases.

d. RNA possesses a single strand helix structure and contains thymine as one of the four bases.

---

47. Match List I with List II : **(+4, -1)**

List I (Quantum Numbers) n, l	**List II (Orbital)**
A. 2, 1	I. 3d
B. 4, 0	II. 2p
C. 5, 3	III. 4s
D. 3, 2	IV. 5f

Choose the correct answer from the options given below :

a. A-II, B-III, C-I, D-IV

b. A-II, B-III, C-IV, D-I

c. A-IV, B-II, C-III, D-I

d. A-I, B-II, C-IV, D-III

48. During Lassaigne's test, the elements present in an organic compound are converted from : **(+4, -1)**

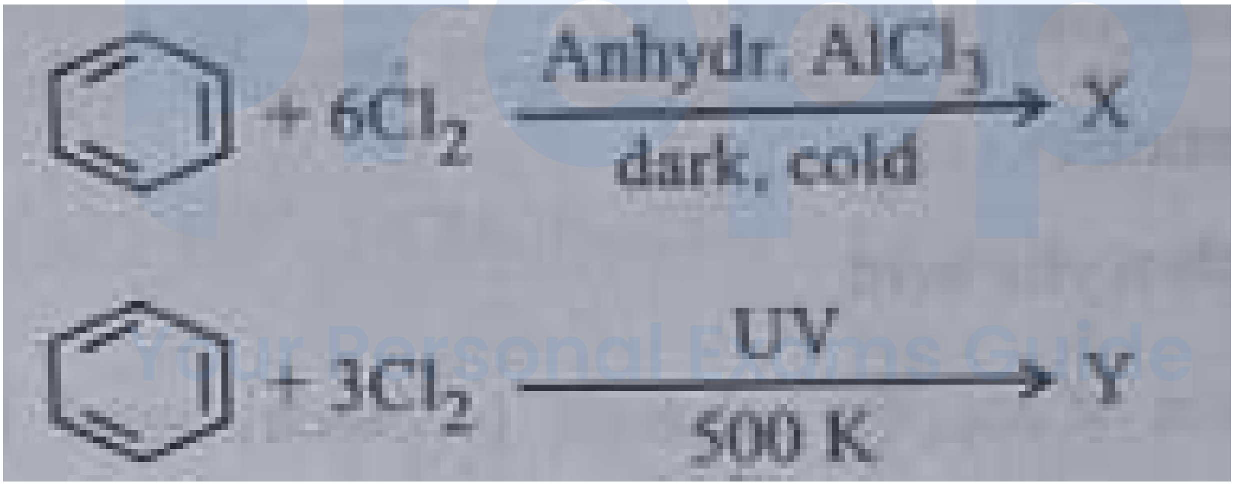
a. covalent form to ionic form

b. covalent form to covalent form

c. ionic form to ionic form

d. ionic form to covalent form

49. The number of chlorine atoms present in the organic products X and Y of the following reactions, respectively, are : **(+4, -1)**



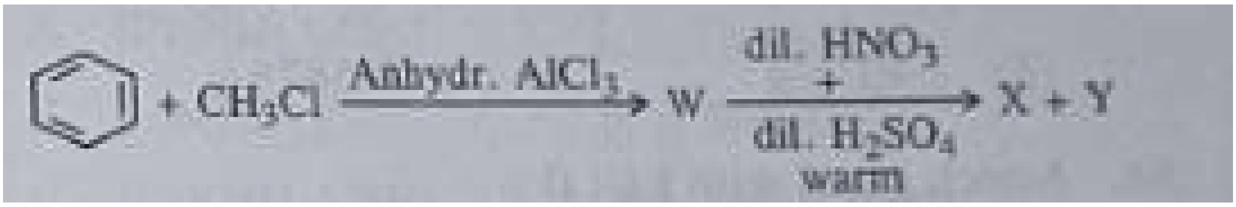
a. 3 and 6

b. 6 and 3

c. 3 and 3

d. 6 and 6

50. Two products X and Y are formed in the following reaction sequence. **(+4, -1)**



The suitable method that can be used for the separation of products X and Y is :

a. Fractional distillation

b. Differential extraction

c. Continuous extraction

d. Sublimation

---

51. Identify the correct statement about ClF_3 from the following options : **(+4, -1)**

a. It has T-shaped geometry with three lone pairs on Cl atom.

b. It has a planar trigonal geometry with two lone pairs on Cl atom.

c. It has T-shaped geometry with two lone pairs on Cl atom.

d. It has a trigonal pyramidal geometry with two lone pairs on Cl atom.

---

52. The functional group that can be identified through phthalein dye test is : **(+4, -1)**

a. Alcohol

b. Aldehyde

c. Phenolic

d. Carboxylic acid

---

53. A solution of copper sulphate is electrolysed for 10 minutes with a current of 1.5 amperes. The mass of copper deposited at cathode is : **(+4, -1)**
(Given : Molar mass of Cu = 63 g mol^{-1}, $1 \text{ F} = 96487 \text{ C mol}^{-1}$)

a. 0.2938 g

b. 1.7018 g

c. 2.4036 g

d. 0.5876 g

---

54. Match List I with List II : **(+4, -1)**

List I (Transition metal/compound)	List II (Catalytic Role)
A. V_2O_5	I. Preparation of ammonia from N_2/H_2 mixture
B. Fe	II. Polymerisation of alkynes
C. $PdCl_2$	III. Preparation of H_2SO_4 from SO_2
D. Ni complex	IV. Oxidation of ethyne to ethanal

Choose the correct answer :

a. A-III, B-I, C-II, D-IV

b. A-III, B-I, C-IV, D-II

c. A-I, B-III, C-IV, D-II

d. A-II, B-III, C-IV, D-III

---

55. Match List I with List II : **(+4, -1)**

List I (Complex/Ion)	List II (Shape/geometry)
A. $[Pt(Cl)_2(NH_3)_2]$	I. Octahedral
B. $[Co(NH_3)_6]Cl_3$	II. Trigonal bipyramidal
C. $[NiCl_4]^{2-}$	III. Square planar
D. $[Fe(CO)_5]$	IV. Tetrahedral

Choose the correct answer :

a. A-IV, B-I, C-III, D-II

b. A-III, B-I, C-IV, D-II

c. A-I, B-III, C-IV, D-II

d. A-III, B-IV, C-I, D-II

---

56. Identify the incorrect statement from the following : **(+4, -1)**

a. The largest and the smallest species among Mg , Mg^2, Al and Al^{3+} are Al and Mg^{2+} respectively

b. The oxidation state and covalency of Al in $[AlCl(H_2O)_5]^{2+}$ are +3 and 6 respectively.

c. The IUPAC name of the element with atomic number 107 is Unnilseptium.

d. The similarity in behaviour of Li with Mg is referred to as 'diagonal relationship"

---

57. Given below is an expression for the rate constant of a first order reaction occurring at a certain temperature, T (K). **(+4, -1)**

$\ln k = 14.34 - \frac{1.25 \times 10^4}{T}$

The energy of activation in $\mathrm{kcal\ mol^{-1}}$ for the reaction is :

(Given : k in $\mathrm{s^{-1}}$, $R = 1.987\ \mathrm{cal\ mol^{-1}\ K^{-1}}$)

a. 14.34

b. 18.63

c. 24.84

d. 12.42

58. Phenolphthalein is used as an indicator for the titration of sodium hydroxide solution against a standard solution of oxalic acid. The colour change that is observed at an alkaline pH close to the equivalence point during this titration is : (+4, -1)

a. pink to colourless

b. pinkish red to yellow

c. colourless to pink

d. yellow to pinkish red

59. Methane reacts with steam at $1273\ \mathrm{K}$ in the presence of nickel catalyst to form : (+4, -1)

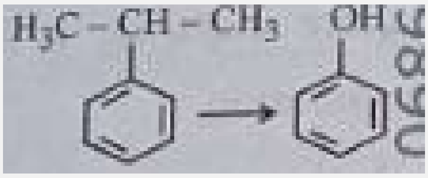
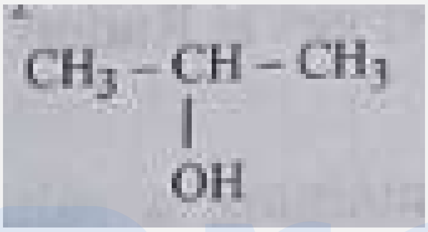
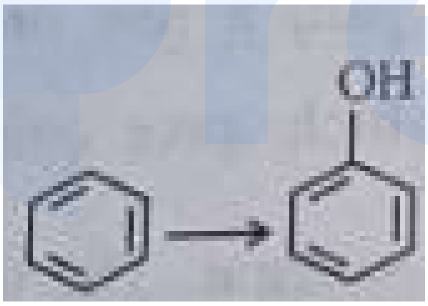
a. CO and H_2

b. CO_2 and H_2

c. CO and H_2O

d. CO_2 and H_2O

60. (+4, -1)

List I (Reactions)	List II (Reagents)
A. 	I. (i) oleum, (ii) $NaOH, \Delta$, (iii) H^+
B. $CH_3COOH \rightarrow CH_3CH_2OH$	II. (i) O_3, (ii) H_2O/H^+
C. $CH_3CH_2CH_2OH \rightarrow$ 	III. (i) CH_3OH, H^+, (ii) H_2, catalyst
D. 	IV. (i) conc. H_2SO_4, Δ, (ii) H_2O

Match List I with List II.

a. A-I, B-III, C-IV, D-II

b. A-II, B-III, C-I, D-IV

c. A-II, B-III, C-IV, D-I

d. A-II, B-IV, C-III, D-I

61. The pair of molecules that are metamers among the following is : (+4, -1)

a. $CH_3OCH_2CH_2CH_3$ and $CH_3CH_2OCH_2CH_3$

b. $CH_3CH_2CH_2CH_3$ and $CH_3CH(CH_3)CH_3$

c. 

d. $CH_3CH_2CH_2OH$ and $CH_3 - CH(OH) - CH_3$

---

62. Match List I with List II : **(+4, -1)**

List I (Complex)	List II (Type of Isomerism)
A. $[Pt(NH_3)_2Cl_2]$	I. Optical
B. $[Co(en)_3]^{3+}$	II. Solvate
C. $[Co(NH_3)_4(NO_2)_2]Cl$	III. Geometrical
D. $[Cr(H_2O)_6]Cl_3$	IV. Linkage

Choose the correct answer :

a. A-II, B-IV, C-III, D-I

b. A-III, B-I, C-II, D-IV

c. A-I, B-III, C-II, D-IV

d. A-III, B-II, C-IV, D-II

---

63. The number of hydrogen atoms present in 5.4 g of urea is : **(+4, -1)**
(Given : Molar mass of urea : $60\ \text{g mol}^{-1}$, $N_A : 6.022 \times 10^{23}$ particles mol^{-1})

a. 2.168×10^{22}

b. 2.168×10^{23}

c. 1.084×10^{22}

d. 1.084×10^{23}

---

64. The calculated 'spin-only' magnetic moment of Ti^{2+} $(3d^2)$ is : **(+4, -1)**

a. 3.87 BM

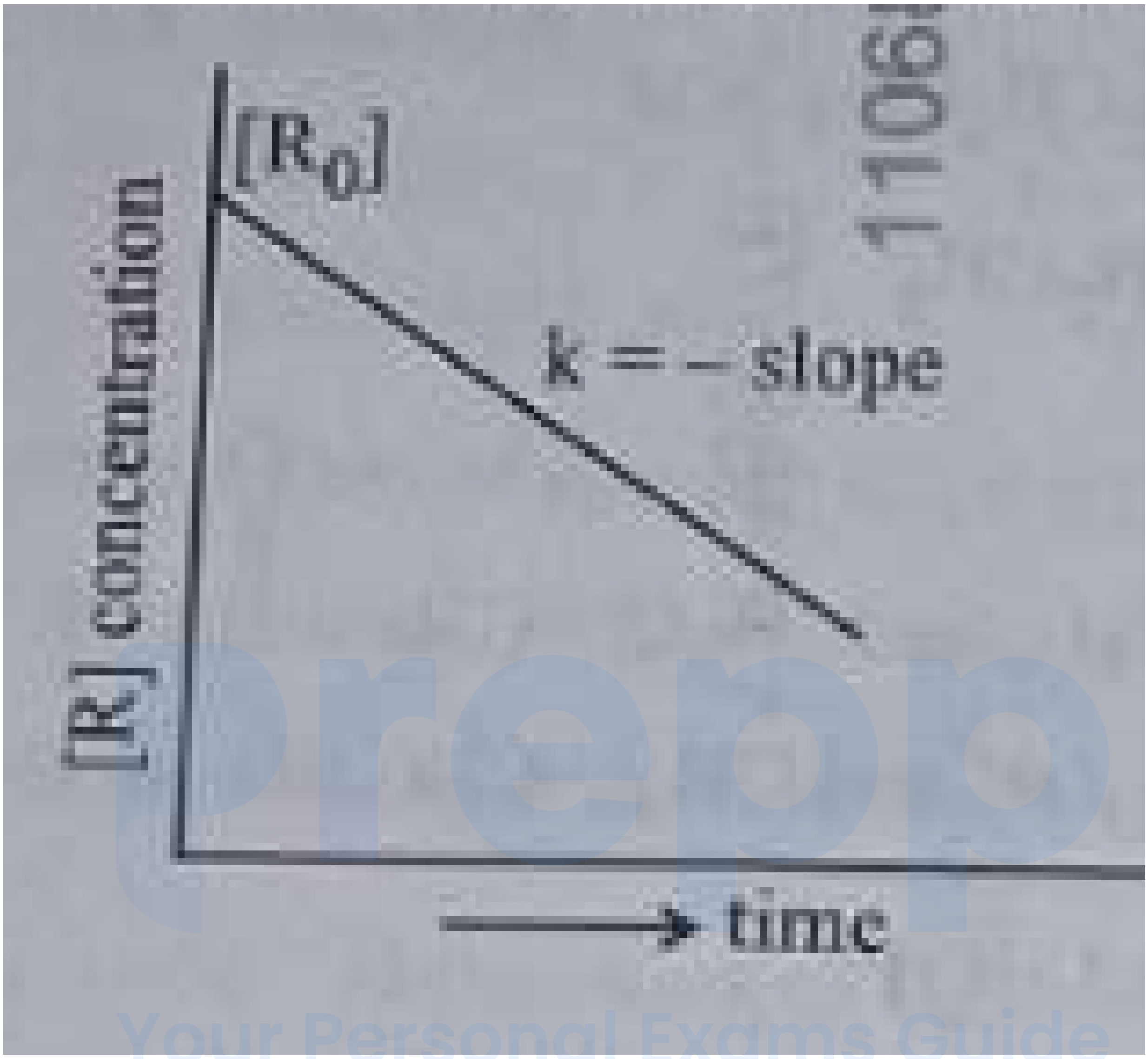
b. 5.92 BM

c. 4.90 BM

d. 2.84 BM

---

65. For a certain reaction R $\rightarrow$ Product, the plot of [R] vs time has a negative slope as shown. The order of reaction is : **(+4, -1)**



a. 1

b. 2

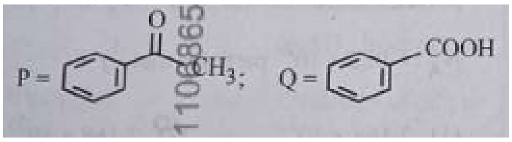
c. 2.5

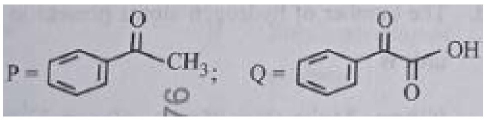
d. 0

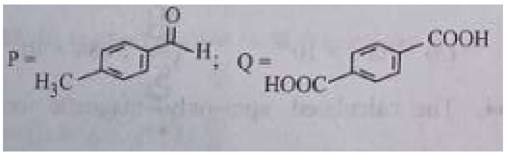
---

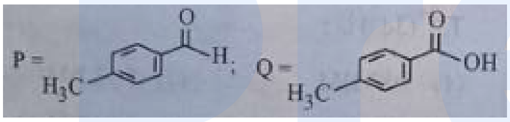
66. Compound P (C_8H_8O) gives a red orange precipitate with 2,4-DNP reagent and it does not reduce Fehling's reagent. On drastic oxidation with chromic acid, P gives an aromatic product Q that produces effervescence on treating with aq. $NaHCO_3$. Compounds P and Q, respectively, are : **(+4, -1)**

a.



b. 

c. 

d. 

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67. Match List I with List II : **(+4, -1)**

List I (Order of reaction)	List II (Unit of rate constant)
A. Zero order	I. $mol^{-1}Ls^{-1}$
B. First order	II. $mol^{-2}L^{2}s^{-1}$
C. Second order	III. s^{-1}
D. Third order	IV. $molL^{-1}s^{-1}$

Choose the correct answer from the options given below :

a. A-IV, B-III, C-I, D-II

b. A-IV, B-III, C-II, D-I

c. A-IV, B-II, C-I, D-III

d. A-I, B-II, C-III, D-IV

---

68. Although +3 oxidation state is most common in lanthanoids, cerium shows +4 oxidation state because : **(+4, -1)**

a. After losing one more electron, it acquires $4f^{14}$ electronic configuration.

b. Its atomic number is 61.

c. After losing one more electron, it acquires $4f^{0}$ electronic configuration.

d. Its nearest inert gas is Radon.

---

69. In a test tube containing a salt, a few drops of dilute H_2SO_4 was added, which gave colourless vapours having the smell of vinegar. The vapours turned the blue litmus paper red. Identify the correct anion from the following : **(+4, -1)**

a. Carbonate, CO_3^{2-}

b. Sulphate, SO_4^{2-}

c. Acetate, CH_3COO^-

d. Sulphide, S^{2-}

---

70. Calculate emf of the half cell given below : **(+4, -1)**

$$Pt(s)|H_2(g, 2\ \text{atm})|HCl(aq, 0.02\ \text{M})$$

$$E^{\circ}_{H_2/H^+} = 0\ \text{V}$$

(Given : $\frac{2.303RT}{F} = 0.059$, $\log 2 = 0.3010$)

a. 0.035 V

b. -0.035 V

c. -0.109 V

d. 0.109 V

---

71. At 298 K, a certain buffer solution contains equal concentrations of X^- and HX. K_b for X^- is 10^{-10}. What is the pH of this buffer solution ? **(+4, -1)**

a. 4

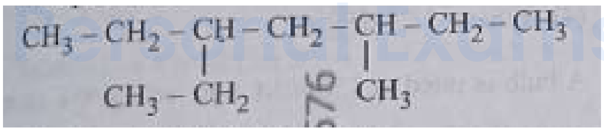
b. 6

c. 2

d. 10

---

72. The correct IUPAC name of the following compound is : **(+4, -1)**



a. 3-methyl-5-ethylhexane

b. 3-ethyl-5-methylheptane

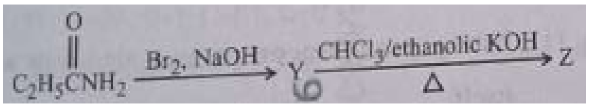
c. 3,5-diethylhexane

d. 2,4-diethylhexane

---

73. The following two reactions give the same foul smelling product Z. **(+4, -1)**

$$C_2H_5Cl \xrightarrow{X} Z$$



X and Z, respectively, are :

a. $X = AgCN$; $Z = C_2H_5CN$

b. $X = KCN$; $Z = C_2H_5CN$

c. $X = KCN$; $Z = C_2H_5NC$

d. $X = AgCN$; $Z = C_2H_5NC$

---

74. Mixture of chloroform and acetone forms a solution with negative deviation from Raoult's law due to **(+4, -1)**

a. increase in escaping tendency of molecules of each component.

b. repulsive forces.

c. stronger intermolecular forces between chloroform molecules than those between chloroform and acetone molecules.

d. formation of hydrogen bonding between acetone and chloroform.

---

75. Identify the correct statements : **(+4, -1)**

A. The molality of a 2.5 g of ethanoic acid (Molar mass : $60 \mathrm{~g~mol^{-1}}$) in 75 g of benzene solution is $0.556 \mathrm{~m}$.

B. The molarity of a solution containing 5 g of NaOH (molar mass : $40 \mathrm{~g~mol^{-1}}$) in 450 mL of solution is 0.278 M at 298 K.

C. Aquatic species are more comfortable in cold water.

D. The solubility of gas increases with decrease in pressure.

E. For a binary mixture of A and B, the number of moles of A and B are n_A and n_B respectively, the mole fraction of B will be $x_B = \frac{n_A}{n_A+n_B}$.
Choose the correct answer from the options given below :

a. A and B only

b. A, D and E only

c. A, B and C only

d. A and C only

---

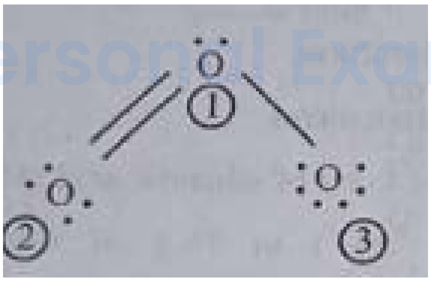
76. Identify the incorrect statement from the following : **(+4, -1)**

a. ECl_3 (E = B and Al) is a monomer when E = B and a dimer when E = Al.

b. The order of catenation property of Group 14 elements is $C \gg Si > Ge \approx Sn$.

c. Oxygen exhibits only - 2 oxidation state.

d. Carbon has the ability to form pπ-pπ multiple bond with itself.



77. The correct formal charges on oxygen atoms numbered 2, 1 and 3 respectively in ozone are : **(+4, -1)**

a. -1, 0, +1

b. 0, 0, 0

c. 0, +1, -1

d. +1, 0, -1

---

78. At a certain temperature T (K), during a process, 500 J is absorbed by the system and work of 200 J is done by the system. Then change in internal energy of the system is : (+4, -1)

a. 500 J

b. 400 J

c. 300 J

d. 700 J

---

79. Consider the following reaction : (+4, -1)
$2A(g) + B(g) \rightarrow 2D(g)$
$\Delta U^{\ominus} = -10 \text{ kJ mol}^{-1}$ and $\Delta S^{\ominus} = -44 \text{ J K}^{-1}$ at 298 K.
Identify the correct option with $\Delta G^{\ominus}$ for the reaction and spontaneity of the reaction at 298 K. (Given : $R = 8.31 \text{ J mol}^{-1}\text{K}^{-1}$)

a. $-0.63568 \text{ kJ mol}^{-1}$, spontaneous

b. $+0.63568 \text{ kJ mol}^{-1}$, non-spontaneous

c. $-1.635 \text{ kJ mol}^{-1}$, spontaneous

d. $+1.635 \text{ kJ mol}^{-1}$, non-spontaneous

---

80. Select the reagents that reduce nitriles to primary amines : (+4, -1)
A. (i) $LiAlH_4$; (ii) H_2O
B. $Sn + HCl$
C. H_2/Ni
D. $Na(Hg)/C_2H_5OH$
E. $Br_2/\text{aq. } NaOH$
Choose the correct answer from the options given below :

a. A, B and C only

b. A, C and D only

c. B, D and E only

d. A, D and E only

81. Which one of the following is an ambidentate ligand ? (+4, -1)

a. Oxalate

b. Ethylenediaminetetraacetate ion

c. Thiocyanate

d. Ethane-1,2-diamine

82. A bulb is rated at 150 watt, converting 8% energy into light. If energy of one photon is 4.42×10^{-19} J, how many photons are emitted by the bulb per second ? (+4, -1)

a. 1.35×10^{19}

b. 2.71×10^{19}

c. 27.2×10^{19}

d. 4.06×10^{19}

83. Identify the incorrect statement from the following : (+4, -1)

a. Nitrogen can form pπ-pπ multiple bonds with itself.

b. $P(C_2H_5)_3$ and $As(C_6H_5)_3$ form dπ-dπ bond with transition metals.

c. Nitrogen can form dπ-pπ bond with oxygen.

d. Phosphorus, arsenic and antimony show catenation property.

84. The correct order of increasing metallic character of Na, Be, P, Mg and Si is : **(+4, -1)**

a. P < Si < Be < Mg < Na

b. Be < Si < P < Mg < Na

c. P < Si < Na < Mg < Be

d. P < Mg < Be < Si < Na

85. In a qualitative analysis, Bi^{3+} is detected by appearance of precipitate of $BiO(OH)(s)$. Calculate pH when the following equilibrium exists at 298 K : **(+4, -1)**

$$BiO(OH)(s) \rightleftharpoons BiO^{+}(aq) + OH^{-}(aq)$$

$K = 4 \times 10^{-10}$
(Given : $\log 2 = 0.3010$)

a. 4.699

b. 9.301

c. 5.286

d. 8.714

86. Match List I with List II : **(+4, -1)**

List I	List II
A. C_2H_4	I. 3 σ bonds, 2 π bond
B. C_2H_2	II. 3 σ bonds, one lone pair
C. CH_4	III. 4 σ bonds
D. NH_3	IV. 5 σ bonds, 1 π bonds

Choose the correct answer from the options given below :

a. A-I, B-II, C-IV, D-III

b. A-II, B-III, C-I, D-IV

c. A-III, B-IV, C-II, D-I

d. A-IV, B-I, C-III, D-II

---

87. Given below are certain reactions. Identify the reaction for which $K_p \neq K_c$. **(+4, -1)**

a. $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

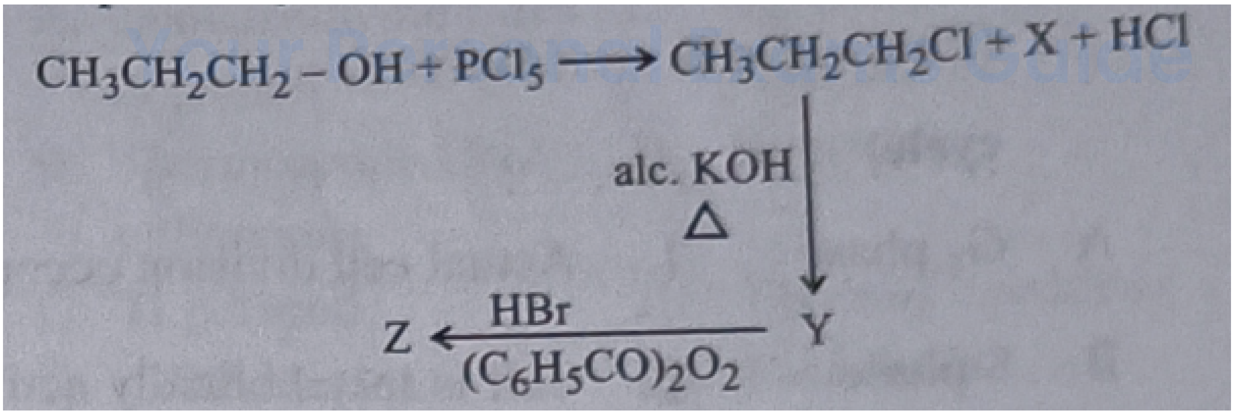
b. $H_2O(g) + CO(g) \rightleftharpoons H_2(g) + CO_2(g)$

c. $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

d. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

---

88. In the following reaction sequence, X and Z respectively are : **(+4, -1)**



a. $X = H_3PO_3;\ Z = CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_3$

b. $X = POCl_3;\ Z = CH_3 - \underset{\displaystyle Br}{\underset{|}{CH}} - CH_3$

c. X = H_3PO_3; Z = $CH_3CH_2CH_2 - Br$

d. X = $POCl_3$; Z = $CH_3CH_2 - Br$

---

89. The major product Z formed in the following sequence of reactions is : **(+4, -1)**

$$C_2H_6 \xrightarrow{Cl_2,\text{UV light}} X \text{ (monochlorinated product)} \xrightarrow{NH_3} Y \xrightarrow{(i)NaNO_2,HCl,(ii)H_2O} Z$$

a. $C_2H_5NO_2$

b. $C_2H_5 - N = N - OH$

c. $C_2H_5NH_2$

d. C_2H_5OH

---

90. When $1\ \text{dm}^3$ of CO_2 gas is passed over hot coke, the volume of gaseous mixture after complete reaction at STP becomes $1.4\ \text{dm}^3$. The composition of the gaseous mixture at STP is : **(+4, -1)**

a. $0.6\ \text{dm}^3$ of CO, $0.8\ \text{dm}^3$ of CO_2

b. $0.6\ \text{dm}^3$ of CO, $0.4\ \text{dm}^3$ of CO_2

c. $0.8\ \text{dm}^3$ of CO, $0.8\ \text{dm}^3$ of CO_2

d. $0.8\ \text{dm}^3$ of CO, $0.6\ \text{dm}^3$ of CO_2

Biology

91. Match List I with List II : **(+4, -1)**

List I (Phase of cell cycle)	List II (Activity)
A. G_1 phase	I. Actual cell division occurs
B. S phase	II. Cell is metabolically active and continuously grows but does not replicate its DNA
C. G_2 phase	III. Synthesis of DNA occurs and the amount of DNA per cell doubles
D. M phase	IV. Proteins are synthesized while cell growth continues

Choose the correct answer from the options given below :

a. A-II, B-III, C-IV, D-I

b. A-IV, B-I, C-II, D-III

c. A-III, B-IV, C-I, D-II

d. A-I, B-II, C-III, D-IV

92. Match List I with List II : **(+4, -1)**

List I	List II
A. Incomplete dominance	I. Human skin colour
B. Co-dominance	II. Inheritance of flower colour in Antirrhinum sp.
C. Pleiotropy	III. Phenylketonuria disease in humans
D. Polygenic inheritance	IV. ABO blood groups

Choose the correct answer from the options given below :

a. A-II, B-IV, C-III, D-I

b. A-I, B-IV, C-III, D-II

c. A-I, B-III, C-II, D-IV

d. A-II, B-I, C-III, D-IV

---

93. Which of the following statements are correct ? **(+4, -1)**

A. The Amazon rainforest being cut and cleared for cultivation of soyabeans is an example of habitat loss.
B. Steller's sea cow and passenger pigeon became extinct due to over-exploitation by humans.
C. The Nile perch introduced into Lake Victoria in East Africa helped in population growth of cichlid fish in the lake.
D. Water hyacinth is an invasive species.
E. When a species becomes extinct, the plant and animal species associated with it are not affected.
Choose the correct answer from the options given below :

a. B, C and D only

b. A, B and D only

c. A, B and E only

d. C, D and E only

---

94. Which of the following statements are correct with reference to a transcription unit ? **(+4, -1)**

A. A transcription unit in DNA is defined primarily by three regions : promoter, structural gene and terminator.
B. The promoter is said to be located towards the 5'-end of the structural gene.
C. The promoter is a DNA sequence that provides binding site for RNA

polymerase.
D. The promoter defines the template and coding strands.
E. The terminator is located towards the 3'-end of the coding strand and it defines the end of the process of transcription.
Choose the correct answer from the options given below :

a. A, C, D and E only

b. A, B, C, D and E

c. A, B, C and D only

d. A, B, C and E only

---

95. Which of the following statements are true with reference to the sex-determination in honeybees ? **(+4, -1)**

A. An offspring formed from the union of a sperm and an egg, develops as a female (queen or worker).
B. An unfertilized egg develops as a male by parthenogenesis.
C. A male has half the number of chromosomes than that of a female.
D. Males produce sperms by meiosis.
E. Honeybees have a haplodiploid sex-determination system.
Choose the correct answer from the options given below :

a. A, B, C and E only

b. B, C, D and E only

c. A, B, C and D only

d. A, B, D and E only

---

96. Match List I with List II : **(+4, -1)**

List I (Process)	List II (Location)
A. Glycolysis	I. Inner mitochondrial membrane
B. ETS	II. Mitochondrial matrix
C. Accumulation of protons	III. Cytoplasm
D. Krebs' cycle	IV. Intermembrane space

Choose the correct answer from the options given below :

a. A-IV, B-II, C-I, D-III

b. A-I, B-IV, C-III, D-II

c. A-II, B-III, C-IV, D-I

d. A-III, B-I, C-IV, D-II

---

97. How many ATP and NADPH molecules are required to make one molecule of glucose through the Calvin pathway ? **(+4, -1)**

a. 18 ATP and 12 NADPH

b. 6 ATP and 12 NADPH

c. 24 ATP and 18 NADPH

d. 12 ATP and 18 NADPH

---

98. Match List I with List II : **(+4, -1)**

List I	List II
A. Genetically modified organism	I. Agrobacterium tumefaciens
B. Thermostable DNA polymerase	II. Bt cotton
C. Ti plasmid	III. Thermus aquaticus
D. pBR322	IV. Escherichia coli

Choose the correct answer from the options given below :

a. A-II, B-III, C-I, D-IV

b. A-I, B-IV, C-II, D-III

c. A-I, B-II, C-IV, D-III

d. A-II, B-I, C-IV, D-III

---

99. In which one of the following, the ovules are not enclosed by an ovary wall and remain exposed ? **(+4, -1)**

a. Funaria

b. Pinus

c. Selaginella

d. Wolffia

---

100. The enzyme required for carboxylation in the Calvin cycle is : **(+4, -1)**

a. Carboxypeptidase

b. PEP carboxylase

c. RuBP carboxylase-oxygenase

d. Hexokinase

101. Match List I with List II : (+4, -1)

List I	List II
A. Trypsin	I. Intercellular ground substance
B. Morphine	II. Lectin
C. Concanavalin A	III. Enzyme
D. Collagen	IV. Alkaloid

Choose the correct answer from the options given below :

a. A-III, B-IV, C-II, D-I

b. A-III, B-II, C-IV, D-I

c. A-I, B-II, C-III, D-IV

d. A-IV, B-III, C-II, D-I

102. Which of the following statements are correct regarding amino acids ? (+4, -1)
A. They are substituted methanes.
B. Serine is an aromatic amino acid.
C. Valine is a neutral amino acid.
D. Lysine is an acidic amino acid.
Choose the correct answer from the options given below :

a. C and D only

b. A and C only

c. B and C only

d. A and B only

---

103. Which one of the following disorders is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule ? (+4, -1)

a. Thalassemia

b. Haemophilia

c. Sickle-cell anaemia

d. Phenylketonuria

---

104. Which one of the following is the site for active ribosomal RNA synthesis ? (+4, -1)

a. Kinetochore

b. Centrosome

c. Chromatin

d. Nucleolus

---

105. Which one of the following statements is not true about the universal rules of binomial nomenclature ? (+4, -1)

a. Both the words in a biological name, when handwritten, are separately underlined or printed in italics.

b. Biological names are generally in Latin.

c. The specific epithet in the biological name starts with a small letter.

d. The first word in the biological name represents the specific epithet, while the second component denotes the genus.

---

106. Match List I with List II : **(+4, -1)**

List I (Growth Regulator)	**List II (Function/Effect)**
A. 2,4-D	I. Brewing industry
B. GA_3	II. Stimulation of stomatal closure
C. Kinetin	III. Herbicide
D. ABA	IV. Nutrient mobilisation

Choose the correct answer from the options given below :

a. A-IV, B-III, C-II, D-I

b. A-I, B-II, C-IV, D-III

c. A-I, B-IV, C-III, D-II

d. A-III, B-I, C-IV, D-II

107. Match List I with List II : **(+4, -1)**

List I	**List II**
A. Conjunctive tissue	I. Specialised cells in the vicinity of guard cells
B. Casparian strips	II. Endodermal cells rich in starch
C. Subsidiary cells	III. Tissue between xylem and phloem
D. Starch sheath	IV. Endodermal cells with suberin deposition

Choose the correct answer from the options given below :

a. A-III, B-IV, C-I, D-II

b. A-IV, B-III, C-I, D-II

c. A-IV, B-III, C-II, D-I

d. A-III, B-IV, C-II, D-I

---

108. Which one of the following types of pollen grains brings genetically different types of pollen grains to the stigma ? **(+4, -1)**

a. Cleistogamy

b. Autogamy

c. Geitonogamy

d. Xenogamy

---

109. Heterophyllous development in response to environment is an example of which of the following phenomena ? **(+4, -1)**

a. Plasticity

b. Dedifferentiation

c. Redifferentiation

d. Elasticity

---

110. Which of the following statements are not true regarding restriction endonucleases ? **(+4, -1)**

A. They are called molecular scissors.
B. These are the enzymes responsible for restricting the growth of bacteriophages in E. coli.
C. They cut the DNA only at the centre of the palindromic sites.
D. They remove nucleotides only from the ends of DNA fragments.
E. They recognise specific palindromic base-pair sequences.
Choose the correct answer from the options given below :

a. C and D only

b. D and E only

c. A and B only

d. C and D only

---

111. Arrange the following steps of DNA fingerprinting in a correct sequence. (+4, -1)
A. Isolation of DNA and its digestion by restriction endonucleases.
B. Hybridisation using a labelled VNTR probe.
C. Transferring of separated DNA fragments to synthetic membranes.
D. Detection of hybridised DNA fragments by autoradiography.
E. Separation of DNA fragments by electrophoresis.
Choose the correct answer from the options given below :

a. A, B, D, C, E

b. A, E, C, B, D

c. A, D, B, E, C

d. A, E, B, C, D

---

112. Find the incorrect statement(s) about photosynthesis from the following : (+4, -1)
A. The water splitting complex is associated with PS II.
B. C_4 plants use the C_3 pathway of CO_2 fixation as the main biosynthetic pathway.
C. In C_4 plants, photorespiration does not occur.
D. C_3 plants exhibit "Kranz" anatomy.
E. ATP synthesis in chloroplast occurs through chemiosmosis.

a. B and C only

b. D only

c. A and D only

d. B and E only

---

113. Arrange the following steps of somatic hybridisation in a correct sequence. **(+4, -1)**

A. Digestion of cell walls.
B. Isolation of naked protoplasts.
C. Fusion of protoplasts to get hybrid protoplast.
D. Isolation of single cells from two different varieties of plants.
E. Growing of hybrid protoplast to form a new plant.

a. E, A, B, C, D

b. D, A, B, C, E

c. E, B, A, D, C

d. D, A, B, E, C

---

114. The main function of bulliform cells in grasses is : **(+4, -1)**

a. to make the leaf impermeable to fungal spores.

b. to perform photosynthesis.

c. to minimize water loss during water stress.

d. to transport water.

---

115. Arrange the following in the correct developmental sequence related to microsporogenesis : **(+4, -1)**

A. Microspore tetrads
B. Sporogenous tissue
C. Pollen grains
D. Pollen mother cells
Choose the correct answer from the options given below :

a. D, A, C, B

b. B, D, C, A

c. A, D, B, C

d. B, D, A, C

116. Which of the following is an in situ conservation method ? (+4, -1)

a. Seed Banks

b. Wildlife Safari Parks

c. Botanical Gardens

d. Sacred Groves

117. Which one of the following is a triploid cell ? (+4, -1)

a. Zygote

b. Central cell

c. Primary endosperm cell

d. Synergid

118. Since the origin and diversification of life on Earth, there have been five episodes of mass extinction of species. How is the sixth extinction, which is in progress, different from the previous episodes ? (+4, -1)

a. The current species extinction rates are far lower than those in previous episodes.

b. The current species extinction rate is nearly 10 times faster than that in previous episodes.

c. The present net species extinction rate is zero.

d. The present species extinction rates are 100 to 1000 times faster than in the pre-human times.

---

119. In the lac operon, the z gene codes for : (+4, -1)

a. permease

b. the repressor of lac operon

c. transacetylase

d. beta-galactosidase

---

120. In racemose inflorescence, _________________________ (+4, -1)

a. flowers are borne in an acropetal succession

b. flowers are solitary

c. the growth is limited

d. the main axis terminates in a flower

---

121. The main criteria used for Five Kingdom Classification proposed by R.H. Whittaker (1969) included : (+4, -1)

A. Cell structure
B. Body organization
C. Presence of flagellum
D. Reproduction
E. Phylogenetic relationships

Choose the correct answer from the options given below :

a. A, B, C, D and E

b. A, B, D and E only

c. B, C and D only

d. A, B and E only

---

122. "The Evil Quartet" of biodiversity loss includes which of the following ? (+4, -1)

a. Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions

b. Over-exploitation; Alien species invasions; Air pollution; Co-extinctions

c. Habitat loss and fragmentation; Air pollution; Water pollution; Co-extinctions

d. Over-exploitation; Alien species invasions; Soil pollution; Co-extinctions

---

123. Identify the correct sequence of steps in each cycle of Polymerase Chain Reaction : (+4, -1)

a. Annealing $\rightarrow$ Denaturation $\rightarrow$ Extension

b. Extension $\rightarrow$ Annealing $\rightarrow$ Denaturation

c. Denaturation $\rightarrow$ Extension $\rightarrow$ Annealing

d. Denaturation $\rightarrow$ Annealing $\rightarrow$ Extension

---

124. $2(C_{51}H_{98}O_6) + 145O_2 \rightarrow 102CO_2 + 98H_2O + \text{energy}$ (+4, -1)
The Respiratory Quotient (RQ) of a biomolecule used for respiration, as per the above equation, would be :

a. 1

b. Less than 0.5

c. Between 0.5 and 0.95

d. Between 1.25 and 2

---

125. Exploring molecular, genetic and species-level diversity for products of economic importance is called : **(+4, -1)**

a. Biomagnification

b. Bioremediation

c. Biofortification

d. Bioprospecting

---

126. Match List I with List II : **(+4, -1)**

List I	List II
A. Decomposition	I. Accumulation of dark coloured amorphous colloidal substance
B. Detritus	II. Release of inorganic nutrients by the activity of microbes in soil
C. Mineralisation	III. Breaking down of complex organic matter into inorganic substances
D. Humification	IV. Dead remains of plants and animals including fecal matter

Choose the correct answer from the options given below :

a. A-III, B-II, C-I, D-IV

b. A-I, B-II, C-III, D-IV

c. A-IV, B-III, C-I, D-II

d. A-III, B-IV, C-II, D-I

127. Identify the correct statements about biomolecules. (+4, -1)

A. Lipids are generally water soluble.

B. Proteins are polypeptides.

C. Polysaccharides are long chains of sugars.

D. Adenine and guanine are substituted pyrimidines.

E. Almost all enzymes are proteins.

Choose the correct answer from the options given below :

a. C, D and E only

b. B, C and E only

c. B, D and E only

d. A, B and C only

128. In angiosperms, root hairs arise from which one of the following regions of the root ? (+4, -1)

a. The region of meristematic activity

b. The root cap zone

c. The region of maturation

d. The region of elongation

129. Which one of the following is **not** a characteristic of plant cells in the phase of elongation ? (+4, -1)

a. New cell wall deposition

b. Increased vacuolation

c. Cell enlargement

d. Large conspicuous nuclei

130. Which of the following floral formula is the correct floral formula of Solanaceae family ? (+4, -1)

a. $\oplus$ ⚥ $K_{(5)}C_{(5)}A_5\underline{G}_{(2)}$

b. $\oplus$ ⚥ $K_5C_{(5)}A_5\underline{G}_{(2)}$

c. $\oplus$ ⚥ $K_5C_5A_5\underline{G}_{(2)}$

d. $\oplus$ ⚥ $K_{(5)}\overparen{C_{(5)}A_5}\underline{G}_{(2)}$

131. Which of the following statements are correct with reference to packaging of DNA helix ? (+4, -1)

A. Histones are organized to form a unit of eight molecules called histone octamer.

B. Histones are negatively charged basic proteins.

C. Histones are rich in the basic amino acid residues - lysine and arginine.

D. The positively charged DNA is wrapped around the histone octamer to form nucleosome.

E. The packaging of chromatin at higher levels requires an additional set of proteins called non-histone chromosomal proteins.

Choose the correct answer from the options given below :

a. A, B and D only

b. B, D and E only

c. A, C and E only

d. C, D and E only

132. Which of the following statements are correct with respect to DNA separation, isolation and visualization ? (+4, -1)

A. The cutting of DNA is done by molecular scissors.

B. The DNA fragments separate according to their size in an agarose gel, upon electrophoresis.

C. The separated DNA fragments can be seen without staining when exposed

to UV light.
D. The separated DNA fragments, when stained with ethidium bromide, can be seen in visible light.
Choose the correct answer from the options given below :

a. B and C only

b. B and D only

c. A and B only

d. A and D only

---

133. Alpha-helix is found in which level of protein structure ? **(+4, -1)**

a. Tertiary structure

b. Quaternary structure

c. Secondary structure

d. Primary structure

---

134. Match List I with List II : **(+4, -1)**

List I	**List II**
A. Productivity	I. Gross primary productivity minus respiration losses
B. Net primary productivity	II. Rate of formation of new organic matter by consumers
C. Gross primary productivity	III. Rate of biomass production
D. Secondary productivity	IV. Rate of production of organic matter during photosynthesis

Choose the correct answer from the options given below :

a. A-III, B-I, C-IV, D-II

b. A-I, B-II, C-III, D-IV

c. A-III, B-I, C-II, D-IV

d. A-I, B-III, C-IV, D-II

---

135. Match List I with List II : (+4, -1)

List I (Placentation)	List II (Example)
A. Marginal	I. Mustard
B. Axile	II. Pea
C. Parietal	III. Marigold
D. Basal	IV. Lemon

Choose the correct answer from the options given below :

a. A-I, B-III, C-II, D-IV

b. A-IV, B-II, C-I, D-III

c. A-II, B-IV, C-I, D-III

d. A-III, B-I, C-IV, D-II

---

136. Choose the correct statements regarding frog's anatomy : (+4, -1)
A. Hepatic portal system is the special venous connection between liver and intestine.
B. There are twelve pairs of cranial nerves arising from the brain.
C. Ureters and oviducts open separately into the cloaca in female frogs.

D. Hind-brain consists of cerebellum, medulla oblongata and optic lobes.
E. Sinus venosus joins the right atrium of heart.
Choose the correct answer from the options given below :

a. B and D only

b. A, B and C only

c. A, C and E only

d. B and C only

---

137. The flightless bird with forelimbs modified as paddle-like structures suited for swimming is known as : (+4, -1)

a. Aptenodytes

b. Neophron

c. Psittacula

d. Struthio

---

138. Male frogs can be distinguished from female frogs due to the presence of : (+4, -1)

a. Copulatory pad on first digit of fore limbs

b. Vocal sacs

c. Webbed digits in feet

d. Bulging eyes

---

139. A group of researchers procured some fish-like animals and upon investigation the following characters were observed : (+4, -1)
A. Endoskeleton is made of cartilage.
B. Ectoparasites, as they were found attached on fish skin with their circular

sucking mouth.
C. Paired fins and scales were absent, but 7 pairs of gill slits were present.
Which of the following species of animals did they consider to fit best with these characters ?

a. Petromyzon sp.

b. Branchiostoma sp.

c. Scoliodon sp.

d. Exocoetus sp.

---

140. In humans, respiration occurs in the following steps. Arrange these steps in the correct order. **(+4, -1)**
A. Diffusion of O_2 and CO_2 between blood and tissues
B. Diffusion of O_2 and CO_2 across alveolar membrane
C. Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out
D. Cellular respiration
E. Transport of gases by the blood
Choose the correct answer from the options given below :

a. A, B, C, D, E

b. E, A, C, D, B

c. C, B, E, A, D

d. C, A, B, E, D

---

141. Non-membrane bound cell organelles found in both prokaryotic and eukaryotic cells are : ____________________ **(+4, -1)**

a. Mitochondria

b. Lysosomes

c. Centrosomes

d. Ribosomes

142. Choose the correct statement regarding GIFT to overcome infertility. (+4, -1)

a. Ova collected from a female donor are transferred to the uterus of an infertile female.

b. Early embryos with up to 8 blastomeres are transferred into the fallopian tube of an infertile female.

c. It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

d. Early embryos with up to 8 blastomeres are transferred to the uterus of an infertile female.

143. Choose the correct statements regarding muscle contraction. (+4, -1)
A. A motor neuron carries a signal sent by the Central Nervous System (CNS) to the sarcolemma of the muscle fibre.
B. The neural signal generates an action potential which causes the release of Ca^{++} into sarcoplasm.
C. Increase in Ca^{++} inactivates the actin for breaking cross bridges.
D. Actin binds to the myosin head to form a cross bridge.
E. Shortening of sarcomere takes place, by pulling actin filaments towards the centre of 'A' band.
Choose the correct answer from the options given below :

a. C and E only

b. A, B, D and E only

c. A and B only

d. C and D only

144. Insertion of a foreign DNA at BamHI site in an E. coli cloning vector pBR322 results in the loss of antibiotic resistance towards : **(+4, -1)**

a. Ampicillin and tetracycline

b. Tetracycline

c. Ampicillin

d. Gentamycin

145. The specific receptors for neurotransmitters in a synapse are present on : **(+4, -1)**

----

a. Post-synaptic membrane

b. Pre-synaptic membrane

c. Myelin sheath

d. Schwann cell

146. Which of the following statements are correct with reference to human endoskeleton ? **(+4, -1)**

A. Human skull is monocondylic.
B. The joint between any two adjoining vertebrae is a cartilaginous joint.
C. In human beings the number of cervical vertebrae is seven.
D. All ribs except the last 2 pairs are bicephalic.
E. The occipital bone of skull is articulated with atlas vertebra.
Choose the correct answer from the options given below :

a. C, D and E only

b. B, C and E only

c. A, B and D only

d. B and E only

---

147. The human protein named α-1-antitrypsin, obtained from transgenic animals, is used for the treatment of : _________________- **(+4, -1)**

a. Alzheimer's disease

b. Emphysema

c. Cystic fibrosis

d. Rheumatoid arthritis

---

148. Select the incorrect statements with reference to Rh grouping. **(+4, -1)**

A. Erythroblastosis foetalis is a condition observed having fetus with Rh^{-ve} blood and mother with Rh^{+ve} blood.

B. Rh antigen is observed on RBCs in the majority of human beings.

C. Before blood transfusion, Rh group should also be matched.

D. Rh incompatibility is observed when a pregnant mother is Rh^{-ve} and the fetus is Rh^{+ve}.

E. Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the second child.

Choose the answer from the options given below :

a. B and C only

b. A and B only

c. A and E only

d. C and D only

---

149. Match List I with List II : **(+4, -1)**

List I (Drug)	List II (Effect)
A. Nicotine	I. Causes sense of euphoria and increased energy
B. Morphine	II. Stimulates adrenal gland to release catecholamines into blood circulation
C. Heroin	III. Effective sedative and painkiller
D. Cocaine	IV. A depressant; slows down body function

Choose the correct answer from the options given below :

a. A-II, B-III, C-IV, D-I

b. A-III, B-II, C-I, D-IV

c. A-III, B-II, C-IV, D-I

d. A-II, B-III, C-I, D-IV

150. Match List I with List II related to muscular/skeletal system : **(+4, -1)**

List I	List II
A. Tetany	I. Inflammation of joints
B. Arthritis	II. Autoimmune disorder affecting neuromuscular junction
C. Myasthenia gravis	III. Wild contraction in muscle due to low Ca^{++} in body fluid
D. Muscular dystrophy	IV. Progressive degeneration of skeletal muscle

Choose the correct answer from the options given below :

a. A-IV, B-II, C-III, D-I

b. A-III, B-I, C-II, D-IV

c. A-I, B-II, C-III, D-IV

d. A-III, B-II, C-I, D-IV

151. Match List I with List II : **(+4, -1)**

List I	**List II**
A. Progestasert	I. Barrier made of rubber used by females
B. Multiload 375	II. Oral contraceptive
C. Diaphragm	III. Hormone releasing IUD
D. Saheli	IV. Copper releasing IUD

Choose the correct answer from the options given below :

a. A-IV, B-II, C-I, D-III

b. A-III, B-IV, C-II, D-I

c. A-III, B-IV, C-I, D-II

d. A-IV, B-III, C-I, D-II

152. Select the correct statements regarding cell membrane in eukaryotic cell. **(+4, -1)**
A. Membrane of human RBCs has approximately 52% protein.
B. The phospholipids are arranged in a bilayer.
C. Extensions of the plasma membrane into the cell form mesosomes.
D. Tails towards the inner part of lipids are hydrophobic and thus protected from aqueous medium.
E. Glycocalyx is present on the outer surface of the plasma membrane.
Choose the correct answer from the options given below :

a. C, D and E only

b. B, C and E only

c. A, C and E only

d. A, B and D only

---

153. Choose the correct statements regarding cell organelles and their inclusions. **(+4, -1)**

A. The endomembrane system includes Golgi complex, endoplasmic reticulum and mitochondria.
B. Rough endoplasmic reticulum bears ribosomes on its surface.
C. Both mitochondria and plastids have circular DNA.
D. A network of microtubules, microfilaments and intermediate filaments present in the cytoplasm is called cytoskeleton.
E. Mitochondrion is a single membrane-bound structure.
Choose the correct answer from the options given below :

a. A, B and C only

b. A and B only

c. C, D and E only

d. B, C and D only

---

154. Match List I with List II related to embryonic development at various months of pregnancy : **(+4, -1)**

List I	List II
A. The foetus movement starts and hair appears on the head	I. 24 weeks of pregnancy
B. The foetus develops limbs and digits	II. 20 weeks of pregnancy
C. The foetus develops external genital organs	III. 8 weeks of pregnancy
D. The foetus body is covered with fine hair; eyelids separate and eyelashes are formed	IV. 12 weeks of pregnancy

Choose the correct answer from the options given below :

a. A-II, B-III, C-IV, D-I

b. A-III, B-II, C-IV, D-I

c. A-II, B-IV, C-III, D-I

d. A-IV, B-II, C-III, D-I

155. Which one of the following is an appropriate example of 'sexual deceit' ? **(+4, -1)**

a. Sea anemone and clown fish

b. Female wasp and fig

c. Ophrys and bumblebee

d. Cuckoo and crow

156. Select the set of fishes which belong to the class Osteichthyes : **(+4, -1)**

a. Saw fish, Fighting fish and Dog fish

b. Devil fish, Cuttlefish and Hagfish

c. Flying fish, Angel fish and Fighting fish

d. Starfish, Hagfish and Cuttlefish

157. Match List I with List II with respect to chronology of evolution of life forms : **(+4, -1)**

List I	**List II**
A. About 65 mya	I. Jawless fish probably evolved
B. About 500 mya	II. The dinosaurs suddenly disappeared from the earth
C. About 350 mya	III. Seaweeds and few plants probably existed
D. About 320 mya	IV. Invertebrates were formed and became active

Choose the correct answer from the options given below :

a. A-II, B-IV, C-I, D-III

b. A-I, B-II, C-III, D-IV

c. A-III, B-IV, C-I, D-II

d. A-II, B-IV, C-III, D-I

158. In which animal do haploid cells divide mitotically to produce gametes ? **(+4, -1)**

a. Male frogs

b. Male honeybees

c. Male grasshoppers

d. Male earthworms

159. The WBC count of a person's blood sample is 8000/cu.mm. How many eosinophils and lymphocytes would be in the same blood sample approximately ? (+4, -1)

a. 300 - 500/cu.mm. and 500 - 700/cu.mm. respectively

b. 300 - 500/cu.mm and 1200 - 1500/cu.mm. respectively

c. 100 - 120/cu.mm and 1600 - 2000/cu.mm. respectively

d. 160 - 240/cu.mm and 1600 - 2000/cu.mm. respectively

160. What is the probability of having children with 'O' blood group, where both mother and father are heterozygous for 'A' and 'B' blood group, respectively ? (+4, -1)

a. 50%

b. 75%

c. 0%

d. 25%

161. Arrange the following events occurring in Renin-Angiotensin mechanism in the correct order : (+4, -1)
A. Increase in blood pressure and Glomerular filtration rate.
B. Reabsorption of Na^+ and water from distal parts of tubule due to Aldosterone.
C. Fall in Glomerular filtration rate.
D. Vasoconstriction by Angiotensin II and release of Aldosterone.
E. Renin converts Angiotensinogen into Angiotensin I, followed by Angiotensin II.
Choose the correct answer from the options given below :

a. C, E, D, B, A

b. A, C, E, B, D

c. A, D, B, E, C

d. C, A, B, D, E

---

162. Choose the correct statements regarding population interactions between two species. **(+4, -1)**
A. In both parasitism and commensalism, only one species benefits and the other species is harmed.
B. Both species benefit in mutualism.
C. Both species benefit in commensalism.
D. In parasitism, only one species is harmed and the other species is unaffected.
E. In amensalism, one species is harmed and the other is unaffected.
Choose the correct answer from the options given below :

a. B, D and E only

b. B and E only

c. A and B only

d. A and D only

---

163. Spermatogonia undergo a series of cell divisions to produce sperms. Select the correct statements from the following : **(+4, -1)**
A. Spermatogonia always undergo meiotic cell division.
B. Primary spermatocytes divide mitotically to produce secondary spermatocytes.
C. Secondary spermatocytes, through their second meiotic division, produce haploid spermatids.
D. Spermatids produce spermatozoa through mitosis.
E. Spermatids transform into spermatozoa by spermiogenesis.
Choose the correct answer from the options given below :

a. C and E only

b. A, C and E only

c. B, C and D only

d. A and E only

---

164. The following are the stages of life cycle of *Plasmodium*. Arrange the stages in the proper order. **(+4, -1)**

A. The parasites reproduce asexually in RBCs, bursting the cells.
B. The parasites reproduce asexually in liver cells, bursting the cells and releasing into blood.
C. Gametocytes develop in RBCs.
D. Sporozoites reach the liver through the blood.
E. Female mosquito injects sporozoites into humans during bite.
Choose the correct answer from the options given below :

a. E, D, B, A, C

b. C, A, B, D, E

c. A, B, C, D, E

d. E, C, D, B, A

---

165. Match List I with List II : **(+4, -1)**

List I (Bioactive molecules)	List II (Importance)
A. Streptokinase	I. Immunosuppressive agent
B. Statins	II. Removal of clots from the blood vessels
C. Lipases	III. Blood cholesterol-lowering agent
D. Cyclosporin A	IV. Detergent formulations

Choose the correct answer from the options given below :

a. A-II, B-III, C-I, D-IV

b. A-IV, B-III, C-II, D-I

c. A-II, B-III, C-IV, D-I

d. A-III, B-II, C-IV, D-I

166. Which of the following is **not** an example of convergent evolution ? (+4, -1)

a. Wings of butterflies and birds

b. Flippers of penguins and dolphins

c. Fore limbs of whales and bats

d. Eyes of octopuses and mammals

167. What is the reason behind production of large holes in 'Swiss Cheese' ? (+4, -1)

a. The production of large amount of CO_2 and H_2 by *Trichoderma polysporum*

b. The production of large amount of CO_2 by *Clostridium butylicum*

c. The production of large amount of CO_2 and H_2 by lactic acid bacteria called *Lactobacillus*

d. The production of large amount of CO_2 by *Propionibacterium sharmanii*

168. Match List I with List II : (+4, -1)

List I	List II
A. Cortisol	I. Stimulates the formation of alveoli in mammary glands
B. Aldosterone	II. Produces anti-inflammatory reactions
C. Cholecystokinin	III. Stimulates reabsorption of Na^+ and water from renal tubule
D. Progesterone	IV. Stimulates secretion of pancreatic enzymes and bile juice

Choose the correct answer from the options given below :

a. A-III, B-II, C-IV, D-I

b. A-IV, B-II, C-I, D-III

c. A-II, B-III, C-I, D-IV

d. A-II, B-III, C-IV, D-I

169. Arrange the following cell layers/structures around the female gamete, from outer to inner side : **(+4, -1)**

A. Zona pellucida
B. Perivitelline space
C. Corona radiata
D. Plasma membrane of ovum

Choose the correct answer from the options given below :

a. C, A, D, B

b. C, A, B, D

c. D, B, A, C

d. A, C, B, D

170. Which of the following equations depicts Verhulst-Pearl logistic population growth ? (+4, -1)

a. $\frac{dN}{dt} = rN\left(\frac{K-N}{N}\right)$

b. $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$

c. $\frac{dN}{dt} = rN\left(\frac{K}{K-N}\right)$

d. $\frac{dN}{dt} = rN\left(\frac{K+N}{K}\right)$

171. The toxin proteins isolated from *Bacillus thuringiensis*, coded by which of the following genes would control cotton bollworms and corn borer, respectively ? (+4, -1)

a. *cryIAc* and *cryIAb*

b. *cryIIAb* and *cryIAc*

c. *cryIAc* and *cryIIIAb*

d. *cryIAc* and *cryIIAb*

172. The JGA (Juxta Glomerular Apparatus) is a special sensitive region formed by cellular modifications in ________ related to the same nephron. (+4, -1)

a. Proximal convoluted tubule and afferent renal arteriole

b. Distal convoluted tubule and efferent renal arteriole

c. Proximal convoluted tubule and efferent renal arteriole

d. Distal convoluted tubule and afferent renal arteriole

173. Match List I with List II : (+4, -1)

List I	List II
A. Molluscs	I. Pulmonary respiration only
B. Reptiles	II. Branchial respiration
C. Adult amphibians	III. Cellular respiration
D. Amoeba	IV. Pulmonary and Cutaneous respiration

Choose the correct answer from the options given below :

a. A-III, B-II, C-I, D-IV

b. A-II, B-I, C-IV, D-III

c. A-II, B-I, C-III, D-IV

d. A-I, B-II, C-IV, D-III

---

174. The sixth mutant codon of beta globin gene causing polymerization of Haemoglobin and change in RBC shape is **(+4, -1)**

a. CAG

b. GUG

c. AUG

d. GAG

---

175. In a population of a grasshopper species, the chromosome number of some members is 23 and some other members possess 24 chromosomes. The 23 and 24 chromosome-bearing members in this species are ________ . **(+4, -1)**

a. females and males, respectively

b. all males

c. males and females, respectively

d. all females

---

176. Select the incorrect statements from the following : (+4, -1)

A. Digestive system in Platyhelminthes is incomplete.
B. Bilateral symmetry is a characteristic feature of adult Echinoderms.
C. Pseudocoelom is possessed by Aschelminthes.
D. Notochord is persistent throughout life in the class Chondrichthyes.
E. Members of class Reptilia maintain a constant body temperature.
Choose the answer from the options given below :

a. A and C only

b. B and E only

c. C and D only

d. B and D only

---

177. The following reaction depicts the activity of a particular class of enzymes : (+4, -1)

$$\begin{array}{cc} \mathrm{X} & \mathrm{Y} \\ | & | \\ \mathrm{C}- & \mathrm{C} \end{array} \xrightarrow{E} \mathrm{X}-\mathrm{Y}+\mathrm{C}=\mathrm{C}$$

(Substrate) (Product) (Product)
Identify the enzyme class 'E' from the following options :

a. Ligases

b. Transferases

c. Lyases

d. Isomerases

---

178. Ecological pyramids represent the relationship between the organisms at different trophic levels and they are generally inverted for : (+4, -1)

a. Pyramid of biomass in grassland

b. Pyramid of biomass in sea

c. Pyramid of number in grassland

d. Pyramid of energy in pond ecosystem

179. Evolution of human appears parallel to the progressive development of brain and language skills. As such, the evolution of individual species in the sequence of their appearance is : (+4, -1)

a. Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens

b. Homo habilis → Homo erectus → Ramapithecus → Neanderthal → Homo sapiens

c. Homo sapiens → Ramapithecus → Homo habilis → Neanderthal → Homo erectus

d. Neanderthal → Ramapithecus → Homo habilis → Homo erectus → Homo sapiens

180. Match List I with List II : (+4, -1)

List I (Respiratory Volume)	List II (Capacity in mL)
A. ERV (Expiratory Reserve Volume)	I. 2500 - 3000 mL
B. RV (Residual Volume)	II. 500 mL
C. IRV (Inspiratory Reserve Volume)	III. 1000 - 1100 mL
D. TV (Tidal Volume)	IV. 1100 - 1200 mL

Choose the correct answer from the options given below :

a. A–I, B–II, C–III, D–IV

b. A–III, B–IV, C–I, D–II

c. A–III, B–I, C–IV, D–II

d. A–I, B–III, C–II, D–IV

Answers

1. **Answer: d**

Explanation:

The question involves identifying the correct velocity-time plot for a ball thrown vertically upwards and falling back down. Here, we analyze the plots based on the motion characteristics of the ball:

1. Initially, the ball is thrown upwards with an initial velocity, so the velocity is positive.
2. As the ball rises, it slows down due to gravity, and the velocity decreases until it reaches zero at the highest point.
3. After reaching the highest point, the ball falls back, gaining speed in the negative direction (downward), so the velocity becomes negative.

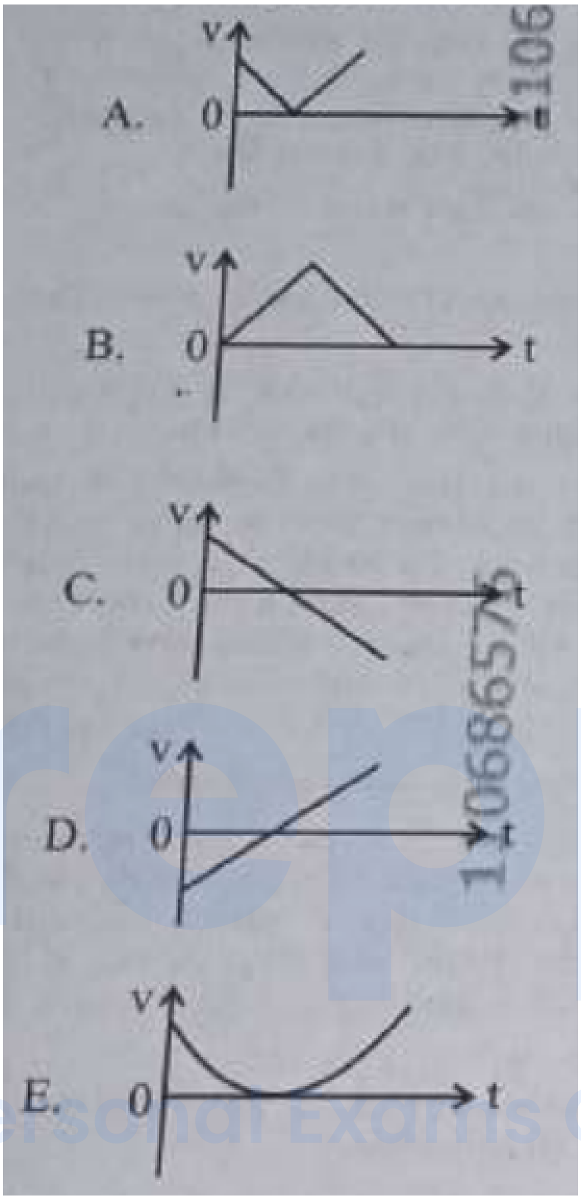
Based on these characteristics, the correct plot should start from a positive velocity, decrease to zero, and then continue into negative values.

Let's examine the options:

1. **Option A**: The plot shows a sudden change in velocity, which is incorrect as the velocity decreases gradually.
2. **Option B**: The plot shows velocity decreasing to zero and remaining constant, which is incorrect as velocity cannot remain zero; it should become negative upon descent.
3. **Option C**: This plot starts with a positive velocity, decreases to zero, and becomes negative. This correctly represents the motion of the ball.
4. **Option D**: The plot suggests a continuous increase in velocity, which does not align with the motion described.
5. **Option E**: The plot shows a continuous curve without a distinct transition to and from zero, which is incorrect for this scenario.

Thus, the correct answer is:

C only



2. **Answer: d**

Explanation:

Understanding the Photoelectric Effect Condition

The photoelectric effect happens only if the energy of the incident photon (E_{photon}) meets or exceeds the metal's work function (ϕ).

Photon energy is calculated using $E_{photon} = \frac{hc}{\lambda}$, where h is Planck's constant, c is the speed of light, and λ is the wavelength.

The condition for the effect is: $\frac{hc}{\lambda} \geq \phi$ This means the effect occurs for wavelengths $\lambda \leq \frac{hc}{\phi}$.

Conversely, the effect *does not* occur if the photon energy is less than the work function: $\frac{hc}{\lambda} < \phi$ This condition implies the effect is absent for wavelengths $\lambda > \frac{hc}{\phi}$. The critical wavelength $\lambda_0 = \frac{hc}{\phi}$ is known as the threshold wavelength.

Calculating the Threshold Wavelength

We are given:

- Work function, $\phi = 6.6 \text{ eV}$
- Planck's constant, $h = 6.6 \times 10^{-34} \text{ J s}$
- Speed of light, $c \approx 3 \times 10^{8} \text{ m/s}$
- Conversion factor, $1 \text{ eV} \approx 1.602 \times 10^{-19} \text{ J}$

First, calculate the value of hc:

$$hc = (6.6 \times 10^{-34} \text{ J s}) \times (3 \times 10^{8} \text{ m/s}) = 1.98 \times 10^{-25} \text{ J m}$$

Next, convert the work function ϕ from electron volts (eV) to Joules (J):

$$\phi = 6.6 \text{ eV} \times (1.602 \times 10^{-19} \text{ J/eV}) \approx 1.05732 \times 10^{-18} \text{ J}$$

Now, calculate the threshold wavelength λ_0 using the formula $\lambda_0 = \frac{hc}{\phi}$:

$$\lambda_0 = \frac{1.98 \times 10^{-25} \text{ J m}}{1.05732 \times 10^{-18} \text{ J}} \approx 1.8726 \times 10^{-7} \text{ m}$$

Convert this threshold wavelength to nanometers (nm):

$$\lambda_0 \approx 1.8726 \times 10^{-7} \text{ m} \times \frac{10^{9} \text{ nm}}{1 \text{ m}} \approx 187.3 \text{ nm}$$

Identifying the Wavelength Without Photoelectric Effect

The photoelectric effect will not occur for incident radiation wavelengths (λ) that are greater than the threshold wavelength (λ_0).

We need to find the option where $\lambda > 187.3 \text{ nm}$. Let's check the given options:

- Option 1: 50 nm $(50 < 187.3)$ - Effect occurs.

- Option 2: 100 nm $(100 < 187.3)$ - Effect occurs.
- Option 3: 150 nm $(150 < 187.3)$ - Effect occurs.
- Option 4: 200 nm $(200 > 187.3)$ - Effect does NOT occur.

Therefore, the wavelength of incident radiation that does not give rise to the photoelectric effect is 200 nm.

---

3. **Answer: c**

Explanation:

Calculate Crane Power

The problem asks for the power of a crane based on the mass it lifts, the height it lifts it to, and the time taken. We are given:

- Mass (m): 1000 kg
- Height (h): 20 m
- Time (t): 10 s
- Acceleration due to gravity (g): 9.8 m/s^2

Power is the rate at which work is done. First, we calculate the work done by the crane in lifting the mass against gravity.

Work Calculation

The work done (W) is equal to the potential energy gained by the mass, calculated using the formula:

$$W = mgh$$

Substituting the given values:

$$W = (1000 \text{ kg}) \times (9.8 \text{ m/s}^2) \times (20 \text{ m})$$

$$W = 196000 \text{ J}$$

Power Calculation

Power (P) is calculated by dividing the work done by the time taken:

$$P = \frac{W}{t}$$

Substituting the calculated work and given time:

$$P = \frac{196000 \text{ J}}{10 \text{ s}}$$

$$P = 19600 \text{ W}$$

Unit Conversion

The options are given in kilowatts (kW). To convert watts to kilowatts, we divide by 1000:

$$P = \frac{19600 \text{ W}}{1000 \text{ W/kW}}$$

$$P = 19.6 \text{ kW}$$

Therefore, the power of the crane is 19.6 kW.

---

4. **Answer: d**

Explanation:

Physics MCQs: Matching Light Concepts

This solution addresses the question of matching concepts related to light and matter waves.

Concepts in List I and List II

List I	List II
A. $E = h\nu$	I. de Broglie wavelength
B. Interference	II. Particle nature of light
C. $\lambda = h/p$	III. Wave nature of light
D. Compton effect	IV. Energy of photon

Matching Process

The matching is performed as follows:

- **A.** $E = h\nu$: This equation defines the energy (E) of a photon based on its frequency (ν) and Planck's constant (h). It correctly matches with **IV. Energy of photon**.
- **B. Interference**: This phenomenon is matched with **II. Particle nature of light**.
- **C.** $\lambda = h/p$: This formula describes the de Broglie wavelength, relating a particle's momentum (p) to its associated wavelength (λ). This matches **I. de Broglie wavelength**.
- **D. Compton effect**: This effect is matched with **III. Wave nature of light**.

Final Answer Derivation

The established pairings are A-IV, B-II, C-I, and D-III.

This combination corresponds to Option D.

---

5. **Answer: b**

Explanation:

Calculate Acceleration Magnitude and Direction for Perpendicular Forces

This solution explains how to find the acceleration's magnitude and direction when forces act perpendicularly on an object.

Given Information

- Mass of the body, $m = 5\ \mathrm{kg}$
- First force, $F_1 = 8\ \mathrm{N}$

- Second force, $F_2 = 6\ \mathrm{N}$
- The forces F_1 and F_2 are mutually perpendicular.

Calculate Net Force

Since the forces are perpendicular, we find the magnitude of the resultant net force (F_{net}) using the Pythagorean theorem:

$F_{net} = \sqrt{F_1^2 + F_2^2}$

Substituting the values:

$F_{net} = \sqrt{(8\ \mathrm{N})^2 + (6\ \mathrm{N})^2}$

$F_{net} = \sqrt{64\ \mathrm{N}^2 + 36\ \mathrm{N}^2}$

$F_{net} = \sqrt{100\ \mathrm{N}^2}$

$F_{net} = 10\ \mathrm{N}$

Calculate Acceleration Magnitude

Using Newton's second law, $F_{net} = m \times a$, we can find the acceleration (a):

$a = \frac{F_{net}}{m}$

Substituting the values:

$a = \frac{10\ \mathrm{N}}{5\ \mathrm{kg}}$

$a = 2\ \mathrm{m\ s^{-2}}$

Determine Acceleration Direction

The acceleration is in the direction of the net force. We find the angle θ the net force makes with the $8\ \mathrm{N}$ force.

The tangent of this angle is the ratio of the perpendicular force (F_2) to the force it's measured against (F_1):

$\tan(\theta) = \frac{F_2}{F_1}$

$\tan(\theta) = \frac{6\,\text{N}}{8\,\text{N}}$

$\tan(\theta) = \frac{3}{4}$

Therefore, the angle is:

$\theta = \tan^{-1}\left(\frac{3}{4}\right)$

The direction is $\tan^{-1}(3/4)$ with the $8\,\text{N}$ force.

Final Result

The magnitude of the acceleration is $2\,\text{m}\,\text{s}^{-2}$ and its direction is $\tan^{-1}(3/4)$ with the $8\,\text{N}$ force.

---

6. **Answer: b**

Explanation:

Pendulum Energy and Speed Calculation

The total mechanical energy (E) of a simple pendulum is the sum of its kinetic energy (KE) and potential energy (PE): $E = KE + PE$

We are given that the total energy is $E = 0.02\,\text{J}$.

At the **equilibrium position**, the potential energy (PE) is minimum, typically considered zero. Therefore, at this point, all the energy is kinetic energy (KE_{max}). $E = KE_{max} + 0$
$KE_{max} = E = 0.02\,\text{J}$

Calculating Speed at Equilibrium

The formula for kinetic energy is $KE = \frac{1}{2}mv^2$, where m is the mass and v is the speed.

Given:

- Total Energy, $E = 0.02\,\text{J}$
- Mass, $m = 20\,\text{g} = 0.02\,\text{kg}$ (converted to kilograms)

At equilibrium, $KE_{max} = \frac{1}{2}mv^2$. We set this equal to the total energy:

$0.02\text{ J} = \frac{1}{2} \times (0.02\text{ kg}) \times v^2$

Now, solve for v:

$0.02 = 0.01 \times v^2$

$v^2 = \frac{0.02}{0.01}$

$v^2 = 2$

$v = \sqrt{2}\text{ m/s}$

Calculating the square root:

$v \approx 1.414\text{ m/s}$

Rounding to two decimal places, the speed is approximately 1.41 m/s.

---

7. **Answer: a**

Explanation:

Maximum Trolley Acceleration Calculation

The problem asks for the maximum horizontal acceleration (a_{max}) a trolley can achieve while a box placed on it remains stationary relative to the trolley. This scenario is governed by the principles of static friction.

Physics Principles Involved

For the box to remain stationary on the accelerating trolley, the static friction force (f_s) acting on the box must provide the necessary centripetal force (or in this case, the force causing linear acceleration).

- According to Newton's second law, the net force on the box horizontally is $F_{net} = m \times a$.
- The static friction force provides this net force: $f_s = m \times a$.

- The static friction force has a maximum value, $f_{s,max}$, given by $f_{s,max} = \mu_s \times N$, where μ_s is the coefficient of static friction and N is the normal force.
- Since the trolley is moving horizontally and the box is on it, the normal force (N) exerted by the trolley on the box equals the weight of the box (mg), assuming no vertical acceleration. Thus, $N = mg$.
- Substituting $N = mg$ into the equation for maximum static friction gives: $f_{s,max} = \mu_s \times mg$.

Determining Maximum Acceleration

To find the maximum acceleration (a_{max}) the trolley can have while keeping the box stationary, we set the required force $(m \times a_{max})$ equal to the maximum available static friction force $(f_{s,max})$:

$$m \times a_{max} = f_{s,max}$$

$$m \times a_{max} = \mu_s \times mg$$

The mass of the box (m) cancels out from both sides:

$$a_{max} = \mu_s \times g$$

Calculation

Given:

- Coefficient of static friction, $\mu_s = 0.12$
- Acceleration due to gravity, $g = 10 \text{ m/s}^2$

Substitute the values into the formula:

$$a_{max} = 0.12 \times 10 \text{ m/s}^2$$

$$a_{max} = 1.2 \text{ m/s}^2$$

Therefore, the maximum acceleration with which the trolley can be moved horizontally while keeping the box stationary is 1.2 m/s^2. This corresponds to Option A.

---

8. Answer: c

Explanation:

Sun Earth Distance Calculation

The problem asks us to find the distance between the Sun and the Earth using a new unit system where the speed of light is considered unity (1). We are given the time it takes for light to travel this distance.

Key Information Provided

- Speed of light (c) = 1 (in the new unit system).
- Time taken (t) = 6 minutes 40 seconds.

Step 1: Convert Time to Seconds

First, we need to express the total time in a single unit, preferably seconds, to be consistent with standard calculations.

$t = (6 \text{ minutes} \times 60 \frac{\text{seconds}}{\text{minute}}) + 40 \text{ seconds}$

$t = 360 \text{ seconds} + 40 \text{ seconds}$

$t = 400 \text{ seconds}$

Step 2: Calculate Distance

We use the fundamental relationship between distance, speed, and time: Distance = Speed × Time.

In this specific problem:

- Speed (c) = 1 (unitless, as speed of light is taken as unity).
- Time (t) = 400 seconds.

Therefore, the distance (d) is:

$d = c \times t$

$d = 1 \times 400 \text{ seconds}$

$d = 400$

Since the speed of light is taken as 1 unit of distance per unit of time (here, seconds), the distance calculated is in these new units. The unit would effectively be 'light-seconds'.

Conclusion

The distance between the Sun and the Earth, in the new unit system where the speed of light is unity, is 400 units (or 400 light-seconds).

---

9. **Answer: b**

Explanation:

To determine the form of the voltage appearing across the diode D in the given circuit, let's analyze the circuit operation:

Circuit Overview:

- The circuit consists of an AC voltage source V_i, a diode D, and a resistor R.

Operation of the Diode:

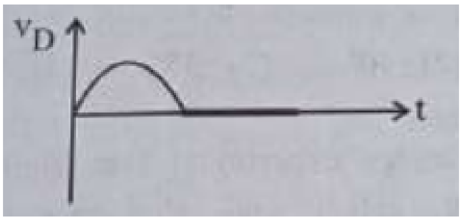
- Diodes allow current to pass through them only in one direction: forward biased and block the current in reverse bias.
- During the positive half-cycle of V_i, the diode becomes forward-biased, and it conducts, allowing the voltage across R and essentially shorting the diode to a very low forward voltage drop.
- During the negative half-cycle of V_i, the diode is reverse-biased, and it does not conduct, resulting in the voltage across the diode equal to the negative of the input voltage.

Voltage Across the Diode V_D:

- When the diode is forward-biased, V_D is approximately 0 (or the forward voltage drop, typically about 0.7 V for silicon diodes, but can be assumed negligible for ideal cases).
- When the diode is reverse-biased, V_D equals $-V_i$.

Conclusion:

The voltage appearing across the diode D will therefore have a waveform similar to the negative half of the input voltage waveform when the diode is non-conducting and be near zero during the positive cycle when conducting.



Hence, the correct answer is the waveform option that represents a negative clipping of the input voltage.

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10. **Answer: a**

Explanation:

Submarine Depth Calculation: Pressure Analysis

The problem asks for the maximum depth a submarine can reach based on its tolerance to absolute pressure.

Key Information Provided

- Maximum Absolute Pressure (P_{abs_max}) the submarine can withstand: 100 atm
- Density of water (ρ): 1000 kg m^{-3}
- Atmospheric pressure conversion: $1 \text{ atm} = 1 \times 10^5 \text{ Pa}$
- Gravitational acceleration (g): 10 m/s^2
- Assume standard atmospheric pressure at the surface (P_{atm}): $1 \text{ atm} = 1 \times 10^5 \text{ Pa}$

Calculating Maximum Depth

The absolute pressure (P_{abs}) at a certain depth (h) in water is given by the formula:

$P_{abs} = P_{atm} + \rho g h$

First, convert the maximum withstandable pressure to Pascals:

$P_{abs\_max} = 100 \text{ atm} \times (1 \times 10^5 \text{ Pa / atm}) = 1 \times 10^7 \text{ Pa}$

Now, set the absolute pressure equal to the maximum withstandable pressure and solve for depth (h):

$1 \times 10^7 \text{ Pa} = (1 \times 10^5 \text{ Pa}) + (1000 \text{ kg m}^{-3}) \times (10 \text{ m/s}^2) \times h$

Subtract the atmospheric pressure from the maximum absolute pressure:

$P_{water} = P_{abs\_max} - P_{atm}$ $P_{water} = (1 \times 10^7 \text{ Pa}) - (1 \times 10^5 \text{ Pa})$ $P_{water} = 10000000 \text{ Pa} - 100000 \text{ Pa} = 9900000 \text{ Pa}$

This remaining pressure is due to the water column. Use the formula $P_{water} = \rho g h$ to find h:

$9900000 \text{ Pa} = (1000 \text{ kg m}^{-3}) \times (10 \text{ m/s}^2) \times h$ $9900000 = 10000 \times h$

Solve for h:

$h = \frac{9900000}{10000}$ $h = 990 \text{ m}$

Therefore, the submarine can go 990 m deep.

---

11. **Answer: d**

Explanation:

Thermodynamics First Law Principle

The First Law of Thermodynamics relates heat, work, and internal energy. For a system, the rate of change of internal energy is equal to the rate at which heat is supplied minus the rate at which work is done by the system.

Mathematically, this is expressed as: $\frac{dU}{dt} = \frac{dQ}{dt} - \frac{dW}{dt}$ Or using rate notation: $\dot{U} = \dot{Q} - \dot{W}$ Where:

- $\dot{U}$ is the rate of change of internal energy (in Watts, W).
- $\dot{Q}$ is the rate of heat supplied to the system (in Watts, W).

- $\dot{W}$ is the rate of work done by the system (in Watts, W or Joules per second, J/s).

Calculating Rate of Internal Energy Increase

Given:

- Rate of heat supplied, $\dot{Q} = 100 \text{ W}$.
- Rate of work done by the system, $\dot{W} = 75 \text{ J/s} = 75 \text{ W}$.

Substitute the given values into the First Law equation:

$\dot{U} = 100 \text{ W} - 75 \text{ W}$ $\dot{U} = 25 \text{ W}$

Therefore, the rate at which the internal energy of the system increases is 25 W.

Answer Selection

The calculated rate of increase in internal energy is 25 W, which corresponds to Option D.

---

12. **Answer: c**

Explanation:

Calculate Coil Current

The magnetic field (B) at the center of a circular coil is related to the current (I) using the formula $B = \frac{\mu_0 N I}{r}$.

Rearranging for current I:

$I = \frac{rB}{\mu_0 N}$

Given values:

- Radius $r = 10 \text{ cm} = 0.1 \text{ m}$
- Magnetic field $B = 3.14 \times 10^{-3} \text{ T}$
- Number of turns $N = 100$
- $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$

Substitute values, using $\pi \approx 3.14$:

$I = \frac{(0.1 \text{ m})\times(3.14\times10^{-3} \text{ T})}{(4\times3.14\times10^{-7} \text{ T m/A})\times100}$

$I = \frac{0.1\times3.14\times10^{-3}}{4\times3.14\times10^{-5}} = \frac{0.1\times10^{-3}}{4\times10^{-5}} = 0.025 \times 10^2 = 2.5 \text{ A}$

Calculate Coil Magnetic Moment

The magnetic moment (μ) is calculated using $\mu = NIA$, where A is the area of the coil ($A = \pi r^2$).

Using the calculated current $I = 2.5 \text{ A}$ and the required magnetic moment $\mu = 2 \text{ A m}^2$ from the option:

The effective area A is determined by $A = \frac{\mu}{NI}$:

$A = \frac{2 \text{ A m}^2}{100\times2.5 \text{ A}} = \frac{2}{250} \text{ m}^2 = 0.008 \text{ m}^2$.

This results in the pair: Current $I = 2.5 \text{ A}$ and Magnetic Moment $\mu = 2 \text{ A m}^2$.

---

13. **Answer: c**

Explanation:

Young's Double Slit Intensity Calculation

In Young's double-slit experiment, the intensity (I) of light at a point on the screen depends on the phase difference (ϕ) between the waves from the two slits. The intensity is given by the formula:

$I = I_{max} \cos^2\left(\frac{\phi}{2}\right)$

where I_{max} is the maximum intensity.

The phase difference (ϕ) is related to the path difference (Δx) by:

$\phi = \frac{2\pi}{\lambda}\Delta x$

Intensity at Path Difference λ

Given:

- Path difference, $\Delta x_1 = \lambda$.
- Intensity, $I_1 = K$.

Calculate the phase difference:

$\phi_1 = \frac{2\pi}{\lambda}(\lambda) = 2\pi$

Now, calculate the intensity:

$I_1 = I_{max}\cos^2\left(\frac{2\pi}{2}\right) = I_{max}\cos^2(\pi) = I_{max}(-1)^2 = I_{max}$

Therefore, $K = I_{max}$.

Intensity at Path Difference $\lambda/3$

Given:

- Path difference, $\Delta x_2 = \lambda/3$.

Calculate the phase difference:

$\phi_2 = \frac{2\pi}{\lambda}\left(\frac{\lambda}{3}\right) = \frac{2\pi}{3}$

Calculate the intensity (I_2):

$I_2 = I_{max}\cos^2\left(\frac{\phi_2}{2}\right) = I_{max}\cos^2\left(\frac{2\pi/3}{2}\right) = I_{max}\cos^2\left(\frac{\pi}{3}\right)$

We know that $\cos(\pi/3) = 1/2$. Substituting this value:

$I_2 = I_{max}\left(\frac{1}{2}\right)^2 = I_{max}\left(\frac{1}{4}\right) = \frac{I_{max}}{4}$

Since $K = I_{max}$, the intensity I_2 is:

$I_2 = \frac{K}{4}$

Conclusion

The intensity of light at a point where the path difference is $\lambda/3$ will be $K/4$. This corresponds to Option C.

14. **Answer: c**

Explanation:

To find the current I in the circuit shown, we need to analyze the configuration of resistors and diodes. All diodes are ideal; therefore, they only allow current to flow in one direction (forward-biased) with no voltage drop across them.

Given the circuit, the diodes are oriented such that they only allow current to pass in the direction depicted by the direction of the current I.

Let's analyze each branch:

1. The diode in the upper branch is forward-biased, allowing current to pass through the 4Ω resistor.
   - Current through the 4Ω resistor: it contributes to the total current with no obstruction.
2. The diode in the middle branch is also forward-biased, allowing current through the 3Ω resistor.
   - Current through the 3Ω resistor: it adds to the net current.
3. The diode in the lower branch is forward-biased, allowing current through the 2Ω and 5Ω resistors in series.
   - Current through the 2Ω + 5Ω resistors: contributes to the total current.

In total, each branch allows a portion of the current to contribute to the net current. Applying Kirchhoff's Voltage Law gives:

- Voltage across each resistor is the same due to diodes being ideal: $V = 10\,\mathrm{V}$.

The equivalent resistance R_{eq} can be calculated by treating each branch separately:

- Upper branch: $4\,\Omega$
- Middle branch: $3\,\Omega$
- Lower branch: equivalent of 2Ω and 5Ω in series = 7Ω

This makes them in parallel:

$\frac{1}{R_{\text{eq}}} = \frac{1}{4} + \frac{1}{3} + \frac{1}{7}$

Calculating this gives:

$R_{\text{eq}} = \frac{84}{41}\,\Omega \approx 2.05\,\Omega$

Now calculating the total current I:

$I = \frac{V}{R_{\text{eq}}} = \frac{10\text{ V}}{\frac{84}{41}\,\Omega} = \frac{410}{84} \approx \frac{15}{2}\text{ A}$

Therefore, the correct answer is:

- $\frac{15}{2}\text{ A}$

---

15. **Answer: b**

Explanation:

Concave Lens Ray Behavior Analysis

Understanding Refraction in Concave Lenses

A concave lens is a diverging lens. It spreads out light rays that pass through it.

According to the rules of ray optics for a concave lens:

- Any light ray travelling parallel to the principal axis, after passing through the concave lens, bends outwards (diverges).
- This diverging ray, when traced backward, appears to originate from a specific point on the principal axis.
- This point is the **first principal focus** (F1) of the concave lens, located on the same side as the incident ray.

Evaluating the Options

Based on the rules:

- **Option 1** is incorrect because passing through the second principal focus is characteristic of parallel rays refracted by a convex lens.
- **Option 2** correctly states that the ray appears to diverge from the first principal focus. This is the defining behavior for a ray parallel to the principal axis in a

concave lens.
- **Option 3** is incorrect. Rays emerge parallel only if they pass through the optical center.
- **Option 4** is incorrect. The ray does not pass through 2F, and 2F is not solely defined as the radius of curvature in this context.

Therefore, the correct description of the light ray's path after refraction is that it appears to diverge from the first principal focus.

---

16. **Answer: d**

Explanation:

Galvanometer to Ammeter Conversion: Shunt Resistance Calculation

This problem involves converting a galvanometer into an ammeter using a shunt resistor. We need to calculate the value of this shunt resistor.

Given Information

- Galvanometer resistance, $G = 100\ \Omega$
- Galvanometer full-scale deflection current, $I_g = 1\ \mathrm{mA} = 1 \times 10^{-3}\ \mathrm{A}$
- Desired ammeter range (maximum current), $I = 10\ \mathrm{A}$

Shunt Resistance Formula

To convert a galvanometer into an ammeter, a low resistance shunt (S) is connected in parallel with the galvanometer. The formula to calculate the shunt resistance is:

$S = \frac{G \times I_g}{I - I_g}$

Calculation Steps

1. Substitute the given values into the formula:

$S = \frac{100\ \Omega \times (1 \times 10^{-3}\ \text{A})}{10\ \text{A} - (1 \times 10^{-3}\ \text{A})}$

2. Simplify the numerator:

   $100\ \Omega \times 0.001\ \text{A} = 0.1\ \text{V}$

3. Simplify the denominator:

   $10\ \text{A} - 0.001\ \text{A} = 9.999\ \text{A}$

4. Calculate the shunt resistance S:

   $S = \frac{0.1\ \text{V}}{9.999\ \text{A}} \approx 0.010001\ \Omega$

5. Rounding to a practical value, the required shunt resistance is approximately $0.01\ \Omega$.

Conclusion

The calculated shunt resistance required to convert the galvanometer into an ammeter with a range of 0-10 A is $0.01\ \Omega$.

---

17. **Answer: b**

Explanation:

Hydrogen Atom Radial Distance Calculation

The problem asks for the approximate radial distance of an electron in the first excited state of a hydrogen atom, given its energy value.

Solution Steps

- **Identify the Quantum State:** The first excited state of an atom corresponds to the principal quantum number $n = 2$. For a hydrogen atom, the energy level is given by the formula $E_n = -\frac{13.6\ \text{eV}}{n^2}$. For $n = 2$, the energy is $E_2 = -\frac{13.6\ \text{eV}}{2^2} = -\frac{13.6\ \text{eV}}{4} = -3.4\ \text{eV}$. This confirms the given energy corresponds to the first excited state $(n = 2)$.

- **Apply Bohr Radius Formula:** The radius of the electron's orbit in the n-th state, according to the Bohr model, is given by $r_n = n^2 a_0$, where a_0 is the Bohr radius. The value of the Bohr radius is approximately $a_0 \approx 0.529 \times 10^{-10}$ m.
- **Calculate the Radius for $n = 2$:** Substitute $n = 2$ into the Bohr radius formula:

  $r_2 = (2)^2 \times a_0$

  $r_2 = 4 \times a_0$

  Now, use the approximate value of the Bohr radius:

  $r_2 \approx 4 \times (0.529 \times 10^{-10}\ \text{m})$

  $r_2 \approx 2.116 \times 10^{-10}\ \text{m}$
- **Conclusion:** The calculated approximate radial distance is 2.116×10^{-10} m. This value closely matches the option 2.1×10^{-10} m.

---

18. **Answer: d**

Explanation:

Calculating Work Done Against Gravity

The work done to move an object against a conservative force, like gravity, is equal to the change in its potential energy.

Potential Energy Formula

The gravitational potential energy (U) of a mass (m) at a distance (r) from the center of the Earth (mass M) is given by:

$U = -\frac{GMm}{r}$

Where G is the gravitational constant.

Initial and Final States

- **Initial State:** At the Earth's surface, the distance from the center is $r_1 = R$. The initial potential energy (U_1) is:

  $U_1 = -\frac{GMm}{R}$

- **Final State:** The mass is raised to a height R above the surface. The final distance from the center is $r_2 = R + R = 2R$. The final potential energy (U_2) is:

  $U_2 = -\frac{GMm}{2R}$

Work Done Calculation

The work done (W) is the change in potential energy:

$W = \Delta U = U_2 - U_1$

$W = \left(-\frac{GMm}{2R}\right) - \left(-\frac{GMm}{R}\right)$

$W = -\frac{GMm}{2R} + \frac{GMm}{R}$

$W = \frac{GMm}{R}\left(1 - \frac{1}{2}\right)$

$W = \frac{GMm}{2R}$

Relating to Surface Gravity

At the Earth's surface, the acceleration due to gravity is g. The force of gravity is mg. This force is also given by Newton's law of gravitation:

$mg = \frac{GMm}{R^2}$

From this, we can express GM as:

$GM = gR^2$

Final Work Done Expression

Substitute $GM = gR^2$ into the work done equation:

$W = \frac{(gR^2)m}{2R}$

$W = \frac{mgR}{2}$

Therefore, the work done to raise the mass is $\frac{mgR}{2}$.

---

19. **Answer: b**

Explanation:

The question asks for the resonance frequency (f_0) of a series AC circuit containing a resistor (R), capacitor (C), and inductor (L).

Resonance Frequency Formula

The resonance frequency in an RLC circuit is the frequency at which the inductive reactance equals the capacitive reactance. It is independent of the resistance and is calculated using the formula:

$f_0 = \frac{1}{2\pi\sqrt{LC}}$

Circuit Component Values

Given values are:

- Inductance, $L = 1 \text{ mH} = 1 \times 10^{-3} \text{ H}$
- Capacitance, $C = 0.1 \ \mu\text{F} = 0.1 \times 10^{-6} \text{ F} = 1 \times 10^{-7} \text{ F}$
- Resistance, $R = 1 \text{ k}\Omega = 1000 \ \Omega$ (Note: Resistance does not affect the resonance frequency itself).

Calculation Steps

1. Substitute the values of L and C into the resonance frequency formula:

   $f_0 = \frac{1}{2\pi\sqrt{(1\times10^{-3} \text{ H})\times(1\times10^{-7} \text{ F})}}$.

2. Simplify the term under the square root:

   $LC = 1 \times 10^{-10} \text{ H} \cdot \text{F}$

   $\sqrt{LC} = \sqrt{1 \times 10^{-10}} = 1 \times 10^{-5} \text{ s}$

3. Calculate the resonance frequency in Hertz (Hz):

$$f_0 = \frac{1}{2\pi \times (1 \times 10^{-5})} = \frac{10^5}{2\pi} \text{ Hz}$$

Using $\pi \approx 3.14159$,

$$f_0 \approx \frac{100000}{2 \times 3.14159} \approx \frac{100000}{6.28318} \approx 15915.5 \text{ Hz}$$

4. Convert the frequency to kilohertz (kHz):

$$f_0 \approx 15.9155 \text{ kHz}$$

Conclusion

The calculated resonance frequency is approximately 15.9 kHz.

---

20. Answer: d

Explanation:

This problem involves calculating the energy lost when a charged capacitor is connected to an identical uncharged capacitor.

Initial State Analysis

We start with two capacitors, each with capacitance $C = 200 \text{ pF}$. One capacitor is charged to an initial voltage $V_0 = 100 \text{ V}$.

- The initial charge Q_0 on the first capacitor is calculated using $Q = CV$: $Q_0 = C \times V_0 = (200 \times 10^{-12} \text{ F}) \times (100 \text{ V}) = 2 \times 10^{-8} \text{ C}$.
- The initial electrostatic energy U_0 stored in this capacitor is calculated using $U = \frac{1}{2}CV^2$: $U_0 = \frac{1}{2}CV_0^2 = \frac{1}{2} \times (200 \times 10^{-12} \text{ F}) \times (100 \text{ V})^2$ $U_0 = \frac{1}{2} \times 200 \times 10^{-12} \times 10000 \text{ J} = 1 \times 10^{-6} \text{ J}$.

Connecting the Capacitors

The charged capacitor is then connected to an identical uncharged capacitor. The total charge Q_0 is conserved and redistributed between the two capacitors.

- Since the capacitors have equal capacitance C, they will share the charge equally after connection. The total capacitance becomes $C_{total} = C + C = 2C$.
- The final voltage V_f across both capacitors will be half the initial voltage: $V_f = \frac{V_0}{2} = \frac{100\text{ V}}{2} = 50\text{ V}$. Alternatively, using total charge and total capacitance: $V_f = \frac{Q_0}{C_{total}} = \frac{2\times10^{-8}\text{ C}}{2\times(200\times10^{-12}\text{ F})} = \frac{2\times10^{-8}}{400\times10^{-12}}\text{ V} = 50\text{ V}$.

Final Energy Calculation

The total electrostatic energy U_f stored in the system of two parallel capacitors is:

- Using the total capacitance and final voltage: $U_f = \frac{1}{2}C_{total}V_f^2 = \frac{1}{2}\times(400\times10^{-12}\text{ F})\times(50\text{ V})^2$ $U_f = \frac{1}{2}\times400\times10^{-12}\times2500\text{ J} = 500000\times10^{-12}\text{ J} = 0.5\times10^{-6}\text{ J}$.
- Alternatively, calculate energy in each capacitor: $U_f = 2\times\left(\frac{1}{2}CV_f^2\right) = CV_f^2 = (200\times10^{-12}\text{ F})\times(50\text{ V})^2$ $U_f = 200\times10^{-12}\times2500\text{ J} = 500000\times10^{-12}\text{ J} = 0.5\times10^{-6}\text{ J}$.

Energy Lost Calculation

The electrostatic energy lost during this process is the difference between the initial and final energies:

- $U_{lost} = U_0 - U_f$ $U_{lost} = (1\times10^{-6}\text{ J}) - (0.5\times10^{-6}\text{ J}) = 0.5\times10^{-6}\text{ J}$.

This energy is dissipated primarily as heat due to the current flow during charge redistribution.

---

21. **Answer: b**

Explanation:

To solve this problem, it is essential to understand the refraction of light through a prism and how the angle of deviation is determined in such scenarios.

Given Data:

- The prism is equilateral, meaning all its angles are $60°$.
- The angle of incidence $i = 50°$.
- The refracted ray QR is parallel to the base BC.

Concept and Calculation:

1. In an equilateral prism, the refracted ray QR being parallel to the base indicates that the angle of refraction at Q and exit angle at R combined will equal the angle of the prism. Therefore, the angle of refraction r at both Q and R is 30°.
2. This situation is depicted by using Snell's Law at the first surface: $n \sin i = \sin r$, where $i = 50^\circ$ and $r = 30^\circ$.
3. The angle of deviation δ is given by the formula: $\delta = i + e - A$, where e is the angle of emergence and A is the angle of the prism.
4. Since $r_1 = r_2$ and the angle of the prism $A = 60^\circ$, the angle of emergence e is also 50°.
5. Substituting the values, we have: $\delta = 50^\circ + 50^\circ - 60^\circ = 40^\circ$.

Therefore, the angle of deviation δ is **40º**.

Conclusion: The correct answer is **40º**, as it matches the angle of deviation calculated.

---

22. Answer: b

Explanation:

In a metre bridge experiment, the setup involves a balance condition for determining unknown resistance using a known resistance. The key components of the meter bridge are the galvanometer (G), the cell (E), and resistors R_1 and R_2 connected across the bridge.

In this particular problem, the positions of the cell (E) and the galvanometer (G) are interchanged. We need to analyze the effect of this interchange on the galvanometer's deflection.

Explanation:

1. When the positions of the galvanometer and the cell are interchanged, the principle of the meter bridge still applies, but the detection of balance point changes.
2. Initially, the galvanometer detects the point of zero current flow when the bridge is at balance. However, with the positions interchanged:

- The galvanometer being used at the position of the cell will cause it to deflect when there is any potential difference across the bridge.
- The cell now at the previous position of the galvanometer will maintain a potential difference across the bridge regardless of the deflection.

3. Due to these changes, two situations arise when using the jockey (sliding contact) across the wire:
   - There will be right-sided and left-sided deflections as the jockey moves across the bridge wire, depending on the current direction.
   - At a particular point (the balance point), the potential across both sides is equal, and the galvanometer shows a zero deflection, indicating balance.

Conclusion:

The correct observation in the galvanometer, given the interchange of the cell and galvanometer positions, is: *Both right-sided and left-sided deflection and at balance point no deflection.*

---

23. **Answer: b**

Explanation:

RMS Speed Ratio Calculation for Argon and Chlorine

The root mean square (RMS) speed of gas molecules is defined by the formula:

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

where R is the ideal gas constant, T is the absolute temperature, and M is the molar mass of the gas.

Ratio of RMS Speeds

For a mixture containing two gases at the same temperature, the ratio of their RMS speeds depends only on their molar masses:

$$\frac{V_{rms}^{Argon}}{V_{rms}^{Chlorine}} = \frac{\sqrt{\frac{3RT}{M_{Ar}}}}{\sqrt{\frac{3RT}{M_{Cl}}}}$$

Since $3R$ and T are constant for both gases, they cancel out:

$$\frac{V_{rms}^{Ar}}{V_{rms}^{Cl}} = \sqrt{\frac{M_{Cl}}{M_{Ar}}}$$

Molar Masses Provided

The problem provides the following molar masses:

- Molar mass of Argon (M_{Ar}) = 40 u
- Molar mass of Chlorine (M_{Cl}) = 70 u

Note: The mass ratio $(2:1)$ and the temperature (27°C) are not required for calculating the ratio of RMS speeds.

Calculating the Final Ratio

Substitute the molar masses into the ratio formula:

$$\frac{V_{rms}^{Ar}}{V_{rms}^{Cl}} = \sqrt{\frac{70}{40}}$$

Simplify the fraction inside the square root:

$$\frac{V_{rms}^{Ar}}{V_{rms}^{Cl}} = \sqrt{\frac{7}{4}}$$

Calculate the square root:

$$\frac{V_{rms}^{Ar}}{V_{rms}^{Cl}} = \frac{\sqrt{7}}{\sqrt{4}} = \frac{\sqrt{7}}{2}$$

---

24. **Answer: a**

Explanation:

To determine the correctness of the given statements, let's analyze each one based on the fundamental concepts of p-n junction diodes:

1. **Statement A:** "When the forward bias voltage across a p-n junction diode increases above a certain threshold voltage, the diode current increases significantly."

- Explanation: In a p-n junction diode, when forward bias voltage is applied, it reduces the barrier potential, allowing charge carriers (electrons and holes) to cross the junction easily. Once the applied voltage exceeds the threshold or knee voltage (usually around 0.7V for silicon diodes, and 0.3V for germanium diodes), the diode starts conducting, and the current increases exponentially. This is a well-understood behavior of diodes.
- Conclusion: Statement A is correct.

2. **Statement B:** "This current is called reverse saturation current."
    - Explanation: Reverse saturation current is the small constant current that flows through the diode when it is reverse biased, due to minority carrier drift. In contrast, the significant current that flows when a p-n junction diode is forward biased above the threshold voltage is simply the forward current, not the reverse saturation current.
    - Conclusion: Statement B is incorrect, as it mislabels the type of current as reverse saturation current.

Conclusion: After analyzing both statements, we can conclude that **Statement A is true but Statement B is false.** Therefore, the correct answer is the option stating, "Statement A is true but Statement B is false."

---

25. **Answer: a**

Explanation:

Travelling Wave Phase Difference Analysis

The equation for the travelling harmonic wave is given as:

$y(x,t) = 2.0 \cos 2\pi(10t - 0.0080x + 0.35)$

In this equation, x and y are in centimeters (cm), and t is in seconds (s).

Phase Difference Calculation Steps

The phase of the wave at any point x and time t is the argument of the cosine function:

$\Phi(x,t) = 2\pi(10t - 0.0080x + 0.35)$

The phase difference $\Delta\Phi$ between two points x_1 and x_2 at the same time t is calculated as:

$$\Delta\Phi = \Phi(x_2, t) - \Phi(x_1, t)$$

Substituting the expression for phase:

$$\Delta\Phi = [2\pi(10t - 0.0080x_2 + 0.35)] - [2\pi(10t - 0.0080x_1 + 0.35)]$$

Simplify the expression by canceling common terms:

$$\Delta\Phi = 2\pi[(10t - 0.0080x_2 + 0.35) - (10t - 0.0080x_1 + 0.35)]$$

$$\Delta\Phi = 2\pi[-0.0080x_2 + 0.0080x_1]$$

Factor out the common term 0.0080:

$$\Delta\Phi = 2\pi \times 0.0080(x_1 - x_2)$$

Distance Unit Conversion

The distance between the two points is given as 0.5 m. Since the variable x in the wave equation is in centimeters, we must convert the distance to centimeters:

$$\Delta x = x_1 - x_2 = 0.5\text{ m} \times \frac{100\text{ cm}}{1\text{ m}} = 50\text{ cm}$$

Final Phase Difference Calculation

Now substitute the distance Δx into the phase difference formula. We consider the magnitude of the phase difference:

$$|\Delta\Phi| = |2\pi \times 0.0080 \times \Delta x|$$

$$|\Delta\Phi| = 2\pi \times 0.0080 \times 50$$

Calculate the product:

$$|\Delta\Phi| = 2\pi \times 0.4$$

$$|\Delta\Phi| = 0.8\pi\text{ rad}$$

The phase difference between the two points separated by 0.5 m is 0.8π radians.

26. **Answer: a**

Explanation:

Using Faraday's law,

emf = B × l × v

Given:
B = 0.3 T
l = 3 cm = 0.03 m
v = 2 cm/s = 0.02 m/s

So,

emf = 0.3 × 0.03 × 0.02
= 1.8×10^{-4} V

Therefore, the induced emf is 1.8×10^{-4} volt.

27. **Answer: d**

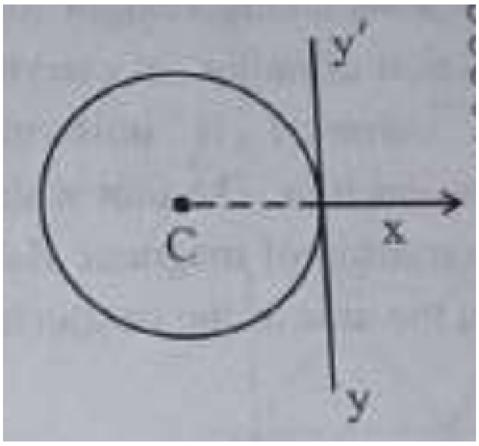
Explanation:

To find the moment of inertia of the ring about an axis yy' (tangent in the plane), we can use the parallel axis theorem and the known formula for the moment of inertia of a circular ring about its center and diameter.

1. First, calculate the radius R of the ring. The circumference of the ring is equal to the length of the wire L, so we have: $L = 2\pi R \Rightarrow R = \frac{L}{2\pi}$
2. The moment of inertia of the ring about its center C (z-axis) is: $I_C = mR^2 = m\left(\frac{L}{2\pi}\right)^2 = \frac{mL^2}{4\pi^2}$
3. The moment of inertia about any diameter (say x-axis) of the ring is half of I_C: $I_x = \frac{I_C}{2} = \frac{mL^2}{8\pi^2}$
4. Using the parallel axis theorem, the moment of inertia about axis yy' is: $I_{yy'} = I_x + mR^2 = \frac{mL^2}{8\pi^2} + \frac{mL^2}{4\pi^2}$

5. Simplify the expression: $I_{yy'} = \frac{mL^2}{8\pi^2} + \frac{2mL^2}{8\pi^2} = \frac{3mL^2}{8\pi^2}$
6. Finally, factor in the mass density m and length L: $I_{yy'} = \frac{3mL^3}{8\pi^2}$

Thus, the moment of inertia of the ring about the tangent axis yy' is $\frac{3mL^3}{8\pi^2}$.



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28. Answer: a

Explanation:

Given Electrical Parameters

The problem provides the following values:

- Battery Electromotive Force (emf), $\mathcal{E} = 12 \text{ V}$
- Internal Resistance, $r = 2\ \Omega$
- Circuit Current, $I = 0.6 \text{ A}$

Formula for Terminal Voltage

The terminal voltage (V_T) of a battery is the actual voltage across its terminals when current is flowing. It can be calculated using the formula:

$V_T = \mathcal{E} - Ir$

Where:

- V_T is the terminal voltage
- $\mathcal{E}$ is the battery emf
- I is the current flowing through the circuit
- r is the internal resistance of the battery

Calculation of Terminal Voltage

Substitute the given values into the formula:

$$V_T = 12\ \text{V} - (0.6\ \text{A} \times 2\ \Omega)$$

First, calculate the voltage drop across the internal resistance:

$$I \times r = 0.6\ \text{A} \times 2\ \Omega = 1.2\ \text{V}$$

Now, subtract this voltage drop from the battery's emf:

$$V_T = 12\ \text{V} - 1.2\ \text{V}$$

$$V_T = 10.8\ \text{V}$$

Result

The terminal voltage of the battery is $10.8\ \text{V}$.

---

29. **Answer: c**

Explanation:

Understanding Ruler Fall Distance and Reaction Time

When a ruler falls vertically from rest, the distance it travels (d) is determined by the time it is in the air. This motion is governed by the laws of kinematics under constant acceleration due to gravity (g).

Kinematic Equation for Distance

The distance fallen by an object starting from rest is given by the equation:

$d = \frac{1}{2}gt^2$

where g is the acceleration due to gravity and t is the time of fall. In this problem, the time of fall is the **reaction time** of each person catching the ruler.

Relationship Between Distance and Reaction Time

Since g is constant $(9.8\ \mathrm{m\ s^{-2}})$, the distance traveled (d) is directly proportional to the square of the reaction time (t^2). For positive reaction times:

- A longer reaction time (t) results in a greater distance traveled $(d \propto t^2)$.
- Therefore, to find the order of distances travelled, we need to order the reaction times from longest to shortest.

Ordering Reaction Times

The given reaction times are:

- Person A: $0.20\ \mathrm{s}$
- Person B: $0.22\ \mathrm{s}$
- Person C: $0.18\ \mathrm{s}$
- Person D: $0.19\ \mathrm{s}$
- Person E: $0.21\ \mathrm{s}$

Ordering these times from largest to smallest:

1. Person B: $0.22\ \mathrm{s}$
2. Person E: $0.21\ \mathrm{s}$
3. Person A: $0.20\ \mathrm{s}$
4. Person D: $0.19\ \mathrm{s}$
5. Person C: $0.18\ \mathrm{s}$

Determining the Order of Distances

Since the distance travelled is directly proportional to the reaction time (or its square), the order of distances will match the order of reaction times from longest to shortest.

Thus, the order of distances travelled is:

Distance for B > Distance for E > Distance for A > Distance for D > Distance for C

This corresponds to the order: **B > E > A > D > C**.

---

30. Answer: b

Explanation:

Flywheel Revolutions Calculation

The problem asks for the total number of revolutions a flywheel completes while its angular speed increases over a specific time period.

Initial Data Conversion

First, convert the initial and final angular speeds from revolutions per minute (rpm) to radians per second (rad/s), as physics calculations require SI units.

- Conversion factor: $1 \text{ rpm} = \frac{2\pi}{60} \text{ rad/s}$
- Initial angular speed (ω_0): $600 \text{ rpm} = 600 \times \frac{2\pi}{60} \text{ rad/s} = 20\pi \text{ rad/s}$
- Final angular speed (ω_f): $1200 \text{ rpm} = 1200 \times \frac{2\pi}{60} \text{ rad/s} = 40\pi \text{ rad/s}$
- Time (t): 10 s

Calculating Angular Acceleration

Determine the **angular acceleration** (α) using the formula $\alpha = \frac{\omega_f - \omega_0}{t}$.

- $\alpha = \frac{40\pi \text{ rad/s} - 20\pi \text{ rad/s}}{10 \text{ s}}$
- $\alpha = \frac{20\pi \text{ rad/s}}{10 \text{ s}} = 2\pi \text{ rad/s}^2$

Calculating Angular Displacement

Use the kinematic equation for angular displacement: $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$.

- $\theta = (20\pi \text{ rad/s})(10 \text{ s}) + \frac{1}{2}(2\pi \text{ rad/s}^2)(10 \text{ s})^2$
- $\theta = 200\pi \text{ rad} + \frac{1}{2}(2\pi \text{ rad/s}^2)(100 \text{ s}^2)$
- $\theta = 200\pi \text{ rad} + 100\pi \text{ rad}$

- $\theta = 300\pi$ radians

Determining Number of Revolutions

Convert the total angular displacement (θ) from radians to revolutions. Since $1 \text{ revolution} = 2\pi \text{ radians}.$

- Number of revolutions = $\frac{\theta}{2\pi}$
- Number of revolutions = $\frac{300\pi \text{ radians}}{2\pi \text{ radians/revolution}}$
- Number of revolutions = 150

Therefore, the flywheel completes 150 revolutions.

---

31. Answer: a

Explanation:

Certainly! Here’s how to respond to the question by explaining the magnetic field variation in a long straight solid wire carrying a steady current.

To find the variation of the magnetic field B with distance r from the axis of a solid wire carrying a steady current I, we can use Ampère's Law and the concept of current distribution.

The given wire has a circular cross-section with radius a, and the current is uniformly distributed across this cross-section. Let's consider two cases based on the distance r from the axis:

For $r \leq a$ (inside the wire):

By symmetry, the magnetic field B at a distance r from the axis will be in the form of concentric circles. Using Ampère's Law:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enclosed}}$$

Here, $I_{\text{enclosed}} = I \frac{\pi r^2}{\pi a^2} = I \left(\frac{r^2}{a^2}\right)$, since the current is uniformly distributed.

Therefore, $B(2\pi r) = \mu_0 I \frac{r^2}{a^2}$.

Solving for B:

$B = \frac{\mu_0 I r}{2\pi a^2}$, which shows that B increases linearly with r.

For $r > a$ (outside the wire):

For any point outside the conductor, the magnetic field is determined by the total current I enclosed, irrespective of how it is distributed inside.

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$$

Thus, $B(2\pi r) = \mu_0 I$.

Solving for B:

$B = \frac{\mu_0 I}{2\pi r}$, which shows that B decreases with $\frac{1}{r}$.

The graph of B with respect to r will initially increase linearly inside the conductor (for $r \leq a$) and then decrease inversely outside the conductor (for $r > a$).

This corresponds to the correct representation of the magnetic field variation. Therefore, the correct option is represented by the image above.

---

32. **Answer: b**

Explanation:

Nuclear Statement Analysis

We need to evaluate the truthfulness of four statements related to nuclear properties.

Statement A Analysis

The radius (R) of a nucleus is empirically found to be proportional to the cube root of the mass number (A): $R \propto A^{1/3}$

The volume (V) of a nucleus is proportional to the cube of its radius: $V \propto R^3$

Substituting the relation for R: $V \propto (A^{1/3})^3 = A$

Therefore, the volume of a nucleus is proportional to A, not $A^{1/3}$. Statement A is false.

Statement B Analysis

As derived above, the volume (V) of a nucleus is directly proportional to the mass number (A). $V \propto A$

Statement B is true.

Statement C Analysis

The mass defect is specifically defined as the difference between the mass of the individual nucleons (protons and neutrons) and the actual measured mass of the nucleus. The mass of electrons in an atom is generally not included in this specific definition of nuclear mass defect. Therefore, this statement is an inaccurate definition. Statement C is false.

Statement D Analysis

The mass defect (Δm) is precisely the difference between the total mass of the constituent nucleons (protons and neutrons) and the measured mass of the nucleus. $\Delta m = (\text{Mass of constituent nucleons}) - (\text{Mass of nucleus})$

Statement D provides the correct definition. Statement D is true.

Conclusion

Based on the analysis:

- Statement A is false.
- Statement B is true.
- Statement C is false.
- Statement D is true.

Therefore, statements B and D are true, while A and C are false.

---

33. Answer: b

Explanation:

Calculating Simple Pendulum Length

To determine the effective length of the simple pendulum, follow these steps:

1. **Calculate the Time Period (T):** The time period is the time taken for one complete oscillation. Given the time for 30 oscillations is 60 s, calculate T.

   $$T = \frac{\text{Total time}}{\text{Number of oscillations}} = \frac{60\ \text{s}}{30} = 2\ \text{s}$$

2. **Use the Simple Pendulum Formula:** The formula for the time period of a simple pendulum is:

   $$T = 2\pi\sqrt{\frac{L}{g}}$$

   Where L is the length of the pendulum and g is the acceleration due to gravity.
3. **Rearrange Formula for Length (L):** To find the length L, rearrange the formula:

   Square both sides: $T^2 = 4\pi^2\frac{L}{g}$

   Isolate L: $L = \frac{gT^2}{4\pi^2}$

4. **Substitute Values and Compute Length:** Use the given values: $g = 9.8\ \text{m/s}^2$, $T = 2\ \text{s}$, and $\pi^2 = 9.8$.

   $$L = \frac{(9.8\ \text{m/s}^2)\times(2\ \text{s})^2}{4\times9.8}$$

   $$L = \frac{9.8\times4\ \text{m}}{4\times9.8}$$

   $$L = 1\ \text{m}$$

Therefore, the calculated length of the simple pendulum is 1 m.

---

34. **Answer: b**

Explanation:

Vernier Callipers Least Count Calculation

This section explains how to calculate the least count (LC) of a vernier callipers using the provided scale division information.

Provided Data Analysis

We are given the following information about the vernier callipers:

- The number of Vernier Scale Divisions (VSD) that coincide is 20.
- These 20 VSD correspond to the length of 16 Main Scale Divisions (MSD).
- The value of one Main Scale Division (MSD) is 1 mm.

Calculating the Least Count (LC)

The least count of a vernier callipers is the smallest measurement that can be accurately taken using the instrument. It is calculated as the difference between one main scale division and one vernier scale division.

1. **Find the length represented by 16 MSD:**

   Since 1 MSD = 1 mm, the length of 16 MSD is:

   $\text{Length of 16 MSD} = 16 \times 1 \text{ mm} = 16 \text{ mm}$
2. **Determine the length of one VSD:**

   We know that 20 VSD have the same length as 16 MSD.

   $\text{Length of 20 VSD} = \text{Length of 16 MSD} = 16 \text{ mm}$

   Therefore, the length of a single VSD is:

   $\text{Length of 1 VSD} = \frac{16 \text{ mm}}{20} = 0.8 \text{ mm}$
3. **Calculate the Least Count (LC):**

   The formula for Least Count is: LC = (Value of 1 MSD) - (Value of 1 VSD)

   $\text{LC} = 1 \text{ mm} - 0.8 \text{ mm} = 0.2 \text{ mm}$
4. **Convert LC to centimeters:**

   To express the least count in centimeters, we use the conversion factor $1 \text{ cm} = 10 \text{ mm}$.

$$LC = \frac{0.2\ \mathrm{mm}}{10\ \mathrm{mm/cm}} = 0.02\ \mathrm{cm}$$

Final Result

The least count of the vernier callipers is $0.02\ \mathrm{cm}$.

---

35. **Answer: b**

Explanation:

Objective: Calculate the density of a metallic cube and express it in the format $X \times 10^3\ \mathrm{kg\ m^{-3}}$, adhering to significant figure rules.

Density Calculation Steps

1. **Convert Units and Calculate Volume:**
    - Given mass $m = 5.580\ \mathrm{kg}$. This has 4 significant figures.
    - Given side length $s = 9.0\ \mathrm{cm}$. This has 2 significant figures.
    - Convert side length to meters: $s = 9.0\ \mathrm{cm} = 0.090\ \mathrm{m}$.
    - Calculate the volume V: $V = s^3 = (0.090\ \mathrm{m})^3 = 0.000729\ \mathrm{m}^3$.
    - Since the side length measurement $(9.0\ \mathrm{m})$ has 2 significant figures, the calculated volume should be rounded to 2 significant figures: $V \approx 0.00073\ \mathrm{m}^3$ or $7.3 \times 10^{-4}\ \mathrm{m}^3$.
2. **Calculate Density:**
    - Use the density formula: $\rho = \frac{m}{V}$.
    - Substitute the values: $\rho = \frac{5.580\ \mathrm{kg}}{0.00073\ \mathrm{m}^3}$.
    - Perform the division: $\rho \approx 7643.83...\ \mathrm{kg\ m^{-3}}$.
    - The result must be rounded to 2 significant figures, as determined by the least precise measurement (side length).
    - Rounding $7643.83...$ to 2 significant figures: The first two digits are 7 and 6. The next digit is 4, which is less than 5, so we keep the '6'.
    - Therefore, $\rho \approx 7600\ \mathrm{kg\ m^{-3}}$.
3. **Determine the value of X:**
    - Express the calculated density in the required format: $\rho = X \times 10^3\ \mathrm{kg\ m^{-3}}$.
    - $7600\ \mathrm{kg\ m^{-3}}$ can be written as $7.6 \times 1000\ \mathrm{kg\ m^{-3}}$, which is $7.6 \times 10^3\ \mathrm{kg\ m^{-3}}$.
    - By comparison, $X = 7.6$.

36. **Answer: c**

Explanation:

Interference and Diffraction: Energy Redistribution

The question concerns the redistribution of energy in light during interference and diffraction phenomena.

Statement A Analysis: Conservation of Energy

Interference and diffraction patterns are formed by the superposition of waves. In these patterns, energy is redistributed. Bright fringes represent regions where constructive interference concentrates energy, while dark fringes represent regions where destructive interference minimizes energy. This redistribution does not create or destroy energy; the total energy remains constant across the entire pattern. Therefore, these phenomena are consistent with the principle of conservation of energy.

Conclusion: Statement A is true.

Statement B Analysis: Wave Characteristics

Interference and diffraction are fundamental characteristics of all wave phenomena, not exclusively light waves. These effects can be observed in other types of waves, such as sound waves and water waves. For instance, sound waves diffract around obstacles, and water waves clearly show interference patterns when two wave sources interact.

Conclusion: Statement B is false.

Final Conclusion

Based on the analysis, Statement A is true, and Statement B is false. This corresponds to Option C.

37. **Answer: d**

Explanation:

Let's analyze the circuit to determine the equivalent capacitance and the charges on each capacitor.

First, observe the arrangement of capacitors:

1. Capacitors C_1, C_2, C_3, and C_4 are connected in series.
2. Series combination is then in parallel with C_5.

We are given $C_1 = C_2 = C_3 = C_4 = 10\ \mu\text{F}$ and $C_5 = 2.5\ \mu\text{F}$.

Step 1: Find the equivalent capacitance of $C_1, C_2, C_3, \text{and } C_4$ in series:

The formula for equivalent capacitance, C_{series}, of capacitors in series is:

$$C_{\text{series}} = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \frac{1}{C_4}\right)^{-1}$$

Substituting values:

$$C_{\text{series}} = \left(\frac{1}{10} + \frac{1}{10} + \frac{1}{10} + \frac{1}{10}\right)^{-1}\ \mu\text{F}$$

$$= \left(\frac{4}{10}\right)^{-1}\ \mu\text{F} = \frac{10}{4}\ \mu\text{F} = 2.5\ \mu\text{F}$$

Step 2: Find the total equivalent capacitance of the entire circuit:

The equivalent capacitance of capacitors in parallel is:

$$C_{\text{eq}} = C_{\text{series}} + C_5$$

Substituting the values:

$$C_{\text{eq}} = 2.5\ \mu\text{F} + 2.5\ \mu\text{F} = 5\ \mu\text{F}$$

Step 3: Calculate the charge on each capacitor:

Using the formula for charge: $Q = C \cdot V$

Total charge for the equivalent capacitor:

$$Q_{\text{total}} = C_{\text{eq}} \cdot V = 5\ \mu\text{F} \cdot 50\ \text{V} = 250\ \mu\text{C}$$

For series capacitors $C_1, C_2, C_3, \text{and } C_4$, the charge is the same:

$Q_1 = Q_2 = Q_3 = Q_4 = Q_{\text{series}} = \frac{Q_{\text{total}}}{2} = 125\ \mu\text{C}$

For parallel capacitor C_5:

$Q_5 = C_5 \cdot V = 2.5\ \mu\text{F} \cdot 50\ \text{V} = 125\ \mu\text{C}$

Thus, the answer is that the equivalent capacitance is $5\ \mu\text{F}$, and the charge on each capacitor is $125\ \mu\text{C}$.

Option: $5\ \mu\text{F}, 125\ \mu\text{C}$ on all capacitors.

---

38. **Answer: b**

Explanation:

Simple Pendulum Kinetic Energy vs. Time Graph Analysis

Understanding Kinetic Energy in Simple Harmonic Motion

For a simple pendulum undergoing Simple Harmonic Motion (SHM), the kinetic energy ($K.E.$) varies cyclically with time (t). Key characteristics of this variation are:

- $K.E.$ is maximum at the mean (equilibrium) position where the pendulum's speed is greatest.
- $K.E.$ is zero at the extreme positions (maximum displacement) where the pendulum's speed momentarily drops to zero.
- Since $K.E. = \frac{1}{2}mv^2$, it is always non-negative ($K.E. \geq 0$).
- The kinetic energy pattern repeats every half of the time period ($T/2$). This means the frequency of $K.E.$ variation is double the frequency of the pendulum's oscillation. The mathematical form is often related to $\sin^2(\omega t)$ or $\cos^2(\omega t)$.

Analyzing the K.E. vs. Time Graph

The correct graph representing $K.E.$ versus t must satisfy these conditions:

- **Non-Negative Values:** The graph must always lie above or on the time axis ($K.E. \geq 0$). Graphs showing negative $K.E.$ are incorrect.

- **Zero Points:** $K.E.$ should be zero at times corresponding to the extreme positions of the pendulum.
- **Maximum Points:** $K.E.$ should reach its peak value at times corresponding to the mean position.
- **Periodicity:** The energy cycle (e.g., from zero to maximum and back to zero) should repeat twice within one full oscillation period (T), indicating a period of $T/2$ for the $K.E.$ variation itself.

Considering these points, if the pendulum starts at an extreme position $(t = 0)$, its $K.E.$ is 0. It increases to a maximum at $t = T/4$ (mean position), decreases to 0 at $t = T/2$ (other extreme), increases to a maximum again at $t = 3T/4$, and returns to 0 at $t = T$. Option B accurately displays this behavior, showing a wave that starts at zero, rises to a peak, falls back to zero, and repeats this pattern, staying entirely within the non-negative domain and exhibiting the required periodicity.

---

39. **Answer: d**

Explanation:

Calculating Heater Power Consumption at Reduced Voltage

The problem asks us to find the new power consumed by a room heater when the supply voltage decreases, given its original rating. We need to use the relationship between power, voltage, and resistance.

Heater Resistance Calculation

First, we determine the resistance of the heater, which remains constant. The rated power (P_{rated}) is 400 W and the rated voltage (V_{rated}) is 220 V. The formula relating power, voltage, and resistance is:

$P = \frac{V^2}{R}$

We can rearrange this to find the resistance (R):

$R = \frac{V_{rated}^2}{P_{rated}}$

Substituting the given values:

$$R = \frac{(220\ \text{V})^2}{400\ \text{W}} = \frac{48400}{400}\ \Omega = 121\ \Omega$$

So, the resistance of the heater is $121\ \Omega$.

New Power Consumption

Now, the supply voltage drops to $V_{new} = 200\ \text{V}$. We use the same resistance $(R = 121\ \Omega)$ and the new voltage to calculate the new power consumed (P_{new}):

$$P_{new} = \frac{V_{new}^2}{R}$$

Substituting the values:

$$P_{new} = \frac{(200\ \text{V})^2}{121\ \Omega} = \frac{40000}{121}\ \text{W}$$

$$P_{new} \approx 330.58\ \text{W}$$

Rounding this to the nearest watt, the approximate power consumed is $331\ \text{W}$.

---

40. **Answer: b**

Explanation:

AC Current Analysis: Time to Peak Value

The question asks for the time (t) it takes for an alternating current (AC) to reach its peak value (I_p) starting from zero. We are given the peak current $I_p = 5\ \text{A}$ and the frequency $f = 60\ \text{Hz}$.

Alternating Current Formula

The instantaneous current $I(t)$ in an AC circuit can be represented as:

$$I(t) = I_p \sin(\omega t)$$

Where:

- I_p is the peak current.
- ω is the angular frequency.
- t is the time.

The angular frequency ω is related to the frequency f by:

$\omega = 2\pi f$

Calculating Time to Peak

We need to find the time t when the current $I(t)$ reaches its peak value I_p.

Set $I(t) = I_p$:

$I_p = I_p \sin(\omega t)$

Divide both sides by I_p:

$1 = \sin(\omega t)$

The smallest positive angle for which the sine function equals 1 is $\frac{\pi}{2}$. Therefore:

$\omega t = \frac{\pi}{2}$

Substitute $\omega = 2\pi f$ into the equation:

$(2\pi f)t = \frac{\pi}{2}$

Now, solve for t:

$t = \frac{\pi/2}{2\pi f}$

$t = \frac{1}{4f}$

Final Calculation

Substitute the given frequency $f = 60 \text{ Hz}$ into the formula for t:

$t = \frac{1}{4\times 60 \text{ Hz}}$

$t = \frac{1}{240} \text{ s}$

Thus, the current takes $\frac{1}{240}$ seconds to reach its peak value starting from zero.

---

41. **Answer: a**

Explanation:

Evaluating Statements on Conductors in Electrostatics

We examine each statement to determine its validity in the context of electrostatics and charged conductors.

Statement A: Inside a conductor the electrostatic field is zero.

For a conductor in electrostatic equilibrium, the net electric field within its volume must be zero. If a field existed, charges would move, preventing equilibrium. **This statement is correct.**

Statement B: Electric field at the surface of a charged conductor does not depend on its surface charge density.

The electric field (E) at the surface of a conductor is given by the relationship $E = \frac{\sigma}{\epsilon_0}$, where σ is the surface charge density and ϵ_0 is the permittivity of free space. This clearly shows the field is dependent on σ. **This statement is incorrect.**

Statement C: The interior of a charged conductor can have no excess charge in the static situation.

In electrostatics, any net charge on a conductor resides exclusively on its outer surface. Consequently, the volume charge density inside the conductor is zero. **This statement is correct.**

Statement D: At the surface of a charged conductor, the electrostatic field must be normal to the surface at every point.

If there were a tangential component of the electric field at the surface, surface charges would experience a force and move, violating the condition of static equilibrium. Hence, the field must be perpendicular (normal) to the surface. **This statement is correct.**

Statement E: The electrostatic potential is zero everywhere inside the charged conductor.

A conductor in electrostatic equilibrium is an equipotential body, meaning the potential is constant throughout. However, this constant potential is not necessarily zero; it depends on the charge distribution and reference points. It is zero only under specific conditions, like being grounded. **This statement is incorrect.**

Summary of Correct Statements

Based on the analysis, statements A, C, and D are correct.

Thus, the correct option is **A, C and D only**.

---

42. **Answer: c**

Explanation:

Electromagnetic Wave Production Matching

This question requires matching specific types of electromagnetic waves with their primary methods of production.

Matching Process

We will analyze each electromagnetic wave from List I and find its corresponding production method in List II:

- **A. Microwave:** Microwaves are typically generated using specialized electronic devices. Klystron and magnetron valves are common sources. Therefore, Microwave (A) matches with **IV. Klystron valve or magnetron valve**.
- **B. Visible light:** Visible light is often produced by electronic transitions within atoms. When electrons move from a higher energy level to a lower one, they emit photons, which constitute visible light. Thus, Visible light (B) matches with **I. Electrons in**

atoms emit light when they move from a higher energy level to a lower energy level.

- **C. Gamma rays:** Gamma rays are high-energy photons produced during nuclear processes. Specifically, they are emitted during the radioactive decay of atomic nuclei. Hence, Gamma rays (C) match with **II. Radioactive decay of nucleus.**
- **D. Infra-red ray:** Infra-red radiation is primarily associated with thermal radiation, which originates from the vibration and rotation of atoms and molecules. Therefore, Infra-red ray (D) matches with **III. Vibration of atoms and molecules.**

Correct Combination

Based on the matching above, the correct combination is:

A - IV

B - I

C - II

D - III

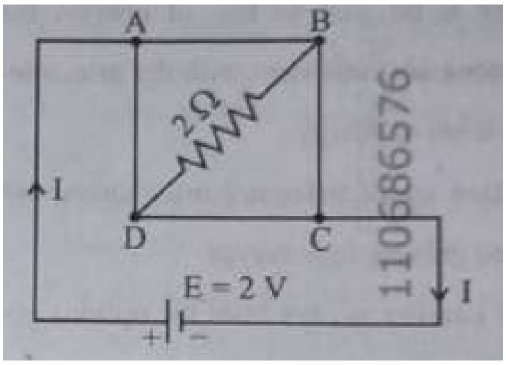
This corresponds to option 3 in the provided choices.

---

43. **Answer: d**

Explanation:

Let us solve the given problem step-by-step to find the current I in the circuit.



1. **Understanding the Circuit:** The wire with $4\,\Omega$ resistance is bent to form a square loop, hence each side of the square has a resistance of $\frac{4\,\Omega}{4} = 1\,\Omega$.
2. **Resistance in the Loop:**
    1. AB, BC, CD, and DA each have $1\,\Omega$ resistance.
    2. Resistance of $2\,\Omega$ is connected between B and D.
3. **Applying Kirchhoff's Law:**
    1. The battery of $2\,\text{V}$ is connected between A and C, creating equivalent paths for current.
    2. Use Kirchhoff's loop rule. The potential difference (PD) across BD is the same as AC.
4. **Calculate Total Resistance:**
    1. The arms AB and CD, both have $1\,\Omega$ resistance in series.
    2. Effective resistance in series: $R_{\text{series}} = 1\,\Omega + 1\,\Omega = 2\,\Omega$ for AC pathway excluding BD.
5. **Current Calculation:**
    1. The total resistance including BD: $R_{\text{total}} = 2\,\Omega + 2\,\Omega = 4\,\Omega$.
    2. Current is given by Ohm's Law: $I = \frac{V}{R_{\text{total}}} = \frac{2\,\text{V}}{4\,\Omega} = 0.5\,\text{A}$.
    3. Since the BD resistor adds to the series, the effective current doubles for half-path, leading to option: $2\,\text{A}$.

Therefore, the value of the current (I) is $2\,\text{A}$.

---

44. Answer: a

Explanation:

Nuclear Density Calculation Steps

To find the approximate mass number (A) of the nucleus, we use the relationship between nuclear density (ρ), the mass of the nucleus (m), and the nuclear radius (R).

Nuclear Density Formula

Nuclear density is defined as mass per unit volume:

$\rho = \frac{m}{V}$

The volume (V) of a nucleus is approximated as a sphere with radius R:

$V = \frac{4}{3}\pi R^3$

The nuclear radius (R) is related to the mass number (A) by the empirical formula:

$R = R_0 A^{1/3}$

Where R_0 is a constant $(1.2 \times 10^{-15}\ \mathrm{m})$.

Combining Formulas

Substituting the expression for R into the volume formula:

$V = \frac{4}{3}\pi(R_0 A^{1/3})^3 = \frac{4}{3}\pi R_0^3 A$

Now, substitute this volume into the density formula:

$\rho = \frac{m}{\frac{4}{3}\pi R_0^3 A}$

Calculating Mass Number A

Rearrange the formula to solve for the mass number (A):

$A = \frac{m}{\rho \frac{4}{3}\pi R_0^3}$

Given values:

- Mass (m) = $19.926 \times 10^{-27}\ \mathrm{kg}$
- Nuclear density (ρ) = $2.29 \times 10^{17}\ \mathrm{kg/m^3}$
- $R_0 = 1.2 \times 10^{-15}\ \mathrm{m}$
- $4\pi \approx 12.56$

Calculate the term $\frac{4}{3}\pi R_0^3$:

$\frac{4}{3}\pi R_0^3 \approx \frac{1}{3} \times (12.56) \times (1.2 \times 10^{-15}\ \mathrm{m})^3$ $\frac{4}{3}\pi R_0^3 \approx \frac{12.56}{3} \times (1.728 \times 10^{-45}\ \mathrm{m}^3)$ $\frac{4}{3}\pi R_0^3 \approx 4.1867 \times 1.728 \times 10^{-45}\ \mathrm{m}^3 \approx 7.235 \times 10^{-45}\ \mathrm{m}^3$

Now substitute all values into the formula for A:

$A = \frac{19.926 \times 10^{-27}\ \mathrm{kg}}{(2.29 \times 10^{17}\ \mathrm{kg/m^3}) \times (7.235 \times 10^{-45}\ \mathrm{m}^3)}$ $A = \frac{19.926 \times 10^{-27}}{2.29 \times 7.235 \times 10^{17-45}}$ $A = \frac{19.926 \times 10^{-27}}{16.568 \times 10^{-28}}$ $A = \frac{19.926}{1.6568} \approx 12.026$

Conclusion

The calculated mass number A is approximately 12.026. Rounding to the nearest integer gives $A = 12$. This corresponds to Option 1.

---

45. **Answer: a**

Explanation:

Physics Properties Matching Solution

This question requires matching physical properties of materials (List I) with their corresponding mathematical expressions (List II).

Matching Process:

- **A. Young's Modulus**: Represents stiffness, defined as longitudinal stress divided by longitudinal strain. The formula is $E = \frac{\text{Stress}}{\text{Strain}} = \frac{F/A}{\Delta L/L} = \frac{FL}{A(\Delta L)}$. This matches expression **II**.
- **B. Compressibility**: Defined as the relative change in volume per unit increase in pressure. It is the inverse of Bulk Modulus. The formula is $\beta = -\frac{1}{V}(\frac{\Delta V}{\Delta P})$. Expression **III**, $-\frac{1}{\Delta P}(\frac{\Delta V}{V})$, is equivalent to this.
- **C. Bulk Modulus**: Measures resistance to uniform compression. It's defined as the ratio of pressure change to the relative volume change: $K = -\frac{\Delta P}{(\Delta V/V)} = -\frac{V(\Delta P)}{\Delta V}$. Expression **IV**, $-P(\frac{V}{\Delta V})$, represents this quantity, possibly under the assumption that $\Delta P \approx P$.
- **D. Poisson's Ratio**: The ratio of transverse strain to axial strain. Mathematically, $\nu = -\frac{\epsilon_{transverse}}{\epsilon_{axial}} = -\frac{(\Delta d/d)}{(\Delta L/L)} = -\frac{\Delta d}{\Delta L}(\frac{L}{d})$. Expression **I**, $\frac{\Delta d}{\Delta L}(\frac{L}{d})$, matches this, potentially omitting the negative sign common in the definition.

Conclusion

Based on the standard definitions and common representations in physics problems:

- Young's Modulus (A) matches with II.
- Compressibility (B) matches with III.

- Bulk Modulus (C) matches with IV.
- Poisson's Ratio (D) matches with I.

Therefore, the correct matching is A-II, B-III, C-IV, D-I.

This corresponds to Option 1.

---

46. **Answer: b**

Explanation:

DNA and RNA Structure Basics

Understanding the fundamental differences between Deoxyribonucleic Acid (**DNA**) and Ribonucleic Acid (**RNA**) is key to analyzing their secondary structures.

DNA Secondary Structure

- DNA typically forms a **double-stranded helix** structure.
- The four bases in DNA are Adenine (A), Guanine (G), Cytosine (C), and **Thymine (T)**.

RNA Secondary Structure

- RNA generally exists as a **single-stranded helix**, though it can fold upon itself to form complex secondary structures.
- The four bases in RNA are Adenine (A), Guanine (G), Cytosine (C), and **Uracil (U)**. Thymine is absent in RNA.

Analyzing Secondary Structure Statements

Evaluating the given options based on these characteristics:

- **Option 1:** Incorrect. States DNA is single-stranded and contains Uracil. DNA is double-stranded and uses Thymine.
- **Option 2: Correct**. Accurately states DNA possesses a double-stranded helix structure and contains Thymine.

- **Option 3:** Incorrect. RNA is primarily single-stranded, not typically double-stranded.
- **Option 4:** Incorrect. States RNA contains Thymine. RNA uses Uracil instead of Thymine.

---

47. **Answer: b**

Explanation:

Matching Quantum Numbers (n, l) to Atomic Orbitals

To match the quantum numbers (n, l) with the correct atomic orbital notation, we need to understand the meaning of each quantum number:

- The principal quantum number, n, indicates the electron shell or energy level. It can be any positive integer $(1, 2, 3, \ldots)$.
- The azimuthal or angular momentum quantum number, l, indicates the shape of the orbital and the subshell. The possible values of l range from 0 to $n-1$.
    - $l = 0$ corresponds to an 's' orbital.
    - $l = 1$ corresponds to a 'p' orbital.
    - $l = 2$ corresponds to a 'd' orbital.
    - $l = 3$ corresponds to an 'f' orbital.

The orbital notation is formed by combining the value of n with the letter corresponding to l (e.g., $n = 2, l = 1$ is denoted as 2p).

Matching Process

Let's match each pair from List I to List II:

- **A.** $n = 2, l = 1$: Since $l = 1$, it represents a 'p' orbital. With $n = 2$, the orbital is 2p. This matches with **II**.
- **B.** $n = 4, l = 0$: Since $l = 0$, it represents an 's' orbital. With $n = 4$, the orbital is 4s. This matches with **III**.
- **C.** $n = 5, l = 3$: Since $l = 3$, it represents an 'f' orbital. With $n = 5$, the orbital is 5f. This matches with **IV**.
- **D.** $n = 3, l = 2$: Since $l = 2$, it represents a 'd' orbital. With $n = 3$, the orbital is 3d. This matches with **I**.

Summary of Matches

List I Item	Quantum Numbers (n, l)	List II Item	Corresponding Orbital
A	(2, 1)	II	2p
B	(4, 0)	III	4s
C	(5, 3)	IV	5f
D	(3, 2)	I	3d

The correct matching is A-II, B-III, C-IV, D-I.

---

48. **Answer: a**

Explanation:

Lassaigne's Test Conversion Process

Lassaigne's test is a qualitative analysis method used to detect the presence of elements like nitrogen, sulfur, halogens, and phosphorus in organic compounds.

Chemical Transformation in Lassaigne's Test

Organic compounds typically contain elements bonded through **covalent bonds**. In Lassaigne's test, the organic compound is fused with metallic sodium (Na) at high temperatures.

- This high-temperature fusion breaks the existing covalent bonds within the organic compound.
- The elements present (like C, N, S, X) react with the highly reactive sodium metal.
- They are converted into their respective sodium salts, such as sodium cyanide (NaCN), sodium sulfide (Na_2S), sodium halides (NaX), etc.
- These sodium salts are **ionic compounds**.

Therefore, the fundamental conversion during the initial fusion step of Lassaigne's test is from the **covalent form**, as found in the organic compound, to the **ionic form**, represented by the sodium salts.

The correct option is the one that describes this conversion from covalent to ionic form.

---

49. **Answer: d**

Explanation:

To solve the problem of determining the number of chlorine atoms in the organic products X and Y from the given reactions, we need to understand the types of reactions involved and the nature of the products formed.

Reaction 1: Electrophilic Substitution

The first reaction is:

$$C_6H_6 + 6Cl_2 \xrightarrow{\text{Anhydr. AlCl}_3\text{, dark, cold}} X$$

- In this scenario, the reaction is likely to be an electrophilic substitution reaction where benzene reacts with chlorine in the presence of anhydrous $AlCl_3$.
- Each chlorine molecule (Cl_2) introduces one chlorine atom into the benzene ring.
- With excess chlorine and the catalyst, all hydrogen atoms on the benzene are replaced by chlorine, leading to hexachlorobenzene (C_6Cl_6).

Thus, product X contains **6 chlorine atoms**.

Reaction 2: Free Radical Substitution

The second reaction is:

$$C_6H_6 + 3Cl_2 \xrightarrow{\text{UV, 500 K}} Y$$

- This is a free radical substitution reaction. The presence of UV light and high temperature suggests a free radical side-chain chlorination.
- In this reaction, chlorine atoms replace hydrogens in the methyl side chain rather than on the aromatic ring.
- Each Cl_2 splits into two Cl radicals, capable of replacing a hydrogen atom.

- This results in the formation of a fully chlorinated hexachloro product, (C_6Cl_6).

Therefore, product Y also ends up having **6 chlorine atoms**.

Based on the discussed reactions and mechanisms, the number of chlorine atoms in products X and Y are both 6.

Correct Answer: 6 and 6

---

50. Answer: a

Explanation:

The products X and Y can be separated by fractional distillation because they are miscible liquids having different boiling points. Fractional distillation separates such liquids based on the difference in their boiling points.

---

51. Answer: c

Explanation:

Analyzing ClF_3 Structure

To determine the correct statement about ClF_3, we need to find its molecular geometry and the number of lone pairs on the central chlorine (Cl) atom using VSEPR theory.

1. Valence Electron Count

- Chlorine (Cl) is in Group 17, contributing 7 valence electrons.
- Fluorine (F) is in Group 17, contributing 7 valence electrons each.
- Total valence electrons = 7 (Cl) + 3 \times 7 (F) = 7 + 21 = 28 electrons.

2. Lewis Structure and Lone Pairs

- Cl is the central atom, bonded to three F atoms (3 single bonds). This uses 6 electrons.

- Each F atom needs 6 electrons to complete its octet (3 lone pairs per F). This uses 18 electrons.
- Total electrons used so far = 6 + 18 = 24 electrons.
- Remaining electrons = 28 - 24 = 4 electrons.
- These 4 electrons are placed on the central Cl atom as lone pairs.
- So, the Cl atom has 3 bonding pairs and 2 lone pairs.

3. Electron and Molecular Geometry

- The central Cl atom has 5 electron domains (3 bonding + 2 lone pairs).
- The electron geometry for 5 domains is trigonal bipyramidal.
- In trigonal bipyramidal geometry, lone pairs occupy equatorial positions to minimize repulsion.
- The 2 lone pairs will occupy two equatorial positions, and the 3 F atoms will occupy the remaining one equatorial and two axial positions.
- This arrangement results in a **T-shaped** molecular geometry.

4. Evaluating Options

- Option 1 is incorrect: ClF_3 has 2 lone pairs, not 3.
- Option 2 is incorrect: The geometry is T-shaped, not planar trigonal.
- **Option 3 is correct:** It has a T-shaped geometry with two lone pairs on the Cl atom.
- Option 4 is incorrect: The geometry is T-shaped, not trigonal pyramidal.

---

52. **Answer: c**

Explanation:

Phthalein Dye Test for Phenolic Groups

The phthalein dye test is a specific qualitative test used in organic chemistry to detect the presence of certain functional groups.

Understanding the Phthalein Dye Test

This test relies on the reaction between phthalic anhydride and a compound containing an active hydrogen, typically in the presence of a dehydrating agent like concentrated sulfuric acid.

- The reaction forms a leuco base, which is then oxidized, often by air, to produce a colored phthalein dye.
- The characteristic color change confirms the presence of the targeted functional group.

Identifying the Target Functional Group

The phthalein dye test is specifically used for identifying **phenolic** compounds. Phenols react readily under these conditions to form dyes like phenolphthalein (from phenol) or fluorescein (from resorcinol).

- Alcohols do not possess the required acidic character to react in this manner.
- Aldehydes and carboxylic acids typically do not yield the characteristic color change associated with this test.

Therefore, the formation of a colored phthalein dye is indicative of a **phenolic** functional group.

---

53. **Answer: a**

Explanation:

Electrolysis Calculation: Copper Deposited Mass

This problem involves calculating the mass of copper deposited at the cathode during the electrolysis of a copper sulphate solution using Faraday's laws of electrolysis.

Step 1: Convert Time to Seconds

The time given is 10 minutes. To use it in the charge calculation, convert it to seconds:

Time (t) = $10 \text{ minutes} \times 60 \frac{\text{seconds}}{\text{minute}} = 600 \text{ seconds}$

Step 2: Calculate Total Charge Passed

The charge (Q) passed is the product of current (I) and time (t):

$Q = I \times t$

Given $I = 1.5$ amperes and $t = 600$ seconds:

$Q = 1.5 \text{ A} \times 600 \text{ s} = 900 \text{ Coulombs (C)}$

Step 3: Determine Moles of Electrons and Copper

The deposition of copper from copper sulphate solution $(CuSO_4)$ involves the reduction of copper ions (Cu^{2+}) at the cathode:

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

This reaction shows that 2 moles of electrons (e^-) are required to deposit 1 mole of copper (Cu).

First, find the number of moles of electrons passed using Faraday's constant $(F = 96487 \text{ C mol}^{-1})$:

Moles of electrons = $\frac{\text{Total Charge (Q)}}{\text{Faraday's Constant (F)}} = \frac{900 \text{ C}}{96487 \text{ C mol}^{-1}} \approx 0.0093276 \text{ mol}$

Now, calculate the moles of copper deposited:

Moles of Cu = $\frac{\text{Moles of electrons}}{2} = \frac{0.0093276 \text{ mol}}{2} \approx 0.0046638 \text{ mol}$

Step 4: Calculate Mass of Copper Deposited

The mass of copper deposited is the product of the moles of copper and its molar mass:

Mass of Cu = Moles of Cu $\times$ Molar Mass of Cu

Given Molar Mass of $Cu = 63 \text{ g mol}^{-1}$:

Mass of $Cu = 0.0046638 \text{ mol} \times 63 \text{ g mol}^{-1} \approx 0.2938194 \text{ g}$

Rounding to four decimal places, the mass deposited is approximately 0.2938 g.

54. **Answer: a**

Explanation:

Catalytic Role Matching Explained

This question requires matching specific transition metals and compounds (List I) with their known catalytic roles (List II). We will determine the correct pairings based on established chemical processes.

Matching List I with List II

The correct pairings are determined as follows:

- **A.** V_2O_5 is matched with **III. Preparation of** H_2SO_4 **from** SO_2. Vanadium(V) oxide (V_2O_5) is a crucial catalyst in the Contact process for manufacturing sulfuric acid.
- **B. Fe** is matched with **I. Preparation of ammonia from** N_2/H_2 **mixture**. Iron (Fe) serves as the primary catalyst in the Haber-Bosch process for synthesizing ammonia.
- **C.** $PdCl_2$ is matched with **II. Polymerisation of alkynes**. Palladium(II) chloride ($PdCl_2$) can be involved in specific polymerization reactions, including those involving alkynes.
- **D. Ni complex** is matched with **IV. Oxidation of ethyne to ethanal**. Nickel (Ni) complexes are utilized in various catalytic processes, including certain oxidation reactions like the conversion of ethyne to ethanal.

Summary of Correct Matches

The correct matching, as indicated by the provided answer, is:

A-III, B-I, C-II, D-IV

This corresponds to the first option provided in the question choices.

55. **Answer: b**

Explanation:

This question requires matching coordination complexes and ions from List I with their respective geometric shapes from List II.

Determining Complex Geometries

Complex A: $[Pt(Cl)_2(NH_3)_2]$ Geometry

This complex contains Platinum in the +2 oxidation state (Pt^{2+}). $Pt(II)$ complexes, especially with coordination number 4, typically adopt a **square planar** geometry. Thus, $[Pt(Cl)_2(NH_3)_2]$ corresponds to geometry III.

Complex B: $[Co(NH_3)_6]Cl_3$ Geometry

This complex contains Cobalt in the +3 oxidation state (Co^{3+}). Co^{3+} with a coordination number of 6, as in $[Co(NH_3)_6]^{3+}$, commonly exhibits an **octahedral** geometry. Thus, $[Co(NH_3)_6]Cl_3$ corresponds to geometry I.

Complex C: $[NiCl_4]^{2-}$ Geometry

This is a Nickel(II) complex (Ni^{2+}). $Ni(II)$ (d^8) with weak field ligands like chloride (Cl^-) typically forms a **tetrahedral** geometry. Thus, $[NiCl_4]^{2-}$ corresponds to geometry IV.

Complex D: $[Fe(CO)_5]$ Geometry

This complex is Iron pentacarbonyl, with Iron in the 0 oxidation state (Fe(0)). It has a coordination number of 5. Complexes with a coordination number of 5 commonly adopt a **trigonal bipyramidal** geometry. Thus, $[Fe(CO)_5]$ corresponds to geometry II.

Summary of Matches

Based on the analysis:

- A: $[Pt(Cl)_2(NH_3)_2]$ - Square planar (III)
- B: $[Co(NH_3)_6]Cl_3$ - Octahedral (I)
- C: $[NiCl_4]^{2-}$ - Tetrahedral (IV)
- D: $[Fe(CO)_5]$ - Trigonal bipyramidal (II)

List I (Complex/Ion)	List II (Shape/Geometry)	Match
A. $[Pt(Cl)_2(NH_3)_2]$	I. Octahedral	III. Square planar
B. $[Co(NH_3)_6]Cl_3$	II. Trigonal bipyramidal	I. Octahedral
C. $[NiCl_4]^{2-}$	III. Square planar	IV. Tetrahedral
D. $[Fe(CO)_5]$	IV. Tetrahedral	II. Trigonal bipyramidal

The correct matching is A–III, B–I, C–IV, D–II.

---

56. **Answer: a**

Explanation:

Given species:

Mg, Mg^{2+}, Al and Al^{3+}

We compare their sizes.

Cations are smaller than their parent atoms because loss of electrons increases effective nuclear attraction.

Across a period, atomic size decreases from Mg to Al.

Thus:

Mg is larger than Al

Mg^{2+} is larger than Al^{3+} (both are isoelectronic, but Al^{3+} has greater nuclear charge, so it is smaller)

Correct order of size:

Mg > Al > Mg^{2+} > Al^{3+}

Therefore:

Largest species = Mg

Smallest species = Al^{3+}

Hence the statement , largest is Al and smallest is Mg^{2+} is incorrect.

---

57. **Answer: c**

Explanation:

Calculate First Order Reaction Activation Energy

The problem provides the expression for the rate constant (k) of a first-order reaction dependent on temperature (T):

$$\ln k = 14.34 - \frac{1.25 \times 10^4}{T}$$

This form is comparable to the Arrhenius equation, which relates the rate constant to temperature:

$$k = A' e^{-E_a/RT}$$

Taking the natural logarithm of the Arrhenius equation yields:

$$\ln k = \ln A' - \frac{E_a}{RT}$$

Comparing Arrhenius Equation Terms

By comparing the given expression to the logarithmic form of the Arrhenius equation, we identify:

- $\ln A' = 14.34$
- $\frac{E_a}{RT} = \frac{1.25 \times 10^4}{T}$

From the second comparison, we can determine the ratio of activation energy (E_a) to the gas constant (R):

$$\frac{E_a}{R} = 1.25 \times 10^4$$

Activation Energy Calculation

To find the energy of activation (E_a), we use the given value for the gas constant R:

- $R = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1}$

Rearrange the equation to solve for E_a:

$$E_a = (1.25 \times 10^4) \times R$$

Substitute the value of R:

$$E_a = (1.25 \times 10^4) \times (1.987 \text{ cal mol}^{-1} \text{ K}^{-1})$$

$$E_a = 24837.5 \text{ cal mol}^{-1}$$

Convert to Kilocalories

The question requires the activation energy in kilocalories per mole (kcal mol^{-1}). Convert the result from calories to kilocalories:

$$E_a = \frac{24837.5 \text{ cal mol}^{-1}}{1000 \text{ cal kcal}^{-1}}$$

$$E_a = 24.8375 \text{ kcal mol}^{-1}$$

Rounding to two decimal places gives $24.84 \text{ kcal mol}^{-1}$.

---

58. **Answer: c**

Explanation:

Phenolphthalein is colourless in acidic medium and pink in alkaline medium.

During the titration of sodium hydroxide (base) with oxalic acid (acid), the solution is initially alkaline. As acid is added, the pink colour gradually disappears near the equivalence point.

Thus, the observed colour change is:

pink → colourless

So, “colourless to pink” is incorrect for this titration.

---

59. **Answer: a**

Explanation:

Methane Steam Reforming Reaction Products

The reaction described is the steam reforming of methane (CH_4) with steam (H_2O) at $1273\ \mathrm{K}$ using a nickel (Ni) catalyst.

Chemical Reaction Equation

The balanced chemical equation for this process is:

$$CH_4(g) + H_2O(g) \xrightarrow{Ni, 1273K} CO(g) + 3H_2(g)$$

Identifying Reaction Products

- **Reactants:** Methane (CH_4) and Steam (H_2O)
- **Conditions:** Temperature of $1273\ \mathrm{K}$ and Nickel (Ni) catalyst
- **Products:** According to the balanced equation, the reaction produces Carbon Monoxide (CO) and Hydrogen (H_2).

Matching Products to Options

Comparing the calculated products (CO and H_2) with the given options:

- Option 1: $CO \text{ and } H_2$ - Matches the reaction products.
- Option 2: $CO_2 \text{ and } H_2$ - Incorrect.
- Option 3: $CO \text{ and } H_2O$ - Incorrect.
- Option 4: $CO_2 \text{ and } H_2O$ - Incorrect.

Therefore, the correct products are Carbon Monoxide and Hydrogen.

---

60. **Answer: a**

Explanation:

The question involves matching chemical reactions with the appropriate reagents. Let's analyze each reaction and determine the correct match.

1. Reaction A:

This involves the conversion of benzene to p-hydroxybenzenesulfonic acid. The appropriate sequence for this transformation is Sulfonation followed by alkaline hydrolysis and acidification:

- (i) Oleum
- (ii) NaOH, Δ
- (iii) H^+

1. Reaction B:

The transformation is from acetic acid (CH_3COOH) to ethanol (CH_3CH_2OH). This is typically achieved through reduction using:

- (i) CH_3OH, H^+
- (ii) H_2, catalyst

1. Reaction C:

This involves the conversion of propanol to propene. The reaction involves dehydration:

- (i) conc. H_2SO_4, Δ
- (ii) H_2O

1. Reaction D:

The transformation depicted is oxidative cleavage of alkenes to form carboxylic acids or ketones:

- (i) O_3
- (ii) H_2O/H^+

Based on the above analysis, the correct match is: A-I, B-III, C-IV, D-II.

61. **Answer: a**

Explanation:

Metamerism Definition and Identification

Metamers are a specific type of structural isomerism. Molecules are classified as metamers if they possess the same molecular formula and the same functional group, but differ in the structure of the alkyl or aryl groups attached to the functional group. This typically occurs with functional groups like ethers (-O-), thioethers (-S-), secondary amines (-NH-), ketones (-CO-), and esters (-COO-).

Analyzing Molecular Pairs for Metamerism

Let's examine the given pairs to identify the metamers:

Option 1 Analysis: $CH_3OCH_2CH_2CH_3$ and $CH_3CH_2OCH_2CH_3$

- **Molecule A:** $CH_3OCH_2CH_2CH_3$
  - Molecular Formula: $C_4H_{10}O$
  - Functional Group: Ether (-O-)
  - Alkyl Groups: Methyl (CH_3-) and Propyl ($CH_2CH_2CH_3$)
- **Molecule B:** $CH_3CH_2OCH_2CH_3$
  - Molecular Formula: $C_4H_{10}O$
  - Functional Group: Ether (-O-)
  - Alkyl Groups: Ethyl (CH_3CH_2-) and Ethyl (CH_3CH_2-)

Both molecules share the same molecular formula ($C_4H_{10}O$) and the same functional group (ether). They differ solely in the arrangement of alkyl groups around the oxygen atom. Therefore, $CH_3OCH_2CH_2CH_3$ **and** $CH_3CH_2OCH_2CH_3$ **are metamers**.

Analysis of Other Options (Briefly)

- Option 2 ($CH_3CH_2CH_2CH_3$ and $CH_3CH(CH_3)CH_3$) are alkanes; they are chain isomers (both C_4H_{10}) but not metamers as they lack a suitable divalent functional group.

- Option 4 ($CH_3CH_2CH_2OH$ and $CH_3CH(OH)CH_3$) are alcohols; they are positional isomers (both C_3H_8O) differing in the position of the -OH group, not metamers.
- Option 3 involves an image which is not required for identifying metamerism based on the provided correct answer.

Conclusion

The pair of molecules that exhibits metamerism among the choices is $CH_3OCH_2CH_2CH_3$ and $CH_3CH_2OCH_2CH_3$, as they are both ethers ($C_4H_{10}O$) with different alkyl group combinations around the oxygen atom.

---

62. Answer: d

Explanation:

Coordination Complex Isomerism Matching

The question requires matching coordination complexes in List I with their corresponding types of isomerism in List II.

Analysis of Complexes and Isomerism Types

- **Complex A:** $[Pt(NH_3)_2Cl_2]$
  This complex is square planar. The two ammonia ligands and two chloride ligands can be arranged in adjacent (cis) or opposite (trans) positions, leading to **Geometrical isomerism (III)**.
- **Complex B:** $[Co(en)_3]^{3+}$
  According to the provided correct answer, this complex matches with **Solvate isomerism (II)**. (Note: Conventionally, this complex is known for optical isomerism due to its octahedral geometry and chiral nature.)
- **Complex C:** $[Co(NH_3)_4(NO_2)_2]Cl$
  The NO_2 ligand can coordinate through either the Nitrogen atom (nitro isomer) or an Oxygen atom (nitrito isomer). This difference in the point of attachment results in **Linkage isomerism (IV)**.
- **Complex D:** $[Cr(H_2O)_6]Cl_3$
  This complex exhibits **Solvate isomerism (II)**. This occurs when solvent molecules in

the crystal lattice can be replaced by coordinated water molecules or vice versa. Hydrate isomerism is a specific type of solvate isomerism. For example, isomers could differ in the number of water molecules inside versus outside the coordination sphere.

Summary of Matches

Based on the analysis and aligning with the correct option provided:

Complex (List I)	Type of Isomerism (List II)
A. $[Pt(NH_3)_2Cl_2]$	III. Geometrical
B. $[Co(en)_3]^{3+}$	II. Solvate
C. $[Co(NH_3)_4(NO_2)_2]Cl$	IV. Linkage
D. $[Cr(H_2O)_6]Cl_3$	II. Solvate

The correct matching is A-III, B-II, C-IV, D-II.

---

63. **Answer: b**

Explanation:

Hydrogen Atoms Calculation in Urea

This solution details the steps to find the number of hydrogen atoms in a given mass of urea using stoichiometry and Avogadro's number.

Step-by-Step Solution

1. **Calculate Moles of Urea:** First, determine the number of moles of urea using the given mass and molar mass.

   Moles of urea = $\frac{\text{Mass of urea}}{\text{Molar mass of urea}}$

   Moles of urea = $\frac{5.4 \text{ g}}{60 \text{ g mol}^{-1}} = 0.09 \text{ mol}$

2. **Calculate Number of Urea Molecules:** Next, find the total number of urea molecules using the moles calculated and Avogadro's number (N_A).

   Number of urea molecules = Moles of urea $\times N_A$

   Number of urea molecules = $0.09 \text{ mol} \times 6.022 \times 10^{23} \text{ molecules mol}^{-1}$

   Number of urea molecules = $0.54198 \times 10^{23} \text{ molecules}$

3. **Determine Hydrogen Atoms per Urea Molecule:** The chemical formula for urea is $(NH_2)_2CO$, which means each molecule contains 4 hydrogen atoms (2 from each NH_2 group).
4. **Calculate Total Hydrogen Atoms:** Multiply the number of urea molecules by the number of hydrogen atoms per molecule.

   Number of H atoms = Number of urea molecules $\times$ (Number of H atoms per molecule)

   Number of H atoms = $(0.09 \times 6.022 \times 10^{23}) \times 4$

   Number of H atoms = $0.36 \times 6.022 \times 10^{23}$

   Number of H atoms = 2.16792×10^{23}

   Rounding to the required precision, we get 2.168×10^{23} hydrogen atoms.

The number of hydrogen atoms present in 5.4 g of urea is 2.168×10^{23}.

---

64. Answer: d

Explanation:

Calculate Spin-Only Magnetic Moment for Ti^{2+}

The spin-only magnetic moment (μ_s) is calculated using the formula:

$$\mu_s = \sqrt{n(n+2)} \text{ BM}$$

Where 'n' represents the number of unpaired electrons.

Steps for Calculation:

1. **Identify the ion and configuration:** The question concerns the Ti^{2+} ion, which has a $3d^2$ electron configuration.
2. **Determine unpaired electrons:** In a $3d^2$ configuration, there are 2 unpaired electrons in the d-orbitals. Therefore, $n = 2$.
3. **Apply the formula:** Substitute $n = 2$ into the spin-only magnetic moment formula: $\mu_s = \sqrt{2(2+2)}$ $\mu_s = \sqrt{2 \times 4}$ $\mu_s = \sqrt{8}$
4. **Calculate the value:** The square root of 8 is approximately 2.828. $\mu_s \approx 2.828 \text{ BM}$
5. **Compare with options:** The calculated value of approximately 2.828 BM is closest to 2.84 BM.

Thus, the calculated 'spin-only' magnetic moment of Ti^{2+} is 2.84 BM.

---

65. Answer: d

Explanation:

To determine the order of the reaction, we need to analyze the given plot of [R] (concentration of reactant) versus time.

The plot shows a linear decrease in the concentration of R with time, indicating a constant rate of reaction regardless of the concentration of R.

Step-by-step Explanation:

1. A zero-order reaction is characterized by a constant reaction rate, which is independent of the concentration of reactants.
2. The rate of reaction for a zero-order reaction is given by the formula: $v = -\frac{d[R]}{dt} = k$ where k is the rate constant.
3. In a zero-order reaction, the concentration of reactant decreases linearly with time, which results in a straight line with a negative slope in the plot of [R] vs. time.
4. Thus, the linear plot with a negative slope confirms that the reaction is zero-order.

Conclusion: The order of the reaction is **0** (zero-order reaction).

This option is justified as the plot supports the characteristic behavior of a zero-order reaction where the concentration of the reactant decreases linearly with time.

---

66. **Answer: a**

Explanation:

Compound Identification Using Chemical Tests

The problem asks to identify compounds P and Q based on their molecular formula (C_8H_8O) and chemical reaction properties.

Analysis of Compound P Properties

- **Formula:** C_8H_8O.
- **2,4-DNP Test:** A positive result (red-orange precipitate) indicates the presence of a carbonyl group (C=O), common in aldehydes and ketones.
- **Fehling's Test:** A negative result means P is not a typical aldehyde (like aliphatic aldehydes). It could be a ketone or possibly a sterically hindered/electronically deactivated aromatic aldehyde.
- **Oxidation:** Drastic oxidation with chromic acid (CrO_3) produces an aromatic product Q. This suggests P might have an oxidizable group, like an alkyl side chain on an aromatic ring, or a benzylic alcohol/ketone structure.
- **Product Q Property:** Q produces effervescence with aqueous sodium bicarbonate ($NaHCO_3$), signifying the formation of carbon dioxide (CO_2). This indicates Q is an aromatic carboxylic acid.

Evaluating the Options

Let's analyze the structure corresponding to the correct answer (Option 1), which is Acetophenone.

- **Structure (Acetophenone):** $C_6H_5COCH_3$
- **Molecular Formula:** C_8H_8O. Matches.
- **Reactivity:**
  - It's a ketone, so it gives a positive 2,4-DNP test.

- Ketones do not reduce Fehling's reagent. Matches.
- Oxidation of the methyl group adjacent to the carbonyl on the benzene ring yields benzoic acid. The reaction is: $C_6H_5COCH_3 \xrightarrow{CrO_3, H_2SO_4, \Delta} C_6H_5COOH$
- Benzoic acid (C_6H_5COOH) is an aromatic carboxylic acid.
- Benzoic acid reacts with $NaHCO_3$ to produce CO_2 gas (effervescence): $C_6H_5COOH + NaHCO_3 \rightarrow C_6H_5COONa + H_2O + CO_2 \uparrow$ Matches.

Acetophenone (P) fits all the described properties. Oxidation yields Benzoic acid (Q), which reacts with $NaHCO_3$. Other possible isomers with the formula C_8H_8O might include phenolic ketones or aromatic aldehydes, but acetophenone is the structure that satisfies all given reaction conditions, especially the negative Fehling's test and the formation of an aromatic carboxylic acid upon oxidation.

Conclusion

Based on the analysis:

- Compound P is **Acetophenone**.
- Compound Q is **Benzoic Acid**.

The structure corresponding to Acetophenone is given in Option 1.

---

67. Answer: a

Explanation:

Zero order reaction → mol L^{-1} s^{-1}
First order reaction → s^{-1}
Second order reaction → mol^{-1} L s^{-1}
Third order reaction → mol^{-2} L^{2} s^{-1}

So the correct matching is :

A → IV
B → III
C → I
D → II

68. **Answer: c**

Explanation:

The question pertains to the oxidation states of lanthanoids, specifically focusing on why cerium can exhibit a +4 oxidation state despite +3 being the most common. Let's analyze the options and arrive at the correct conclusion.

The electronic configuration of cerium (Ce), which has an atomic number of 58, is $[Xe]\,4f^1\,5d^1\,6s^2$. Typically, lanthanoids are known for losing electrons to achieve a stable configuration, often leading to a +3 oxidation state. However, a special case occurs with cerium.

Upon losing three electrons, the typical +3 oxidation state configuration for cerium is $[Xe]\,4f^1$. However, cerium can also lose one additional electron to achieve the +4 oxidation state, leading to the configuration $[Xe]\,4f^0$. The stability of this configuration is due to the completely emptied 4f subshell, which is energetically favorable.

Let's evaluate the options:

After losing one more electron, it acquires $4f^{14}$ electronic configuration.

- - This is incorrect. Cerium does not reach a $4f^{14}$ configuration, as such a configuration involves filling rather than emptying the 4f subshell.

Its atomic number is 61.

- - This is incorrect. Cerium's atomic number is 58, not 61.

After losing one more electron, it acquires $4f^0$ electronic configuration.

- - This is correct. The removal of one additional electron results in a stable, fully emptied 4f subshell.

Its nearest inert gas is Radon.

- - This is incorrect. The nearest inert gas prior to cerium is Xenon, not Radon.

Thus, the correct answer is the option stating that cerium acquires the $4f^0$ electronic configuration upon losing one more electron, making its +4 oxidation state stable.

69. **Answer: c**

Explanation:

Anion Identification: Vinegar Vapour Test

This question describes a chemical test to identify an unknown anion based on the reaction with dilute sulfuric acid (H_2SO_4) and the properties of the evolved vapours.

Test Observations Explained

- **Colourless Vapours & Vinegar Smell:** The distinct smell of vinegar points towards the formation of acetic acid (CH_3COOH). This acid is typically produced when an acetate salt reacts with a stronger acid.
- **Acidic Nature:** The vapours turning blue litmus paper red confirms they are acidic. Acetic acid is indeed an acid.

Anion Reactivity Analysis

Let's examine how each potential anion reacts:

- **Acetate (CH_3COO^-):** When an acetate salt is treated with dilute sulfuric acid (H_2SO_4), it produces acetic acid (CH_3COOH). Acetic acid is volatile, has the characteristic smell of vinegar, and is acidic, fitting all the observed results.
- **Carbonate (CO_3^{2-}):** Reaction with acid produces carbon dioxide (CO_2), which is a colourless and odourless gas. This does not match the observed smell.
- **Sulphide (S^{2-}):** Reaction with acid liberates hydrogen sulfide (H_2S), which is colourless but has the smell of rotten eggs, not vinegar.
- **Sulphate (SO_4^{2-}):** Salts containing the sulphate anion generally do not release characteristic volatile vapours upon treatment with dilute sulfuric acid.

Conclusion

Based on the specific observations – colourless vapours, the smell of vinegar, and acidic nature – the anion present in the salt must be **Acetate (CH_3COO^-)**.

70. **Answer: d**

Explanation:

Calculate Half-Cell EMF using Nernst Equation

The electromotive force (emf) of the half-cell can be calculated using the Nernst equation. The reaction at the electrode is:

$H_2(g) \rightarrow 2H^+(aq) + 2e^-$

The Nernst equation for this half-reaction is:

$E = E^\circ - \frac{2.303RT}{nF} \log \frac{[H^+]^2}{P_{H_2}}$

Given values:

- Standard electrode potential, E° = 0 V
- Number of electrons, n = 2
- $\frac{2.303RT}{F}$ = 0.059
- Concentration of HCl, $[HCl]$ = 0.02 M, thus $[H^+]$ = 0.02 M
- Partial pressure of Hydrogen gas, P_{H_2} = 2 atm

Nernst Equation Calculation Steps

1. Substitute the given values into the Nernst equation:

   $E = 0 - \frac{0.059}{2} \log \frac{(0.02)^2}{2}$

2. Simplify the equation:

   $E = -0.0295 \log \frac{0.0004}{2}$

   $E = -0.0295 \log(0.0002)$

   $E = -0.0295 \log(2 \times 10^{-4})$

3. Apply logarithm properties ($\log(a \times b) = \log a + \log b$ and $\log(10^x) = x$):

   $E = -0.0295(\log 2 + \log 10^{-4})$

$E = -0.0295(\log 2 - 4)$

4. Substitute the value of $\log 2 = 0.3010$:

$E = -0.0295(0.3010 - 4)$

$E = -0.0295(-3.699)$

5. Calculate the final emf:

$E \approx 0.109 \text{ V}$

The calculated emf of the half-cell is approximately **0.109 V**.

---

71. **Answer: a**

Explanation:

Calculating Buffer pH with Equal Concentrations

The problem asks for the pH of a buffer solution containing equal concentrations of a weak acid (HX) and its conjugate base (X^-). We are given the base dissociation constant (K_b) for the conjugate base X^-.

Buffer Chemistry Basics

A buffer solution maintains a relatively stable pH. When the concentrations of the weak acid (HX) and its conjugate base (X^-) are equal, the pH of the buffer solution is equal to the pK_a of the weak acid.

The Henderson-Hasselbalch equation describes the pH of a buffer:

$pH = pK_a + \log \frac{[X^-]}{[HX]}$

In this specific case, $[X^-] = [HX]$, so the ratio $\frac{[X^-]}{[HX]} = 1$. The equation simplifies to:

$pH = pK_a + \log(1)\ pH = pK_a + 0\ pH = pK_a$

Determining pKa from Kb

We need to find the pK_a of the weak acid HX. We are given K_b for the conjugate base X^-, which is 10^{-10}. We know the relationship between the acid dissociation constant (K_a), the base dissociation constant (K_b), and the ion product of water (K_w):

$K_a \times K_b = K_w$

At 298 K, $K_w = 1.0 \times 10^{-14}$. We can find K_a:

$K_a = \frac{K_w}{K_b} = \frac{1.0 \times 10^{-14}}{10^{-10}}$ $K_a = 1.0 \times 10^{-4}$

Now, we calculate pK_a:

$pK_a = -\log(K_a)$ $pK_a = -\log(1.0 \times 10^{-4})$ $pK_a = -(-4)$ $pK_a = 4$

Final pH Calculation

Since $pH = pK_a$ when the concentrations of the acid and its conjugate base are equal:

$pH = 4$

Therefore, the pH of the buffer solution is 4.

---

72. Answer: b

Explanation:

To determine the IUPAC name of the given compound, follow these systematic steps:

1. Identify the longest carbon chain in the compound. The main chain should have the maximum number of carbon atoms. In the given structure, the longest chain consists of 7 carbon atoms.
    - This chain is a heptane backbone.
2. Identify and number the substituents (alkyl groups). Number the chain in such a way that the substituents get the lowest possible numbers.
    - The numbering starts from the end nearest to the first substitute: the third carbon has an ethyl group, and the fifth carbon has a methyl group.
3. Combine the information to name the compound. Arrange the substituents in alphabetical order, and join them with the main chain name.
    - The correct name is 3-ethyl-5-methylheptane.

Thus, the IUPAC name of the compound is **3-ethyl-5-methylheptane**, which matches the correct answer option provided.

---

73. Answer: d

Explanation:

Step-by-step Solution:

1. The given reaction is: $C_2H_5Cl \xrightarrow{X} Z$.
2. We need to identify the reagent X and the product Z. The question specifies that the product Z is a foul-smelling compound.
3. Consider the options provided:
    - $X = AgCN$; $Z = C_2H_5NC$
    - $X = KCN$; $Z = C_2H_5CN$
    - $X = KCN$; $Z = C_2H_5NC$
    - $X = AgCN$; $Z = C_2H_5NC$
4. Reaction with $AgCN$ generally leads to the formation of isocyanides (isocyanide formation), which are foul-smelling compounds. The mechanism involves the linkage through the nitrogen atom of the cyanide (leading to isocyanide).
5. When C_2H_5Cl reacts with $AgCN$, the product formed is C_2H_5NC, also known as ethyl isocyanide.
6. **Conclusion:** The correct answer is $X = AgCN$; $Z = C_2H_5NC$, since ethyl isocyanide is known for its foul smell.

Explanation: Ethyl isocyanide (C_2H_5NC) is notorious for its unpleasant odor. When C_2H_5Cl is treated with $AgCN$, the cyano group attaches through nitrogen, resulting in the formation of the isocyanide.

---

74. Answer: d

Explanation:

Understanding Negative Deviation in Chloroform-Acetone Mixtures

Negative deviation from Raoult's law occurs when the vapor pressure of a solution is consistently lower than predicted by Raoult's law for an ideal solution. This phenomenon indicates that the intermolecular attractive forces between the solute and solvent molecules in the mixture are stronger than the forces between molecules of the pure components.

Analyzing Interactions in Chloroform-Acetone Mixtures

To understand the negative deviation in a mixture of chloroform ($CHCl_3$) and acetone (CH_3COCH_3), consider the intermolecular forces:

- In pure components, molecules interact via dipole-dipole forces and van der Waals forces.
- When chloroform and acetone are mixed, a new type of interaction occurs: **hydrogen bonding**. The hydrogen atom in chloroform, bonded to chlorine, becomes partially positive. This hydrogen can form a hydrogen bond with the partially negative oxygen atom of the carbonyl group (C=O) in acetone.
- This specific **formation of hydrogen bonding** between $CHCl_3$ and CH_3COCH_3 molecules creates stronger attractions between the unlike molecules compared to the attractions between like molecules ($CHCl_3$-$CHCl_3$ or CH_3COCH_3-CH_3COCH_3).
- Because the intermolecular forces in the mixture are stronger, molecules have a reduced tendency to escape from the liquid phase into the vapor phase.
- This leads to a lower vapor pressure than theoretically predicted by Raoult's law, resulting in negative deviation.

The key factor driving this negative deviation is the stabilization of the mixture through the formation of hydrogen bonds between chloroform and acetone.

---

75. **Answer: c**

Explanation:

Statement A Analysis: Molality Calculation

To verify statement A, we calculate the molality of the ethanoic acid solution.

- Mass of solute (ethanoic acid) = 2.5 g
- Molar mass of solute = 60 g mol^{-1}
- Mass of solvent (benzene) = 75 g = 0.075 kg
- Moles of solute = $\frac{\text{Mass of solute}}{\text{Molar mass of solute}} = \frac{2.5 \text{ g}}{60 \text{ g mol}^{-1}} \approx 0.04167 \text{ mol}$
- Molality (m) = $\frac{\text{Moles of solute}}{\text{Mass of solvent (kg)}} = \frac{0.04167 \text{ mol}}{0.075 \text{ kg}} \approx 0.5556 \text{ m}$

The calculated molality is approximately 0.556 m, making statement A **correct**.

Statement B Analysis: Molarity Calculation

To verify statement B, we calculate the molarity of the NaOH solution.

- Mass of solute (NaOH) = 5 g
- Molar mass of solute = 40 g mol^{-1}
- Volume of solution = 450 mL = 0.450 L
- Moles of solute = $\frac{\text{Mass of solute}}{\text{Molar mass of solute}} = \frac{5 \text{ g}}{40 \text{ g mol}^{-1}} = 0.125 \text{ mol}$
- Molarity (M) = $\frac{\text{Moles of solute}}{\text{Volume of solution (L)}} = \frac{0.125 \text{ mol}}{0.450 \text{ L}} \approx 0.2778 \text{ M}$

The calculated molarity is approximately 0.278 M, making statement B **correct**.

Statement C Analysis: Aquatic Life Comfort

The solubility of gases in water is temperature-dependent. Gases are more soluble in colder water than in warmer water. Aquatic species, like fish, require dissolved oxygen, which is more abundant in cold water. Therefore, aquatic species are more comfortable in cold water.

Statement C is **correct**.

Statement D Analysis: Gas Solubility vs Pressure

According to Henry's Law, the solubility of a gas in a liquid is directly proportional to the partial pressure of that gas above the liquid. This means solubility increases with increasing pressure.

Statement D, which claims solubility increases with decreasing pressure, is **incorrect**.

Statement E Analysis: Mole Fraction Formula

For a binary mixture of components A and B, with moles n_A and n_B respectively:

- Total moles = $n_A + n_B$
- The mole fraction of B (x_B) is defined as: $x_B = \frac{n_B}{n_A+n_B}$
- The formula provided in the statement, $x_B = \frac{n_A}{n_A+n_B}$, actually represents the mole fraction of A (x_A).

Statement E is **incorrect**.

Conclusion

Based on the analysis, the correct statements are A, B, and C.

Therefore, the correct option is **A, B and C only**.

---

76. **Answer: c**

Explanation:

Identifying the Incorrect Chemistry Statement

The question asks to identify the statement about chemical properties that is incorrect.

Analysis of Statements

- **Statement 1:** This statement discusses the structure of ECl_3 where E is Boron (B) or Aluminum (Al).
  - Boron trichloride (BCl_3) is electron deficient and exists as a monomer.
  - Aluminum trichloride $(AlCl_3)$ typically exists as a dimer (Al_2Cl_6) in the solid and gaseous states at lower temperatures, becoming monomeric at higher temperatures.

  Therefore, the description of BCl_3 as a monomer and $AlCl_3$ as a dimer is generally considered correct under standard conditions.

- **Statement 2:** This concerns the catenation property of Group 14 elements. Catenation is the ability of atoms to link together covalently to form chain or ring structures.
  - Carbon (C) exhibits the strongest catenation due to its strong C-C single and multiple bonds.
  - Silicon (Si) also shows catenation, forming polysilanes, but the Si-Si bonds are weaker than C-C bonds.
  - Germanium (Ge) and Tin (Sn) exhibit limited catenation due to progressively weaker M-M bonds.

  The order $C \gg Si > Ge \approx Sn$ accurately reflects this trend. This statement is correct.
- **Statement 3:** This statement claims Oxygen exhibits *only* a -2 oxidation state.
  - Oxygen typically has an oxidation state of -2 in most compounds (e.g., H_2O, CO_2).
  - However, Oxygen exhibits other oxidation states:
    - -1 in peroxides (e.g., H_2O_2).
    - -1/2 in superoxides (e.g., KO_2).
    - 0 in elemental form (O_2).
    - Positive oxidation states, like +1 (in OF_2, where F is considered) and +2 (in OF_2, where F is more electronegative). Note: OF_2 has O as +2, O_2F_2 has O as +1.

  Since Oxygen can exhibit oxidation states other than -2, the claim that it exhibits *only* -2 is incorrect.
- **Statement 4:** This statement addresses the ability of Carbon to form $p\pi - p\pi$ multiple bonds with itself.
  - Carbon atoms readily form double bonds ($C = C$) and triple bonds ($C \equiv C$) with each other.
  - These multiple bonds involve the sideways overlap of parallel p-orbitals, known as $p\pi - p\pi$ bonding.

  This capability is fundamental to organic chemistry and is a correct statement.

Conclusion

Based on the analysis, the statement that Oxygen exhibits only a -2 oxidation state is factually incorrect because Oxygen displays various oxidation states depending on the compound.

77. **Answer: c**

Explanation:

Step-by-step Solution:

To determine the formal charges on the oxygen atoms in ozone (O_3), we need to analyze its structure and use the formula for calculating formal charge:

$$\text{Formal Charge} = \text{Valence Electrons} - \text{Non-bonding Electrons} - \frac{\text{Bonding Electrons}}{2}$$

1. **Structure of Ozone:** The ozone molecule has a bent shape with a resonance structure. The central oxygen atom is bonded to two other oxygen atoms.
   - Oxygen number 1 (central): Forms a double bond with one oxygen and a single bond with another.
   - Oxygen number 2: Forms a single bond with the central oxygen.
   - Oxygen number 3: Forms a double bond with the central oxygen.
2. **Calculate Formal Charge:**
   - **Oxygen 2:**
     - Valence electrons = 6 (for oxygen)
     - Non-bonding electrons = 6
     - Bonding electrons = 2 (single bond with central oxygen)
     - $\text{Formal Charge} = 6 - 6 - \frac{2}{2} = 0$
   - **Oxygen 1 (central):**
     - Valence electrons = 6
     - Non-bonding electrons = 4
     - Bonding electrons = 6 (in a single and a double bond)
     - $\text{Formal Charge} = 6 - 4 - \frac{6}{2} = +1$
   - **Oxygen 3:**
     - Valence electrons = 6
     - Non-bonding electrons = 4
     - Bonding electrons = 4 (double bond with central oxygen)
     - $\text{Formal Charge} = 6 - 4 - \frac{4}{2} = -1$
3. **Conclusion:** The correct formal charges on the oxygen atoms numbered 2, 1, and 3 in ozone are 0, +1, and -1 respectively. Thus, the correct answer is: 0, +1, -1.

78. **Answer: c**

Explanation:

Understanding First Law of Thermodynamics

This question relates to the First Law of Thermodynamics, which describes the conservation of energy. The law states that the change in internal energy (ΔU) of a system is equal to the heat added to the system (Q) minus the work done by the system (W).

Applying the Formula

The formula for the First Law of Thermodynamics is:

$$\Delta U = Q - W$$

- **Heat Absorbed (Q):** The system absorbs 500 J, so $Q = +500$ J.
- **Work Done (W):** The work is done *by* the system, which is 200 J. In the formula $\Delta U = Q - W$, work done *by* the system is taken as positive. So, $W = +200$ J.

Calculating Internal Energy Change

Substitute the given values into the First Law equation:

$$\Delta U = 500 \text{ J} - 200 \text{ J}$$

$$\Delta U = 300 \text{ J}$$

Therefore, the change in internal energy of the system is 300 J.

---

79. **Answer: b**

Explanation:

Calculating Gibbs Free Energy and Reaction Spontaneity

This section details the calculation of the standard Gibbs free energy change ($\Delta G^\ominus$) and determination of the reaction's spontaneity.

Reaction Data Summary

- Reaction: $2A(g) + B(g) \rightarrow 2D(g)$
- Standard Internal Energy Change: $\Delta U^\ominus = -10 \text{ kJ mol}^{-1}$
- Standard Entropy Change: $\Delta S^\ominus = -44 \text{ J K}^{-1} \text{ mol}^{-1}$
- Temperature: $T = 298 \text{ K}$
- Gas Constant: $R = 8.31 \text{ J mol}^{-1}\text{K}^{-1}$

Calculating Enthalpy Change ($\Delta H^\ominus$)

First, find the change in moles of gas (Δn_g):

- $\Delta n_g = (\text{moles of gaseous products}) - (\text{moles of gaseous reactants})$
- $\Delta n_g = 2 - (2 + 1) = -1$

Next, calculate $\Delta H^\ominus$ using the relation $\Delta H^\ominus = \Delta U^\ominus + \Delta n_g RT$. Convert $\Delta U^\ominus$ to Joules for calculation:

- $\Delta U^\ominus = -10 \text{ kJ mol}^{-1} = -10000 \text{ J mol}^{-1}$
- $\Delta H^\ominus = -10000 \text{ J mol}^{-1} + (-1) \times (8.31 \text{ J mol}^{-1}\text{K}^{-1}) \times (298 \text{ K})$
- $\Delta H^\ominus = -10000 \text{ J mol}^{-1} - 2476.58 \text{ J mol}^{-1}$
- $\Delta H^\ominus = -12476.58 \text{ J mol}^{-1}$

Calculating Gibbs Free Energy Change ($\Delta G^\ominus$)

Use the Gibbs Free Energy equation: $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$. Use values in Joules for calculation consistency:

- $\Delta H^\ominus = -12476.58 \text{ J mol}^{-1}$
- $\Delta G^\ominus = (-12476.58 \text{ J mol}^{-1}) - (298 \text{ K} \times -44 \text{ J K}^{-1} \text{ mol}^{-1})$
- $\Delta G^\ominus = -12476.58 \text{ J mol}^{-1} + 13112 \text{ J mol}^{-1}$
- $\Delta G^\ominus = 635.42 \text{ J mol}^{-1}$

Convert the result back to kilojoules per mole:

- $\Delta G^\ominus = \frac{635.42}{1000} \text{ kJ mol}^{-1}$
- $\Delta G^\ominus \approx 0.635 \text{ kJ mol}^{-1}$

Determining Reaction Spontaneity

The spontaneity of a reaction depends on the sign of $\Delta G^\ominus$:

- If $\Delta G^\ominus > 0$, the reaction is non-spontaneous.
- If $\Delta G^\ominus < 0$, the reaction is spontaneous.

Since the calculated $\Delta G^\ominus$ is positive $(+0.635 \text{ kJ mol}^{-1})$, the reaction is **non-spontaneous** at 298 K.

---

80. **Answer: b**

Explanation:

Nitriles are reduced to primary amines by strong reducing agents.

A. $LiAlH_4$ followed by H_2O reduces nitriles to primary amines. (True)

B. Sn + HCl does not reduce nitriles to primary amines. (False)

C. H_2/Ni reduces nitriles to primary amines by catalytic hydrogenation. (True)

D. $Na(Hg)/C_2H_5OH$ also reduces nitriles to primary amines. (True)

E. Br_2/aq. NaOH is used in Hofmann bromamide reaction, not for reduction of nitriles. (False)

Therefore, the correct answer is:

A, C and D only.

---

81. **Answer: c**

Explanation:

Defining Ambidentate Ligands

An **ambidentate ligand** is a ligand that can form a coordinate bond with a central metal atom through **two different donor atoms**. The specific atom involved in bonding depends on the metal ion and reaction conditions.

Analyzing Ligand Options

- **Option 1: Oxalate ($C_2O_4^{2-}$)**

  Oxalate is a **bidentate** ligand. It coordinates to a metal ion using **two oxygen atoms** simultaneously.

- **Option 2: Ethylenediaminetetraacetate ion (EDTA)**

  EDTA is a powerful **chelating** ligand, typically acting as **hexadentate**. It binds through two nitrogen atoms and four oxygen atoms.

- **Option 3: Thiocyanate (SCN^-)**

  Thiocyanate is an **ambidentate** ligand. It can coordinate to a metal ion either through the **sulfur atom** (M-SCN) or the **nitrogen atom** (M-NCS).

- **Option 4: Ethane-1,2-diamine ($C_2H_8N_2$)**

  Ethane-1,2-diamine (often abbreviated as 'en') is a common **bidentate** ligand. It coordinates using its **two nitrogen atoms**.

Conclusion on Ambidentate Ligands

Based on the analysis, the thiocyanate ion (SCN^-) is the only ligand among the options that exhibits ambidentate behaviour, capable of coordinating through two distinct donor atoms (S or N).

---

82. **Answer: b**

Explanation:

This problem asks us to calculate the number of photons emitted per second by a bulb, given its power rating, light conversion efficiency, and the energy of a single photon.

Calculating Light Power Output

First, we determine the actual power converted into light. The bulb is rated at **150 watt**, which means it consumes 150 Joules of energy per second (150 J/s). However, only 8% of this energy is converted into light.

Power converted to light (P_{light}) is calculated as:

$$P_{light} = P_{total} \times \text{efficiency}$$

Where:

- $P_{total} = 150 \text{ W} = 150 \text{ J/s}$
- Efficiency $= 8\% = 0.08$

Substituting the values:

$$P_{light} = 150 \text{ J/s} \times 0.08 = 12 \text{ J/s}$$

So, the bulb emits 12 Joules of light energy every second.

Determining Photons Per Second

Next, we use the energy of a single photon to find out how many photons make up this light energy emitted each second.

The energy of one photon (E_{photon}) is given as 4.42×10^{-19} J.

The number of photons emitted per second (N) can be found by dividing the total light power (P_{light}) by the energy per photon (E_{photon}):

$$N = \frac{P_{light}}{E_{photon}}$$

Substitute the known values:

$$N = \frac{12 \text{ J/s}}{4.42 \times 10^{-19} \text{ J/photon}}$$

Now, perform the calculation:

$N = \frac{12}{4.42} \times 10^{19}$ photons/s

$N \approx 2.7149 \times 10^{19}$ photons/s

Rounding to two decimal places, we get approximately 2.71×10^{19} photons per second.

---

83. **Answer: c**

Explanation:

Bonding Properties of N, P, As Analysis

This section identifies the incorrect statement among the given options regarding the chemical bonding characteristics of Nitrogen (N), Phosphorus (P), and Arsenic (As).

Evaluating Chemical Statements

- **Statement 1:** Nitrogen forms pπ-pπ multiple bonds with itself. This is **correct**. Nitrogen utilizes its 2p orbitals for strong multiple bonding, like the N_2 triple bond ($N \equiv N$).
- **Statement 2:** $P(C_2H_5)_3$ and $As(C_6H_5)_3$ form dπ-dπ bonds with transition metals. This statement is considered **correct**. Phosphine ligands can participate in back-bonding involving metal d-orbitals and ligand orbitals, potentially including d-orbitals for P and As.
- **Statement 3:** Nitrogen forms dπ-pπ bonds with oxygen. This statement is **incorrect**. Nitrogen (Period 2) lacks accessible d-orbitals essential for dπ bonding. Bonds between N and O, such as in NO_3^-, typically involve pπ-pπ interactions.
- **Statement 4:** Phosphorus, arsenic, and antimony show catenation property. This is **correct**. These elements can form chains with themselves (e.g., P-P bonds), though the tendency decreases down the group.

Identifying the Incorrect Statement

The statement claiming Nitrogen can form dπ-pπ bonds is chemically inaccurate due to the absence of valence d-orbitals in Nitrogen.

---

84. **Answer: a**

Explanation:

Metallic Character Trend Analysis

Metallic character is defined as an element's tendency to lose electrons and form a positive ion (cation). This property is governed by periodic trends.

Understanding Periodic Trends

- **Across a Period (Left to Right):** Metallic character generally **decreases**. This is because the effective nuclear charge increases, making it harder to lose electrons.
- **Down a Group (Top to Bottom):** Metallic character generally **increases**. The valence electrons are farther from the nucleus and shielded by more electrons, making them easier to lose.

Element Analysis

Let's analyze the positions and types of the given elements (Na, Be, P, Mg, Si):

- **P** (Phosphorus): Group 15, Period 3. A **non-metal**. Expected to have the lowest metallic character.
- **Si** (Silicon): Group 14, Period 3. A **metalloid**. More metallic than P, less metallic than metals in the same period.
- **Be** (Beryllium): Group 2, Period 2. An **alkaline earth metal**.
- **Mg** (Magnesium): Group 2, Period 3. An **alkaline earth metal**. Below Be in the same group.
- **Na** (Sodium): Group 1, Period 3. An **alkali metal**. To the left of Mg in the same period.

Determining the Order

Applying the periodic trends:

- Comparing non-metals and metalloids: P is less metallic than Si. So, $P < Si$.
- Comparing metals and metalloids: Metals are generally more metallic than metalloids. Be is a metal, Si is a metalloid. So, $Si < Be$.

- Comparing elements in Group 2: Be is above Mg. Metallic character increases down a group. So, $Be < Mg$.
- Comparing elements in Period 3: Na is in Group 1, Mg is in Group 2. Metallic character decreases from left to right across a period. Thus, Na is more metallic than Mg. So, $Mg < Na$.

Combining these inequalities gives the order of increasing metallic character:

$P < Si < Be < Mg < Na$

---

85. Answer: b

Explanation:

Equilibrium Calculation for $BiO(OH)$ Dissolution

The dissolution of $BiO(OH)(s)$ establishes the following equilibrium:

$BiO(OH)(s) \rightleftharpoons BiO^+(aq) + OH^-(aq)$

The equilibrium constant (K) expression is given by:

$K = [BiO^+][OH^-]$

We are given $K = 4 \times 10^{-10}$.

Determining Hydroxide Ion Concentration

Since the solid $BiO(OH)$ does not appear in the equilibrium expression, and assuming initial concentrations of BiO^+ and OH^- are zero before dissolution:

- Let $[OH^-] = x$.
- From the stoichiometry, $[BiO^+] = [OH^-] = x$.
- Substituting into the K expression: $K = x \times x = x^2$.

Therefore:

$x^2 = 4 \times 10^{-10}$ $x = \sqrt{4 \times 10^{-10}} = 2 \times 10^{-5}\,\mathrm{M}$

So, the hydroxide ion concentration is $[OH^-] = 2 \times 10^{-5}$ M.

Calculating pH from pOH

First, calculate the pOH:

$\text{pOH} = -\log_{10}[OH^-]$ $\text{pOH} = -\log_{10}(2 \times 10^{-5})$ $\text{pOH} = -(\log_{10} 2 + \log_{10} 10^{-5})$ $\text{pOH} = -(\log_{10} 2 - 5)$

Using the given value $\log_{10} 2 = 0.3010$:

$\text{pOH} = -(0.3010 - 5)$ $\text{pOH} = 5 - 0.3010$ $\text{pOH} = 4.699$

At 298 K, the relationship between pH and pOH is:

$\text{pH} + \text{pOH} = 14$

Now, calculate the pH:

$\text{pH} = 14 - \text{pOH}$ $\text{pH} = 14 - 4.699$ $\text{pH} = 9.301$

---

86. Answer: d

Explanation:

Chemical Bonding Analysis: Matching Molecular Structures

This solution matches the given chemical species in List I (C_2H_4, C_2H_2, CH_4, NH_3) with their respective bonding characteristics in List II.

Analyzing C_2H_4 (Ethene) Bonding

- Structure: $H_2C = CH_2$. It has one carbon-carbon double bond and four carbon-hydrogen single bonds.
- Bonding Breakdown: The $C = C$ double bond contains one sigma (σ) bond and one pi (π) bond. Each $C - H$ single bond is a sigma (σ) bond.

- Total Bonds: There are 1 (σ from $C = C$) + 4 (σ from $C - H$) = 5 sigma bonds. There is 1 pi bond from the $C = C$ bond.
- Match: This matches List II - IV (5 σ bonds, 1 π bond).

Analyzing C_2H_2 (Ethyne) Bonding

- Structure: $HC \equiv CH$. It has one carbon-carbon triple bond and two carbon-hydrogen single bonds.
- Bonding Breakdown: The $C \equiv C$ triple bond contains one sigma (σ) bond and two pi (π) bonds. Each $C - H$ single bond is a sigma (σ) bond.
- Total Bonds: There are 1 (σ from $C \equiv C$) + 2 (σ from $C - H$) = 3 sigma bonds. There are 2 pi bonds from the $C \equiv C$ bond.
- Match: This matches List II - I (3 σ bonds, 2 π bonds).

Analyzing CH_4 (Methane) Bonding

- Structure: CH_4 has a central carbon atom bonded to four hydrogen atoms via single bonds in a tetrahedral arrangement.
- Bonding Breakdown: All four $C - H$ bonds are single sigma (σ) bonds. There are no pi bonds.
- Total Bonds: 4 sigma bonds.
- Match: This matches List II - III (4 σ bonds).

Analyzing NH_3 (Ammonia) Bonding

- Structure: NH_3 has a central nitrogen atom bonded to three hydrogen atoms via single bonds. The nitrogen atom also has one lone pair of electrons.
- Bonding Breakdown: The three $N - H$ bonds are single sigma (σ) bonds.
- Total Bonds: 3 sigma bonds. The lone pair is not counted as a sigma or pi bond.
- Match: This matches List II - II (3 σ bonds, one lone pair).

Final Matching Summary

- A (C_2H_4) matches IV.
- B (C_2H_2) matches I.
- C (CH_4) matches III.
- D (NH_3) matches II.

The correct combination derived is A-IV, B-I, C-III, D-II.

---

87. **Answer: d**

Explanation:

Understanding K_p and K_c Relationship

The equilibrium constants K_p and K_c relate to reactions involving gases. K_p is expressed in terms of partial pressures, while K_c is expressed in terms of molar concentrations.

The relationship between K_p and K_c is given by the equation:

$K_p = K_c(RT)^{\Delta n}$

where:

- R is the ideal gas constant.
- T is the absolute temperature in Kelvin.
- Δn is the change in the number of moles of gaseous products and reactants: $\Delta n = (\text{Sum of moles of gaseous products}) - (\text{Sum of moles of gaseous reactants})$.

For K_p to be equal to K_c, the value of Δn must be zero $\left(K_p = K_c(RT)^0 = K_c\right)$. Therefore, the reaction for which $K_p \neq K_c$ is the one where $\Delta n \neq 0$.

Analyzing Gaseous Moles in Reactions

Let's calculate Δn for each given reaction:

- **Option 1:** $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

  Reactants: 1 mole N_2 + 1 mole O_2 = 2 moles of gas

  Products: 2 moles NO = 2 moles of gas

  $\Delta n = 2 - 2 = 0$. Thus, $K_p = K_c$.

- **Option 2:** $H_2O(g) + CO(g) \rightleftharpoons H_2(g) + CO_2(g)$

Reactants: 1 mole H_2O + 1 mole CO = 2 moles of gas

Products: 1 mole H_2 + 1 mole CO_2 = 2 moles of gas

$\Delta n = 2 - 2 = 0$. Thus, $K_p = K_c$.

- **Option 3:** $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

  Reactants: 1 mole H_2 + 1 mole I_2 = 2 moles of gas

  Products: 2 moles HI = 2 moles of gas

  $\Delta n = 2 - 2 = 0$. Thus, $K_p = K_c$.

- **Option 4:** $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

  Reactants: 1 mole N_2 + 3 moles H_2 = 4 moles of gas

  Products: 2 moles NH_3 = 2 moles of gas

  $\Delta n = 2 - 4 = -2$. Since $\Delta n \neq 0$, $K_p \neq K_c$.

Conclusion

The reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ is the one for which $K_p \neq K_c$ because the change in the number of moles of gas (Δn) is not zero.

---

88. **Answer: d**

Explanation:

Let's analyze the given reaction sequence step-by-step:

1. The first step involves the reaction of $CH_3CH_2CH_2OH$ (propanol) with PCl_5 (phosphorus pentachloride).
   - This reaction is a typical conversion of an alcohol to an alkyl chloride. The products are $CH_3CH_2CH_2Cl$, HCl, and the by-product $POCl_3$.
2. Thus, in the first step, $X = POCl_3$.
3. Next, $CH_3CH_2CH_2Cl$ reacts with alcoholic KOH:

- This is a dehydrohalogenation reaction leading to the formation of an alkene, $CH_3CH = CH_2$ (propene), labeled as Y.

4. In the final step, Y (propene) undergoes Markovnikov addition with HBr in the presence of a peroxide to form $Z = CH_3CH_2Br$ (ethyl bromide).

Hence, the correct answer is $X = POCl_3$ and $Z = CH_3CH_2Br$.

---

89. **Answer: d**

Explanation:

Reaction Pathway Analysis

The question asks for the major product (Z) formed from a sequence of reactions starting with ethane (C_2H_6).

Step 1: Monochlorination of Ethane

Ethane (C_2H_6) reacts with chlorine (Cl_2) under UV light via a free-radical substitution mechanism. This produces monochlorinated ethane (X).

Reaction: $C_2H_6 + Cl_2 \xrightarrow{UV\ \text{light}} C_2H_5Cl + HCl$

Thus, X is monochlorinated ethane, C_2H_5Cl.

Step 2: Ammonolysis of Ethyl Chloride

Monochlorinated ethane $(X = C_2H_5Cl)$ reacts with ammonia (NH_3) through nucleophilic substitution (ammonolysis) to form ethylamine (Y).

Reaction: $C_2H_5Cl + NH_3 \rightarrow C_2H_5NH_2 + HCl$

Thus, Y is ethylamine, $C_2H_5NH_2$. This is a primary amine.

Step 3: Reaction of Primary Amine with Nitrous Acid

The primary amine $(Y = C_2H_5NH_2)$ is treated with sodium nitrite $(NaNO_2)$ and hydrochloric acid (HCl) at low temperatures. This generates nitrous acid $(HONO)$ in situ.

Reaction: $NaNO_2 + HCl \rightarrow HONO + NaCl$

Primary aliphatic amines react with nitrous acid $(HONO)$ to form unstable diazonium salts, which immediately decompose in the aqueous acidic medium to yield alcohols, nitrogen gas, and regenerate the acid.

Reaction: $C_2H_5NH_2 + HONO \xrightarrow{HCl} [C_2H_5N_2]^+Cl^- \xrightarrow{H_2O,\text{heat}} C_2H_5OH + N_2 + HCl$

Therefore, the final product Z is ethanol, C_2H_5OH.

Conclusion

The major product Z formed in the sequence is ethanol (C_2H_5OH).

---

90. **Answer: d**

Explanation:

Reaction Analysis: CO2 with Hot Coke

The reaction between carbon dioxide (CO_2) gas and hot coke (carbon, C) produces carbon monoxide (CO) gas. The balanced chemical equation is:

$CO_2(g) + C(s) \rightarrow 2CO(g)$

This equation shows that 1 volume of CO_2 reacts to produce 2 volumes of CO. This results in a net increase in the volume of gaseous products compared to the reactant consumed.

Volume Change Calculation

Let the initial volume of CO_2 gas be $V_{CO_2,initial} = 1 \text{ dm}^3$. Let x represent the volume of CO_2 that reacts completely.

- Volume of CO_2 consumed = $x \text{ dm}^3$.
- Volume of CO produced = $2x \text{ dm}^3$ (based on the stoichiometry).

- Volume of CO_2 remaining unreacted = $V_{CO_2,initial} - x = (1 - x)\ \mathrm{dm}^3$.

The total final volume of the gaseous mixture (V_{final}) is the sum of the unreacted CO_2 and the produced CO:

$$V_{final} = (\text{Volume of } CO_2 \text{ remaining}) + (\text{Volume of } CO \text{ produced})$$

$$V_{final} = (1 - x) + 2x$$

$$V_{final} = 1 + x$$

The problem states that the final volume after the complete reaction is $1.4\ \mathrm{dm}^3$. Therefore:

$$1 + x = 1.4\ \mathrm{dm}^3$$

Solving for x gives:

$$x = 1.4 - 1 = 0.4\ \mathrm{dm}^3$$

Final Gas Composition

Now, we calculate the volume of each gas present in the final mixture:

- Volume of CO produced = $2x = 2 \times 0.4 = 0.8\ \mathrm{dm}^3$.
- Volume of CO_2 remaining = $1 - x = 1 - 0.4 = 0.6\ \mathrm{dm}^3$.

Thus, the composition of the gaseous mixture at STP is $0.8\ \mathrm{dm}^3$ of CO and $0.6\ \mathrm{dm}^3$ of CO_2.

---

91. **Answer: a**

Explanation:

Cell Cycle Phase Activities

This question requires matching the phases of the cell cycle (List I) with their specific biological activities (List II).

Matching Cell Cycle Phases

- **A. G_1 phase** is a period of cell growth and metabolic activity where DNA replication does not occur. This corresponds to activity **II** (Cell is metabolically active and continuously grows but does not replicate its DNA).
- **B. S phase** is characterized by the synthesis of DNA, during which the amount of DNA per cell doubles. This matches activity **III** (Synthesis of DNA occurs and the amount of DNA per cell doubles).
- **C. G_2 phase** involves the synthesis of proteins necessary for cell division while the cell continues to grow. This corresponds to activity **IV** (Proteins are synthesized while cell growth continues).
- **D. M phase** represents the actual cell division stage, including mitosis and cytokinesis. This matches activity **I** (Actual cell division occurs).

Correct Combination

Based on the matching:

- A $\rightarrow$ II
- B $\rightarrow$ III
- C $\rightarrow$ IV
- D $\rightarrow$ I

This results in the combination A-II, B-III, C-IV, D-I.

---

92. **Answer: a**

Explanation:

This question requires matching different modes of genetic inheritance (List I) with their respective examples (List II).

Matching Genetic Inheritance Concepts

Explanation of Matches:

- **A. Incomplete dominance:** This occurs when neither allele is completely dominant over the other, resulting in a blended heterozygous phenotype. A classic example is the inheritance of flower colour in *Antirrhinum sp.* (snapdragons), where crossing red and white flowers produces pink offspring. Thus, A matches with **II. Inheritance of flower colour in Antirrhinum sp.**

- **B. Co-dominance:** In co-dominance, both alleles for a trait are expressed simultaneously in the heterozygous state. The human ABO blood group system is a prime example, where individuals with genotype $I^A I^B$ express both A and B antigens. Thus, B matches with **IV. ABO blood groups**.
- **C. Pleiotropy:** Pleiotropy describes the phenomenon where a single gene influences multiple distinct phenotypic traits. Phenylketonuria (PKU) is a genetic disorder caused by a mutation in a single gene, leading to various health issues including intellectual disability and seizures. Thus, C matches with **III. Phenylketonuria disease in humans**.
- **D. Polygenic inheritance:** This mode of inheritance involves multiple genes contributing to a single trait, leading to a continuous distribution of phenotypes. Human skin colour is a well-known example, influenced by several genes. Thus, D matches with **I. Human skin colour**.

Correct Option Determination:

Based on the matching:

- A - II
- B - IV
- C - III
- D - I

This corresponds to the option **A-II, B-IV, C-III, D-I**.

---

93. **Answer: b**

Explanation:

Environmental Statements Analysis

Statement A: Habitat Loss

This statement is **correct**. The clearing of the Amazon rainforest for large-scale agriculture, such as soybean cultivation, directly destroys the natural environment of countless species, which is a clear example of **habitat loss**.

Statement B: Extinction Causes

This statement is **correct**. Historical records confirm that the Steller's sea cow was hunted to extinction by the 18th century, and the passenger pigeon, once incredibly abundant, was driven to extinction in the early 20th century primarily due to relentless hunting and habitat destruction by humans (**over-exploitation**).

Statement C: Invasive Species Impact

This statement is **incorrect**. The introduction of the Nile perch, a predatory fish, into Lake Victoria had devastating consequences for the native cichlid fish populations. Instead of helping their growth, the Nile perch caused the extinction or endangerment of hundreds of cichlid species.

Statement D: Invasive Species Definition

This statement is **correct**. The water hyacinth is a well-known example of an **invasive species**. It reproduces rapidly, forms dense mats on water bodies, and outcompetes native aquatic plants, disrupting the ecosystem.

Statement E: Impact of Extinction

This statement is **incorrect**. When a species goes extinct, it can have significant ripple effects throughout the ecosystem. Other species that relied on the extinct species for food, pollination, or habitat may be negatively impacted, potentially leading to further declines or even extinctions.

Conclusion

Based on the analysis, the correct statements are A, B, and D.

Therefore, the correct option is **2. A, B and D only**.

---

94. Answer: b

Explanation:

Transcription Unit Components Identified

A transcription unit is the fundamental segment of DNA that directs the synthesis of a functional RNA molecule. It consists of three essential regions.

Evaluating Transcription Unit Statements

- **Statement A:** Correct. A transcription unit is defined by the promoter (initiation site), the structural gene (coding sequence), and the terminator (termination site).
- **Statement B:** Correct. The promoter sequence is located upstream of the structural gene, conventionally towards the 5'-end of the DNA template strand that is being transcribed.
- **Statement C:** Correct. The promoter region serves as the specific binding site for the enzyme RNA polymerase, which is crucial for initiating transcription.
- **Statement D:** Correct. By defining the start site and direction of transcription, the promoter determines which of the two DNA strands will function as the template strand and which as the coding strand.
- **Statement E:** Correct. The terminator sequence is located downstream of the structural gene, towards the 3'-end relative to the coding strand, and it signals the termination point for transcription.

Finalizing Correct Statements

All the statements (A, B, C, D, and E) provide accurate descriptions of the components and functions related to a transcription unit in DNA.

Therefore, the correct choice must include all these valid statements.

---

95. **Answer: a**

Explanation:

Understanding Honeybee Sex Determination

Honeybees exhibit a unique sex-determination system known as haplodiploidy. This system dictates how males and females develop based on chromosome number.

Analysis of Statements

- **Statement A: True**
  Fertilized eggs, resulting from the fusion of sperm (from a male) and an egg (from a female), are diploid (possess '$2n$' chromosomes). These diploid individuals develop into females, specifically either queens or worker bees, depending on environmental factors like nutrition.
- **Statement B: True**
  Unfertilized eggs are haploid (possess 'n' chromosomes). These eggs develop into males (drones) through a process called parthenogenesis, which is the development of an embryo from an unfertilized egg.
- **Statement C: True**
  Females are diploid ('$2n$'), having inherited chromosomes from both parents. Males are haploid ('n'), developing from unfertilized eggs. Therefore, a male bee has exactly half the number of chromosomes compared to a female bee.
- **Statement D: False**
  Male bees are haploid ('n'). To produce sperm, their cells divide mitotically. Meiosis is a process that reduces the chromosome number, which is typically used by diploid organisms to produce haploid gametes. Since males are already haploid, they produce sperm through mitosis, not meiosis.
- **Statement E: True**
  The sex-determination system in honeybees, where females are diploid ('$2n$') and males are haploid ('n'), is called haplodiploidy.

Conclusion

Based on the analysis, statements A, B, C, and E are true. Statement D is false.

Therefore, the correct option includes A, B, C, and E.

---

96. **Answer: d**

Explanation:

This question requires matching key processes in cellular respiration and ATP synthesis with their specific locations within the cell.

Matching Cellular Respiration Locations

- **A. Glycolysis** matches with **III. Cytoplasm**. Glycolysis is the initial breakdown of glucose and occurs in the cell's cytoplasm.
- **B. ETS (Electron Transport System)** matches with **I. Inner mitochondrial membrane**. The ETS components are embedded in the inner mitochondrial membrane where electron transfer and proton pumping occur.
- **C. Accumulation of protons** matches with **IV. Intermembrane space**. Protons are actively pumped from the mitochondrial matrix to the intermembrane space during the ETS.
- **D. Krebs' cycle** matches with **II. Mitochondrial matrix**. The Krebs' cycle reactions, following glycolysis, take place within the mitochondrial matrix.

Therefore, the correct matching is A-III, B-I, C-IV, D-II.

---

97. **Answer: a**

Explanation:

Calvin Pathway ATP & NADPH Requirements for Glucose

The Calvin pathway is the primary route for carbon fixation in photosynthesis, ultimately producing sugars like glucose. It utilizes energy (ATP) and reducing power (NADPH) generated during the light-dependent reactions.

Calvin Cycle Requirements per CO_2

For every molecule of carbon dioxide (CO_2) fixed and processed through the Calvin cycle, the following are consumed:

- 2 molecules of NADPH
- 3 molecules of ATP

This process leads to the formation of glyceraldehyde-3-phosphate (G3P), a 3-carbon sugar.

Calculating Inputs for One Glucose Molecule

A single molecule of glucose is a 6-carbon sugar. To synthesize one molecule of glucose, two molecules of G3P must be produced and leave the cycle. This requires the cycle to effectively fix six molecules of CO_2.

The total ATP and NADPH needed are calculated based on the requirements per CO_2 fixation, multiplied by the number of CO_2 molecules fixed:

ATP Required: $6\ \mathrm{CO_2} \times 3\ \mathrm{ATP/CO_2} = 18\ \mathrm{ATP}$

NADPH Required: $6\ \mathrm{CO_2} \times 2\ \mathrm{NADPH/CO_2} = 12\ \mathrm{NADPH}$

Summary of Requirements

Therefore, to produce one molecule of glucose via the Calvin pathway, 18 ATP molecules and 12 NADPH molecules are required.

---

98. **Answer: a**

Explanation:

Matching Biotechnology Concepts: List I and List II

This question requires matching specific biotechnology terms in List I with their corresponding entities or examples in List II.

Understanding the Matches

Here's a breakdown of the correct matches:

- **A. Genetically modified organism** corresponds to **II. Bt cotton**. Bt cotton is engineered to produce an insecticide, making it a genetically modified organism (GMO).

- **B. Thermostable DNA polymerase** corresponds to **III. *Thermus aquaticus***. This bacterium is the source of Taq polymerase, a heat-resistant DNA polymerase essential for PCR.
- **C. Ti plasmid** corresponds to **I. *Agrobacterium tumefaciens***. The Ti plasmid, naturally found in this bacterium, is widely used for genetic transformation in plants.
- **D. pBR322** corresponds to **IV. *Escherichia coli***. pBR322 is a commonly used plasmid vector, often propagated and manipulated within *E. coli*.

Summary Table of Matches

List I	List II
A. Genetically modified organism	II. Bt cotton
B. Thermostable DNA polymerase	III. *Thermus aquaticus*
C. Ti plasmid	I. *Agrobacterium tumefaciens*
D. pBR322	IV. *Escherichia coli*

Correct Option Identification

Based on the individual matches derived above (A-II, B-III, C-I, D-IV), the correct option is the first one provided.

Final Answer: The correct option is A (A-II, B-III, C-I, D-IV).

---

99. **Answer: b**

Explanation:

Identifying Plants with Exposed Ovules

The question asks to identify the plant where ovules are **exposed** and not enclosed by an ovary wall. This is a defining characteristic of **Gymnosperms**.

Pinus Classification

Pinus is a Gymnosperm. In Gymnosperms, the ovules are borne on the surface of carpels (or modified leaves) and are not enclosed within a protective ovary. They are often referred to as "naked seeds" because they lack an outer fruit wall (derived from the ovary). Therefore, Pinus fits the description of having exposed ovules.

Other Plant Groups

- **Funaria** is a Bryophyte. Bryophytes reproduce via spores and do not possess true flowers, fruits, or ovaries containing ovules.
- **Selaginella** is a Pteridophyte (a type of vascular plant). While some Pteridophytes have seeds, they generally differ from Gymnosperms in reproductive structures, and the term "exposed ovule" most accurately describes Gymnosperms.
- **Wolffia** is an Angiosperm (flowering plant). Angiosperms are characterized by having ovules enclosed within a well-developed ovary, which later matures into a fruit.

Based on this, Pinus is the correct identification for a plant with exposed ovules.

---

100. **Answer: c**

Explanation:

Understanding the **Calvin cycle** is key to identifying the enzyme responsible for **carboxylation**.

Calvin Cycle Carboxylation Enzyme

The **Calvin cycle** is the primary pathway for carbon fixation in photosynthesis. The first step, **carboxylation**, involves attaching carbon dioxide (CO_2) to Ribulose-1,5-bisphosphate (RuBP).

Identifying the Correct Enzyme

The specific enzyme catalyzing this crucial **carboxylation** reaction is **RuBP carboxylase-oxygenase**, commonly known as Rubisco.

- **RuBP carboxylase-oxygenase** directly facilitates the addition of CO_2 to RuBP, initiating the cycle.

Evaluating Other Options

The other options are incorrect because:

- **Carboxypeptidase**: This enzyme breaks down peptides, acting on proteins, not involved in carbon fixation.
- **PEP carboxylase**: While involved in carbon fixation, it's primarily active in C4 and CAM plants, fixing CO_2 into phosphoenolpyruvate (PEP), not directly in the Calvin cycle's initial carboxylation step as described here.
- **Hexokinase**: This enzyme phosphorylates glucose, the first step of glycolysis, unrelated to the Calvin cycle.

Therefore, the enzyme required for **carboxylation** in the **Calvin cycle** is **RuBP carboxylase-oxygenase**.

---

101. **Answer: a**

Explanation:

Matching Biological Molecules: List I and List II

The question requires matching items from List I with their corresponding categories in List II.

Analysis of Matches

- **A. Trypsin** is a protease, which functions as an **Enzyme**. Thus, A matches with III.
- **B. Morphine** is a naturally occurring opiate derived from the opium poppy, classified as an **Alkaloid**. Thus, B matches with IV.
- **C. Concanavalin A** is a protein found in plants that binds carbohydrates, specifically identified as a **Lectin**. Thus, C matches with II.
- **D. Collagen** is a fibrous protein that forms the main structural component of connective tissues and is integral to the **Intercellular ground substance**. Thus, D

matches with I.

Summary of Matches

The correct pairings are:

List I Item	List II Category
A. Trypsin	III. Enzyme
B. Morphine	IV. Alkaloid
C. Concanavalin A	II. Lectin
D. Collagen	I. Intercellular ground substance

This corresponds to the selection A-III, B-IV, C-II, D-I.

---

102. **Answer: b**

Explanation:

Amino Acids: Statement Analysis

Let's evaluate each statement regarding amino acids:

- **Statement A: They are substituted methanes.**

  Amino acids possess a central alpha-carbon atom bonded to an amino group ($-NH_2$), a carboxyl group ($-COOH$), a hydrogen atom ($-H$), and a variable side chain (R-group). Methane is CH_4. By replacing hydrogen atoms on methane with these functional groups, we obtain the basic amino acid structure. Therefore, amino acids are indeed substituted methanes. This statement is **correct**.

- **Statement B: Serine is an aromatic amino acid.**

  Aromatic amino acids contain a ring structure in their side chain (e.g., Phenylalanine, Tyrosine). Serine's side chain is $-CH_2OH$, which contains a hydroxyl group, not an aromatic ring. Thus, Serine is an alcohol-containing amino acid, not an aromatic one. This statement is **incorrect**.

- **Statement C: Valine is a neutral amino acid.**

  Valine has an isopropyl side chain $(-CH(CH_3)_2)$. This is a nonpolar alkyl group, meaning it does not carry a charge at physiological pH. Amino acids with such side chains are classified as neutral. This statement is **correct**.

- **Statement D: Lysine is an acidic amino acid.**

  Acidic amino acids have carboxyl groups in their side chains (e.g., Aspartic acid, Glutamic acid), resulting in a negative charge. Lysine has an additional amino group $(-NH_2)$ in its side chain, making it positively charged and classifying it as a basic amino acid. This statement is **incorrect**.

Conclusion on Correct Statements

Based on the analysis, statements A and C are correct.

The correct option is the one that includes only A and C.

Correct Option: 2. A and C only

---

103. **Answer: c**

Explanation:

Understanding the Sickle-Cell Anaemia Mutation

The question describes a specific genetic mutation affecting the haemoglobin molecule.

- **Mutation Type:** Point mutation.
- **Affected Chain:** Beta-globin chain of haemoglobin.
- **Specific Change:** Substitution of the amino acid Glutamic acid (Glu) with Valine (Val).
- **Location:** Sixth position of the beta-globin chain.

Consequences of the Mutation

This single amino acid substitution alters the properties of the haemoglobin molecule. Under conditions of low oxygen, the altered haemoglobin molecules tend to polymerize, causing the red blood cells to deform into a characteristic sickle shape. This sickling of red blood cells leads to various complications associated with the disorder.

Identifying the Disorder

The disorder caused by this specific Glu to Val substitution in the beta-globin chain is known as Sickle-cell anaemia. The other options are incorrect:

- **Thalassemia:** A group of inherited blood disorders characterized by reduced or absent synthesis of haemoglobin chains.
- **Haemophilia:** A group of inherited bleeding disorders where the blood does not clot properly due to missing clotting factors.
- **Phenylketonuria (PKU):** An inherited metabolic disorder where the body cannot break down an amino acid called phenylalanine.

Therefore, the substitution of Glutamic acid by Valine at the sixth position of the beta-globin chain specifically causes Sickle-cell anaemia.

---

104. **Answer: d**

Explanation:

Site of Active Ribosomal RNA Synthesis

Ribosomal RNA (rRNA) synthesis is a critical process for building ribosomes, the protein-synthesizing machinery of the cell. This process occurs in a specific region within the eukaryotic cell nucleus.

The Nucleolus Function

The **Nucleolus** is a dense structure found inside the nucleus. Its primary function is the synthesis and processing of rRNA, and its assembly with proteins imported from the cytoplasm to form ribosomes. Therefore, the nucleolus is the principal site for active ribosomal RNA synthesis.

Analysis of Other Options

- **Kinetochore:** Involved in chromosome segregation during cell division, connecting chromosomes to spindle fibers.
- **Centrosome:** Organizes microtubules and is crucial for cell division (mitosis and meiosis) in animal cells.
- **Chromatin:** The complex of DNA and proteins (histones) that forms chromosomes within the nucleus. While it contains rRNA genes, the synthesis itself happens within the nucleolus.

Based on cellular function, the **Nucleolus** is the correct site for active ribosomal RNA synthesis.

---

105. **Answer: d**

Explanation:

Binomial Nomenclature Rules Analysis

Binomial nomenclature is the standardized system for naming species. It follows specific universal rules to ensure clarity and consistency in biological classification. Let's examine each statement based on these rules.

Rule Verification

- **Statement 1 (Underlining/Italics):** When handwritten, biological names should be underlined separately. When printed, they should be in italics. This statement is **true**.
- **Statement 2 (Latin Origin):** Biological names are typically derived from Latin or Greek, or are Latinized. This statement is **true**.

- **Statement 3 (Specific Epithet Case):** The first word (genus) starts with a capital letter, and the second word (specific epithet) starts with a lowercase letter. This statement is **true**.
- **Statement 4 (Word Order):** This statement claims the first word is the specific epithet and the second denotes the genus. This contradicts the standard convention.

Identifying the Incorrect Statement

In binomial nomenclature, a scientific name consists of two parts: the genus name followed by the specific epithet.

- The **first word** represents the **Genus**.
- The **second word** represents the **specific epithet**.

Therefore, the statement "The first word in the biological name represents the specific epithet, while the second component denotes the genus" is incorrect. The correct order is Genus followed by specific epithet.

---

106. **Answer: d**

Explanation:

This question requires matching plant growth regulators from List I with their respective functions or effects in List II.

Matching Plant Growth Regulators to Functions

Let's analyze each growth regulator:

- **A. 2,4-D:** This is a synthetic auxin commonly used as a selective **herbicide**.
- **B. GA_3 (Gibberellic acid):** Gibberellins promote seed germination and growth. They are utilized in the **brewing industry** to enhance malting (germination) of barley.
- **C. Kinetin:** Kinetin is a type of cytokinin, known for promoting cell division and delaying senescence. Cytokinins also play a role in **nutrient mobilization** towards stimulated areas, like developing fruits or leaves.

- **D. ABA (Abscisic acid):** Abscisic acid acts as a stress hormone, primarily involved in inhibiting growth and inducing dormancy. It is crucial for the **stimulation of stomatal closure** during water stress to prevent dehydration.

Summary of Matches

Based on the analysis, the correct matches are:

List I Item	Growth Regulator	List II Match	Function/Effect
A	2,4-D	III	Herbicide
B	GA_3	I	Brewing industry
C	Kinetin	IV	Nutrient mobilisation
D	ABA	II	Stimulation of stomatal closure

Therefore, the correct combination is A-III, B-I, C-IV, D-II.

---

107. **Answer: a**

Explanation:

Matching Plant Anatomy Terms

This question requires matching specific plant anatomical terms from List I with their correct descriptions in List II.

Analysis of Matches:

- **A. Conjunctive tissue**: This refers to the parenchyma tissue found between the xylem and phloem in vascular bundles, important for support. It matches **III. Tissue between xylem and phloem**.
- **B. Casparian strips**: These are characteristic bands found in the endodermis of plant roots, composed of suberin, which regulate water movement into the stele. They match **IV. Endodermal cells with suberin deposition**.

- **C. Subsidiary cells**: These are specialized epidermal cells adjacent to guard cells, forming the stomatal apparatus. They match **I. Specialised cells in the vicinity of guard cells**.
- **D. Starch sheath**: This is typically the endodermis layer when its cells are packed with starch granules, playing a role in nutrient transport. It matches **II. Endodermal cells rich in starch**.

Correct Matching:

Based on the analysis, the correct pairings are:

- A - III
- B - IV
- C - I
- D - II

This corresponds to the option **A-III, B-IV, C-I, D-II**.

---

108. **Answer: d**

Explanation:

Pollen Grains and Stigma Interaction

The question asks which pollination type introduces pollen grains that are genetically different from the recipient plant's genetic makeup. Let's examine the options:

Understanding Pollination Types

- **Autogamy:** Pollen transfer occurs within the same flower. This results in genetically similar pollen.
- **Geitonogamy:** Pollen transfer happens between different flowers on the **same plant**. While the flowers are different, the pollen is genetically identical to the parent plant's pollen.
- **Xenogamy:** Pollen transfer occurs between flowers on **different plants** of the same species. This process brings pollen that is genetically different from the pollen normally produced by the recipient plant's flowers.

- **Cleistogamy:** This is a type of autogamy where flowers remain closed, ensuring self-pollination. The pollen is genetically similar.

Identifying the Correct Pollination Type

Based on the definitions, **Xenogamy** is the only type that guarantees the transfer of genetically different pollen grains to the stigma, as it involves pollen from a different individual plant.

Therefore, Xenogamy brings genetically different types of pollen grains to the stigma.

---

109. **Answer: a**

Explanation:

Understanding Heterophylly and Environmental Plasticity

What is Heterophylly?

Heterophylly is the development of different types of leaves on the same plant.

This variation occurs in direct response to environmental factors, such as light, water, or nutrients.

Defining Environmental Plasticity

Environmental plasticity, or phenotypic plasticity, is the ability of a single genotype to produce different physical characteristics (phenotypes) when exposed to different environmental conditions.

It allows organisms to adapt their traits to suit their surroundings without changing their underlying genetic makeup.

Heterophylly as an Example of Plasticity

The occurrence of heterophylly is a prime example of phenotypic plasticity.

A plant's genetic potential allows it to modify its leaf development based on environmental signals, optimizing its survival and function.

Evaluating Other Options

- **Dedifferentiation:** This involves cells losing their specialization and reverting to an immature state. It's crucial for processes like regeneration but doesn't directly explain varied leaf forms due to the environment.
- **Redifferentiation:** This is the process where cells regain specialized functions after dedifferentiation. It's a subsequent step, not the overall phenomenon of environmental response.
- **Elasticity:** This typically refers to the ability of a tissue to deform and return to its original state. It is not related to developmental changes in response to environmental conditions.

Therefore, heterophyllous development in response to the environment directly demonstrates plasticity.

---

110. **Answer: a**

Explanation:

Solution: Analyzing Restriction Endonuclease Statements

This question asks to identify the statements that are **not true** about restriction endonucleases.

Statement Analysis

- **A. They are called molecular scissors.**
  True. Restriction enzymes cut DNA strands at specific recognition sites, acting metaphorically like scissors.
- **B. These are the enzymes responsible for restricting the growth of bacteriophages in E. coli.**
  True. Bacteria like *E. coli* use restriction-modification systems, involving these enzymes, to defend against foreign DNA, such as that from bacteriophages.

- **C. They cut the DNA only at the centre of the palindromic sites.**
  False. While restriction enzymes recognize palindromic sequences, they cut *within* these sequences. The cut site is not always precisely at the center, and different enzymes have different cutting patterns (e.g., leaving sticky or blunt ends). The word "only" makes this statement inaccurate.
- **D. They remove nucleotides only from the ends of DNA fragments.**
  False. Restriction endonucleases are *endo*nucleases, meaning they cleave the DNA backbone from *within* the DNA molecule. Enzymes that remove nucleotides from the ends are called *exo*nucleases.
- **E. They recognise specific palindromic base-pair sequences.**
  True. A key characteristic of restriction enzymes is their ability to recognize and bind to unique, often palindromic, DNA sequences.

Identifying Incorrect Statements

Based on the analysis, statements **C** and **D** are not true regarding restriction endonucleases.

Conclusion

The correct option includes only the false statements, which are C and D.

---

111. **Answer: b**

Explanation:

DNA Fingerprinting Steps Sequence Explained

Understanding the correct order of steps in DNA fingerprinting is crucial for accurate analysis. The process involves several key stages, from extracting DNA to detecting specific fragments.

Correct Sequence of DNA Fingerprinting Steps

The standard procedure for DNA fingerprinting follows this sequence:

1. **Step A: Isolation of DNA and its digestion by restriction endonucleases.** This initial step involves collecting a biological sample (like blood or saliva) and extracting the DNA. The DNA is then cut into smaller fragments using specific enzymes called restriction endonucleases.
2. **Step E: Separation of DNA fragments by electrophoresis.** The digested DNA fragments are separated based on their size using a technique called gel electrophoresis. Smaller fragments move further down the gel than larger ones.
3. **Step C: Transferring of separated DNA fragments to synthetic membranes.** The separated DNA fragments on the gel are transferred onto a solid support, typically a nylon membrane, through a process similar to Southern blotting. This makes the fragments easier to analyze.
4. **Step B: Hybridisation using a labelled VNTR probe.** The membrane is exposed to a radioactive or fluorescently labelled probe. This probe is designed to bind specifically to variable number tandem repeat (VNTR) regions within the DNA fragments, which are unique to individuals.
5. **Step D: Detection of hybridised DNA fragments by autoradiography.** Finally, the locations where the labelled probe has bound to the DNA fragments on the membrane are visualized. This is often done using autoradiography (if radioactive probes are used) or other detection methods, revealing the unique pattern of DNA fragments – the DNA fingerprint.

Therefore, the correct sequence is A, E, C, B, D.

---

112. **Answer: b**

Explanation:

Photosynthesis Statements Analysis

The question asks to identify the incorrect statement(s) about photosynthesis among the given options.

Analyzing Photosynthesis Statements

- **Statement A:** The water splitting complex is associated with PS II. This is **correct**. Photolysis, the splitting of water molecules, occurs at Photosystem II (PSII) during the light-dependent reactions.

- **Statement B:** C_4 plants use the C_3 pathway of CO_2 fixation as the main biosynthetic pathway. This is **correct**. While C_4 plants have an initial CO_2 fixation step, the Calvin cycle (C_3 pathway) remains the primary pathway for synthesizing sugars.
- **Statement C:** In C_4 plants, photorespiration does not occur. This is generally considered **correct**. The CO_2 concentrating mechanism in C_4 plants significantly reduces the oxygenase activity of RuBisCO, thereby minimizing photorespiration.
- **Statement D:** C_3 plants exhibit "Kranz" anatomy. This statement is **incorrect**. Kranz anatomy, characterized by specialized large bundle sheath cells surrounding the vascular tissue, is a distinct feature of C_4 plants, not C_3 plants. C_3 plants lack this specialized structure.
- **Statement E:** ATP synthesis in chloroplasts occurs through chemiosmosis. This is **correct**. The proton gradient generated across the thylakoid membrane during light-dependent reactions drives ATP synthase to produce ATP via chemiosmosis.

Conclusion on Incorrect Statement

Based on the analysis, only statement D is incorrect because Kranz anatomy is characteristic of C_4 plants, not C_3 plants.

---

113. **Answer: b**

Explanation:

Somatic Hybridisation Steps Sequence

Somatic hybridisation is a technique used to create hybrid plants by fusing protoplasts (plant cells without cell walls) from different parent plants. The process involves several key steps that must be performed in a specific order:

1. **D**: Isolation of single cells from two different varieties of plants. This is the starting point, gathering the individual cells that will eventually form the hybrid.
2. **A**: Digestion of cell walls. Enzymes are applied to break down and remove the rigid cell walls surrounding the isolated plant cells.

3. **B**: Isolation of naked protoplasts. After cell wall removal, the resulting cells, now called protoplasts, are isolated.
4. **C**: Fusion of protoplasts to get hybrid protoplast. The isolated protoplasts from the different varieties are induced to fuse together, often using chemical agents like polyethylene glycol (PEG) or electrofusion, resulting in a hybrid protoplast.
5. **E**: Growing of hybrid protoplast to form a new plant. The successfully formed hybrid protoplast is then cultured using appropriate techniques (like tissue culture) to regenerate and grow into a complete, new hybrid plant.

Following this logical progression, the correct sequence of steps for somatic hybridisation is D, A, B, C, E.

---

114. **Answer: c**

Explanation:

Bulliform Cells Function in Grasses

Bulliform cells, also known as motor cells, are specialized large, thin-walled epidermal cells found in the leaves of many grasses (family Poaceae).

Primary Role of Bulliform Cells

- **Structure:** These are large, bubble-shaped cells typically arranged in longitudinal rows on the upper (adaxial) surface of grass leaves.
- **Mechanism:** Bulliform cells absorb water and become turgid, causing the leaf to expand and flatten, exposing more surface area for photosynthesis. Conversely, during conditions of water scarcity or drought stress, they lose water, become flaccid, and collapse.
- **Function:** This collapse causes the grass leaf to fold or roll inwards (involute). This action reduces the surface area exposed to direct sunlight and wind, significantly decreasing transpiration and thus minimizing water loss.

Analysis of Options

- **Option 1:** Making the leaf impermeable to fungal spores is not the primary function; this is usually achieved by the cuticle.
- **Option 2:** Photosynthesis primarily occurs in mesophyll cells, not specialized epidermal cells like bulliform cells, which often lack sufficient chloroplasts.
- **Option 3:** The ability to fold the leaf inwards during water stress to conserve water is the defining function of bulliform cells.
- **Option 4:** Water transport is the role of xylem tissue, not epidermal cells.

Therefore, the main function of bulliform cells is crucial for plant survival under arid conditions.

---

115. **Answer: d**

Explanation:

Microsporogenesis Developmental Sequence

Microsporogenesis is the process of formation of microspores (which develop into pollen grains) within the anthers of a flower. The correct developmental sequence involves specific stages.

Understanding the Stages

1. **Sporogenous tissue (B):** This is the initial diploid tissue present in the microsporangium responsible for producing microspores.
2. **Pollen mother cells (D):** Cells within the sporogenous tissue differentiate into specialized cells called pollen mother cells or microsporocytes. These are also diploid.
3. **Microspore tetrads (A):** Each pollen mother cell undergoes meiosis (a type of cell division reducing chromosome number by half) to form four haploid microspores, arranged in a tetrad.
4. **Pollen grains (C):** The microspores in the tetrad eventually separate and develop into mature haploid pollen grains through mitosis.

Correct Sequence

Based on the developmental process, the correct order is:

Sporogenous tissue (B) → Pollen mother cells (D) → Microspore tetrads (A) → Pollen grains (C)

Therefore, the correct option is B, D, A, C.

---

116. **Answer: d**

Explanation:

In Situ Conservation Method Identification

The question asks to identify an **in situ conservation method**. *In situ* conservation refers to the conservation of species or populations in their natural habitats or ecosystems. It involves protecting the entire ecosystem where the species naturally lives.

Analyzing Conservation Options

Let's examine each option:

- **Seed Banks:** These store seeds *ex situ* (outside their natural habitat). This method preserves genetic material but not the species in its natural environment.
- **Wildlife Safari Parks:** These are enclosures where animals are kept for public display and conservation purposes, often simulating natural habitats but are fundamentally *ex situ*.
- **Botanical Gardens:** These cultivate and display a wide range of plants, usually *ex situ*, preserving species outside their original ecosystems.
- **Sacred Groves:** These are small patches of forest or woodland, often protected by local communities due to religious or cultural beliefs. They represent natural ecosystems where species are conserved within their native environment, making them an *in situ* conservation method.

Conclusion on In Situ Conservation

Based on the definition, **Sacred Groves** are the only option that represents conservation within the species' natural habitat, thus qualifying as an *in situ conservation method*.

117. **Answer: c**

Explanation:

Understanding Cell Ploidy Levels

Ploidy refers to the number of complete sets of chromosomes within a biological cell. Key terms include:

- **Haploid (n):** Contains one set of chromosomes (e.g., gametes).
- **Diploid (2n):** Contains two sets of chromosomes (e.g., zygote, somatic cells).
- **Triploid (3n):** Contains three sets of chromosomes.

Analyzing Options in Angiosperm Reproduction

Let's examine the ploidy of each option during double fertilization in angiosperms:

- **Zygote:** Formed by the fusion of a male gamete (n) and the egg cell (n). Therefore, the zygote is diploid, represented as $n + n = 2n$.
- **Central cell:** Before fertilization, the central cell contains two polar nuclei, making it effectively diploid $(n + n = 2n)$.
- **Primary endosperm cell:** This cell results from the fusion of a male gamete (n) with the central cell's polar nuclei $(2n)$. This fusion, known as triple fusion, results in a triploid nucleus: $n + 2n = 3n$. This cell develops into the endosperm, which nourishes the embryo.
- **Synergid:** These are haploid cells (n) located within the ovule, near the egg cell.

Identifying the Triploid Cell

Based on the analysis:

- The **primary endosperm cell** is formed through triple fusion $(n + 2n)$ and is therefore **triploid** $(3n)$.

Thus, the primary endosperm cell is the triploid cell among the choices.

118. **Answer: d**

Explanation:

Extinction Rate Differences: Sixth vs. Past Events

The ongoing sixth mass extinction event is distinct from the previous five major extinction episodes primarily due to the unprecedented rate of species loss. Scientific consensus indicates that the current rate of species extinction is significantly elevated compared to natural background rates.

Analyzing Extinction Rate Data

Previous mass extinction events were often driven by catastrophic natural phenomena like asteroid impacts or massive volcanic activity, occurring over geological timescales. While devastating, the rate of species loss during those periods is being surpassed by the current extinction event.

- **Option 1 Incorrect:** States current rates are lower, contradicting evidence.
- **Option 2 Incorrect:** While faster, the magnitude is significantly understated.
- **Option 3 Incorrect:** The net species extinction rate is far from zero; it is alarmingly high.
- **Option 4 Correct:** Accurately reflects the scientific understanding that current species extinction rates are estimated to be 100 to 1000 times higher than the natural background rates observed in pre-human times. This accelerated rate is largely attributed to human activities like habitat destruction, climate change, and pollution.

The key difference lies in the speed and the primary driver. The sixth extinction is characterized by exceptionally high species extinction rates, driven by anthropogenic factors, making it distinct from earlier, naturally triggered mass extinctions.

Key Distinction: The current extinction rate is substantially faster than natural background rates and likely exceeds the rates seen in many prior extinction events, although precise comparisons across all five previous events can be complex.

119. **Answer: d**

Explanation:

Lac Operon Gene Function Explained

The lac operon is a genetic unit in bacteria, like *E. coli*, that enables the metabolism of lactose. It contains several genes, including structural genes that code for specific enzymes.

Lac Operon Structural Genes

The structural genes within the lac operon and their functions are:

- **lacZ gene:** Codes for the enzyme **beta-galactosidase**.
- **lacY gene:** Codes for the enzyme permease.
- **lacA gene:** Codes for the enzyme transacetylase.

z Gene Product: Beta-galactosidase

The question specifically asks what the **z gene** codes for. Based on the lac operon structure:

- The **lacZ** gene is responsible for producing **beta-galactosidase**.
- **Beta-galactosidase** is crucial as it breaks down lactose into glucose and galactose, allowing the bacteria to utilize it as an energy source.

Therefore, the **z gene** codes for **beta-galactosidase**.

---

120. **Answer: a**

Explanation:

Racemose Inflorescence Explained

A **racemose inflorescence** is a type of flower cluster where the arrangement follows a specific pattern.

Defining Characteristic: Acropetal Succession

The primary feature that defines a **racemose inflorescence** is the arrangement of flowers in an **acropetal succession**.

- In acropetal succession, the floral axis (main stalk) continues to grow indefinitely.
- Youngest flowers are found towards the apex (tip) of the axis, while the older, mature flowers are located towards the base.

Analysis of Options

- **Option 1: flowers are borne in an acropetal succession** - This correctly describes the pattern of flower development in a racemose inflorescence.
- **Option 2: flowers are solitary** - Solitary flowers grow alone and do not form an inflorescence.
- **Option 3 & 4: the growth is limited / the main axis terminates in a flower** - These characteristics are typical of a *cymose* inflorescence, not a racemose one. In cymose types, the main axis ends in a flower, limiting its growth.

Based on the definition, the continuous growth of the main axis and the acropetal arrangement of flowers are hallmarks of the racemose type.

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121. **Answer: b**

Explanation:

Whittaker's Five Kingdom Classification Criteria

R.H. Whittaker's Five Kingdom Classification system (1969) aimed to group organisms based on fundamental characteristics. The primary criteria used were:

- **Cell Structure:** Differentiating between prokaryotic (Monera) and eukaryotic cells.
- **Body Organization:** Distinguishing between unicellular and multicellular organisms.

- **Mode of Nutrition:** Autotrophic (e.g., plants) versus heterotrophic (e.g., animals, fungi).
- **Life Style/Ecological Role:** Producers, consumers, decomposers.
- **Reproduction:** Methods of reproduction (asexual and sexual).
- **Phylogenetic Relationships:** Evolutionary connections between organisms.

Analyzing the Options

The question asks for the main criteria used by Whittaker. Let's evaluate the provided options:

- **A. Cell structure:** This was a fundamental criterion (prokaryote vs. eukaryote).
- **B. Body organization:** Crucial for distinguishing unicellular (Monera, Protista) from multicellular kingdoms.
- **C. Presence of flagellum:** While flagella are important structures, their presence/absence was not a *main* kingdom-level criterion itself, but rather related to cell structure and motility within certain groups.
- **D. Reproduction:** Mode of reproduction influenced classification.
- **E. Phylogenetic relationships:** Whittaker aimed to reflect evolutionary history in his classification.

Determining the Correct Criteria

Based on Whittaker's proposal, the key criteria were indeed cell structure, body organization, mode of nutrition, reproduction, and phylogenetic relationships. The presence or absence of a flagellum (C) is a specific feature, not a primary overarching criterion for defining the five kingdoms themselves, although it relates to cell structure and motility.

Therefore, the criteria included are A, B, D, and E.

---

122. **Answer: a**

Explanation:

Evil Quartet Causes of Biodiversity Loss

The "Evil Quartet" refers to the four primary factors identified as the major drivers of species extinction and biodiversity loss.

Understanding The Evil Quartet Components

- **Habitat loss and fragmentation:** This involves the destruction, degradation, and division of natural environments, reducing the space and resources available for species.
- **Over-exploitation:** This occurs when resources (like species) are harvested at a rate that exceeds their natural ability to replenish, leading to population decline and potential extinction.
- **Alien species invasions:** The introduction of non-native or invasive species into new environments can disrupt ecosystems, outcompete native species for resources, or introduce diseases.
- **Co-extinctions:** This is the phenomenon where the extinction of one species leads to the extinction of other species that depend on it (e.g., a specific parasite that relies on a host species).

Other factors mentioned in the incorrect options, such as air pollution, water pollution, and soil pollution, contribute to environmental degradation but are not specifically included within the framework of the "Evil Quartet".

---

123. **Answer: d**

Explanation:

Polymerase Chain Reaction (PCR) Cycle Steps

Polymerase Chain Reaction (PCR) is a technique used to amplify specific DNA sequences. It involves repeated cycles, each consisting of three key temperature-dependent steps:

PCR Cycle Sequence

- **Denaturation:** The reaction mixture is heated to a high temperature, typically around $94 - 98°C$. This heat breaks the hydrogen bonds holding the two strands of

the DNA template together, separating them into single strands.
- **Annealing:** The temperature is then lowered, usually to $50 - 65°C$. This allows short DNA sequences called primers to bind (anneal) to complementary bases on the single-stranded DNA templates.
- **Extension:** The temperature is raised to an optimal point for the DNA polymerase enzyme, typically around $72°C$. The polymerase enzyme attaches to the primers and synthesizes a new complementary DNA strand, extending from the primer along the template strand.

These three steps—**Denaturation**, **Annealing**, and **Extension**—constitute one cycle of PCR. This cycle is repeated multiple times (usually 25-35 cycles) to exponentially amplify the target DNA sequence.

Therefore, the correct sequence of steps in each cycle of Polymerase Chain Reaction is Denaturation $\rightarrow$ Annealing $\rightarrow$ Extension.

---

124. **Answer: c**

Explanation:

Understanding Respiratory Quotient (RQ)

The Respiratory Quotient (RQ) is a measure used in cellular respiration. It is defined as the ratio of the volume of carbon dioxide (CO_2) produced to the volume of oxygen (O_2) consumed during the process.

The formula for RQ is:

$$RQ = \frac{\text{Volume of } CO_2 \text{ produced}}{\text{Volume of } O_2 \text{ consumed}}$$

RQ Calculation for the Given Biomolecule

The provided equation is:

$$2(C_{51}H_{98}O_6) + 145O_2 \rightarrow 102CO_2 + 98H_2O + \text{energy}$$

From this balanced equation, we can identify the volumes (or moles) of CO_2 produced and O_2 consumed:

- Volume of CO_2 produced = 102
- Volume of O_2 consumed = 145

Now, we calculate the RQ using the formula:

$$RQ = \frac{102}{145}$$

$$RQ \approx 0.703$$

Determining the Biomolecule Type and RQ Range

The molecule $C_{51}H_{98}O_6$ is a representation of a fat or lipid. Fats are characterized by a lower proportion of oxygen compared to carbohydrates.

- Carbohydrates typically yield an RQ of 1.
- Fats yield an RQ less than 1, usually between 0.5 and 0.95, because they require more oxygen for complete oxidation and produce less CO_2 relative to oxygen consumed.
- Proteins yield an RQ around 0.8 to 0.9.

Our calculated RQ of approximately 0.703 falls within the typical range for fats.

Conclusion on RQ Value

The calculated RQ of approximately 0.703 indicates that the biomolecule undergoing respiration is likely a fat. Therefore, the RQ falls within the range of 0.5 to 0.95.

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125. **Answer: d**

Explanation:

The question asks for the term describing the exploration of molecular, genetic, and species diversity to find products valuable for economic purposes.

Defining Bioprospecting for Economic Products

Bioprospecting is the systematic search for valuable compounds, genes, or other materials within the diversity of life (molecular, genetic, and species levels) that can be used for commercial or economic benefit. This aligns directly with the question's definition.

Analyzing Other Biological Terms

- **Biomagnification**: This refers to the increasing concentration of toxic substances in organisms at successively higher levels in a food chain. It does not relate to discovering products.
- **Bioremediation**: This involves using living organisms, like microbes or plants, to break down or remove pollutants from contaminated environments. It focuses on environmental cleanup, not economic product discovery.
- **Biofortification**: This is the process of increasing the density of micronutrients (like vitamins and minerals) in a staple food crop through agricultural breeding or genetic engineering. While it relates to improving products, it's a specific enhancement process, not the general search for diverse economic products.

Conclusion on Diversity Exploration

Therefore, exploring molecular, genetic, and species diversity for economic products is specifically termed Bioprospecting.

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126. **Answer: d**

Explanation:

Decomposition Process Matching

This question requires matching ecological terms related to decomposition (List I) with their correct definitions (List II). Let's break down each match:

A - III: Decomposition

Decomposition is the fundamental process where complex organic matter is broken down. This aligns perfectly with definition **III**, which states "Breaking down of complex organic matter into inorganic substances".

B - IV: Detritus

Detritus specifically refers to the dead organic material originating from dead plants and animals. Definition **IV**, "Dead remains of plants and animals including fecal matter", accurately describes detritus.

C - II: Mineralisation

Mineralisation is a key stage in decomposition where organic compounds are converted into inorganic nutrients. Definition **II**, "Release of inorganic nutrients by the activity of microbes in soil", captures this transformation.

D - I: Humification

Humification is the process that leads to the formation of humus. Humus is characterized as a dark, stable, amorphous substance. Definition **I**, "Accumulation of dark coloured amorphous colloidal substance", describes this end product.

Final Match

Based on the analysis, the correct matching is:

- A - III
- B - IV
- C - II
- D - I

This corresponds to option 4 (D).

---

127. **Answer: b**

Explanation:

Biomolecule Statements Analysis

Statement A: Lipids Solubility

Lipids are primarily nonpolar molecules, meaning they do not mix well with water, which is a polar solvent. Therefore, lipids are generally **insoluble** in water (hydrophobic).

Statement A is incorrect.

Statement B: Proteins as Polypeptides

Proteins are large biomolecules. They are polymers made up of amino acid monomers. These amino acids are linked together by peptide bonds to form a long chain, which is known as a **polypeptide**. A protein consists of one or more polypeptides.

Statement B is correct.

Statement C: Polysaccharides Structure

Polysaccharides are a type of carbohydrate. They are formed when many simple sugar units (monosaccharides) are joined together by glycosidic bonds. This results in **long chains of sugars**.

Statement C is correct.

Statement D: Adenine and Guanine Classification

Adenine and guanine are two of the five main nitrogenous bases found in nucleic acids. They are classified as **purines**, which have a characteristic double-ring structure. The other bases, cytosine, thymine, and uracil, are classified as pyrimidines, which have a single-ring structure.

Statement D is incorrect.

Statement E: Enzymes Composition

Enzymes are biological catalysts responsible for speeding up chemical reactions in living organisms. The vast majority of known enzymes are **proteins**. While a small number of RNA molecules (ribozymes) also exhibit catalytic activity, proteins are the principal type of enzyme.

Statement E is correct.

Conclusion on Correct Statements

The analysis shows that statements B, C, and E are correct descriptions related to biomolecules.

Summary of Correct Statements

- **B. Proteins are polypeptides.**
- **C. Polysaccharides are long chains of sugars.**
- **E. Almost all enzymes are proteins.**

---

128. **Answer: c**

Explanation:

Angiosperm Root Hairs: Origin Region

Root hairs are crucial extensions of epidermal cells in plant roots, primarily responsible for absorbing water and nutrients from the soil. In angiosperms, their development is specific to a particular zone.

Root Anatomy and Root Hair Formation

- **Meristematic Activity Region:** Contains actively dividing cells responsible for root growth. Cells here are small and undifferentiated.
- **Elongation Region:** Cells in this region increase in length, pushing the root tip further into the soil.
- **Maturation Region:** This is where cells differentiate and mature. It is characterized by the presence of root hairs. These root hairs greatly increase the surface area for absorption.
- **Root Cap:** Protects the root apical meristem as the root grows through the soil.

Root hairs are specialized structures that emerge from the epidermal cells (trichoblasts) only in the **region of maturation**. This region is located just above the region of elongation.

Therefore, root hairs arise from the region where cells have differentiated and specialized, which is the region of maturation.

129. **Answer: d**

Explanation:

The phase of cell elongation in plant growth is characterized by rapid increase in cell size, mainly due to vacuolation and cell wall loosening.

However, **large conspicuous nuclei** is not a characteristic feature of the elongation phase. In fact, during elongation, the nucleus usually becomes less prominent as the cell becomes highly vacuolated and stretched.

So, the correct answer is:

Large conspicuous nuclei (not a characteristic of elongation phase)

---

130. **Answer: a**

Explanation:

The question asks to identify the correct floral formula for the Solanaceae family among the given options.

Floral Formula Breakdown

A floral formula represents the characteristics of a flower using standardized symbols:

- Symmetry: $\oplus$ (Actinomorphic / Radial Symmetry)
- Sexuality: ⚥ (Bisexual Flower)
- Calyx (Sepals): K. The number indicates the count, and parentheses $(())$ denote fusion.
- Corolla (Petals): C. The number indicates the count, and parentheses $(())$ denote fusion.
- Androecium (Stamens): A. The number indicates the count.
- Gynoecium (Carpels): G. The number indicates the count, parentheses $(())$ denote fusion, and an underline ($__$) indicates a superior ovary.

Solanaceae Family Characteristics

The Solanaceae family typically exhibits the following floral characteristics:

- The flowers are generally actinomorphic ($\oplus$).
- Flowers are bisexual (⚥).
- The calyx consists of 5 sepals that are fused (gamosepalous), represented as $K_{(5)}$.
- The corolla consists of 5 petals that are fused (gamopetalous), represented as $C_{(5)}$.
- There are 5 stamens, usually free and inserted on the corolla (epipetalous), represented as A_5.
- The gynoecium is bicarpellary (2 carpels), fused (syncarpous), and has a superior ovary, represented as $\underline{G}_{(2)}$.

Evaluating the Options

Based on the typical characteristics of Solanaceae:

- Option 1: $\oplus$ ⚥ $K_{(5)}\ C_{(5)}\ A_5\ \underline{G}_{(2)}$ - This formula accurately reflects the fused calyx, fused petals, free stamens, and fused carpels with a superior ovary.
- Option 2: $\oplus$ ⚥ $K_5\ C_{(5)}\ A_5\ \underline{G}_{(2)} - Incorrectly shows free sepals (K_5$).
- Option 3: $\oplus$ ⚥ $K_5\ C_5\ A_5\ \underline{G}_{(2)}$ - Incorrectly shows free sepals and free petals.
- Option 4: $\oplus$ ⚥ $K_{(5)}$ C(5) A5 $\underline{G}_{(2)} - The notation$ C(5) A5 implies fusion between petals and stamens, which is less precise than simply listing the stamens as A_5. It doesn't represent the standard formula.

Conclusion

The floral formula that correctly represents the Solanaceae family is $\oplus$ ⚥ $K_{(5)}\ C_{(5)}\ A_5\ \underline{G}_{(2)}$.

---

131. **Answer: c**

Explanation:

DNA Packaging: Analyzing Statement Accuracy

This question assesses understanding of DNA packaging within eukaryotic cells, focusing on the role of histone proteins and chromatin structure.

Statement Analysis

- **A. Histones form histone octamers:** This statement is **correct**. Histones assemble into a core structure containing eight protein molecules (two each of H2A, H2B, H3, and H4), known as the histone octamer.
- **B. Histones are negatively charged basic proteins:** This statement is **incorrect**. Histones are **positively** charged basic proteins.
- **C. Histones rich in lysine and arginine:** This statement is **correct**. Histones contain a high proportion of basic amino acids, specifically **lysine** and **arginine**, which are positively charged.
- **D. Positively charged DNA wraps around histone octamer:** This statement is **incorrect**. DNA carries a **negative** charge due to its phosphate backbone. The **positively charged** histone octamer interacts with and wraps the DNA.
- **E. Non-histone proteins in higher-level packaging:** This statement is **correct**. Further condensation and organization of the chromatin fiber involve additional proteins known as **non-histone chromosomal proteins**.

Conclusion on Correct Statements

Based on the analysis, statements A, C, and E are correct regarding DNA helix packaging.

Identifying the Correct Option

The option that includes only the correct statements (A, C, and E) is Option 3.

---

132. **Answer: c**

Explanation:

Understanding DNA Separation and Visualization Techniques

This question assesses knowledge of common molecular biology techniques used for DNA manipulation and observation.

Analyzing Statement A: DNA Cutting

Statement A claims that "molecular scissors" are used to cut DNA. This refers to restriction enzymes, which are proteins that cleave DNA molecules at specific recognition nucleotide sequences. They act like precise scissors for DNA. Therefore, statement A is considered correct in the context of molecular biology.

Analyzing Statement B: DNA Separation by Electrophoresis

Statement B states that DNA fragments separate based on size in an agarose gel during electrophoresis. This is the fundamental principle of agarose gel electrophoresis. DNA molecules carry a net negative charge due to their phosphate backbone. When an electric field is applied, DNA fragments migrate towards the positive electrode. The gel matrix acts as a sieve, impeding the movement of larger fragments more than smaller ones. Thus, smaller DNA fragments travel faster and further through the gel than larger fragments. Statement B is correct.

Analyzing Statement C: Visualization Without Staining

Statement C suggests that separated DNA fragments are visible without staining when exposed to UV light. DNA itself does not inherently fluoresce strongly enough to be seen under UV light. Specific stains, such as ethidium bromide, are required to intercalate into the DNA and fluoresce under UV illumination, making the DNA bands visible. Therefore, statement C is incorrect.

Analyzing Statement D: Visualization in Visible Light

Statement D proposes that DNA fragments stained with ethidium bromide can be seen in visible light. While ethidium bromide does fluoresce, this fluorescence is typically excited by ultraviolet (UV) light, not visible light. Specialized equipment emitting UV light is needed to visualize the stained DNA bands. Therefore, statement D is incorrect.

Conclusion on Correct Statements

Based on the analysis:

- Statement A is correct (Restriction enzymes act as molecular scissors).
- Statement B is correct (Electrophoresis separates DNA by size in agarose gel).
- Statement C is incorrect (Staining is required for UV visualization).
- Statement D is incorrect (Ethidium bromide visualization requires UV light, not visible light).

The correct statements are A and B.

---

133. **Answer: c**

Explanation:

Protein Structure Levels & Alpha-Helix

Proteins fold into specific three-dimensional shapes, organized into four structural levels:

- **Primary Structure:** The linear sequence of amino acids.
- **Secondary Structure:** Localized, repeating folding patterns like alpha-helices and beta-sheets, stabilized by hydrogen bonds between backbone atoms.
- **Tertiary Structure:** The overall 3D conformation of a single polypeptide chain.
- **Quaternary Structure:** The assembly of multiple polypeptide chains.

Alpha-Helix Formation

The **alpha-helix** is a common motif characterized by its helical shape. This structure arises from hydrogen bonding between the carbonyl oxygen of one amino acid residue and the amide hydrogen of the amino acid residue four positions earlier in the polypeptide chain. These regular, repeating local folding patterns are hallmarks of protein **secondary structure**.

Therefore, the **alpha-helix** is found in the **secondary structure** level of protein organization.

---

134. **Answer: a**

Explanation:

Productivity Concepts Defined

Understanding ecological productivity involves matching specific terms with their accurate definitions.

- **Gross Primary Productivity (GPP):** The total rate at which producers convert light energy into chemical energy via photosynthesis.
- **Net Primary Productivity (NPP):** The energy remaining after GPP minus the energy used by producers for their own respiration (R). Formula: NPP = GPP - R.
- **Secondary Productivity:** The rate at which consumers assimilate and accumulate energy by consuming other organisms.
- **Productivity (General):** The rate at which biomass is generated.

Matching Productivity Terms

Match the items from List I to the definitions in List II:

- **A. Productivity** corresponds to **III. Rate of biomass production**. This is a general measure of organic matter creation.
- **B. Net primary productivity** corresponds to **I. Gross primary productivity minus respiration losses**. This is the definition of NPP.
- **C. Gross primary productivity** corresponds to **IV. Rate of production of organic matter during photosynthesis**. This captures the total photosynthetic output.
- **D. Secondary productivity** corresponds to **II. Rate of formation of new organic matter by consumers**. This defines productivity in heterotrophs.

Final Productivity Match

The correct pairings are A-III, B-I, C-IV, D-II.

This set of matches aligns with Option A.

---

135. **Answer: c**

Explanation:

Placentation Examples Matching

This section explains the matching between different types of **placentation** in flowers and their common plant examples.

Understanding Placentation Types

- **Marginal Placentation (A):** Here, the placenta runs along the ventral suture of the ovary. Ovules are arranged in a single row on this placenta. A typical example is **Pea**.
- **Axile Placentation (B):** In axile placentation, the ovary is divided into multiple compartments or lobes by septa. Ovules are attached to the central axis formed where these septa meet. **Lemon** is a representative example.
- **Parietal Placentation (C):** With parietal placentation, ovules develop on the inner wall of the ovary. The ovary is typically unilocular, or if multilocular, the septa might have disappeared. **Mustard** showcases this type.
- **Basal Placentation (D):** This type is characterized by a single ovule attached at the base of the ovary. Plants like **Marigold** exhibit basal placentation.

Deriving the Correct Match

Based on the definitions and examples:

- A. Marginal Placentation matches with **Pea** (II).
- B. Axile Placentation matches with **Lemon** (IV).
- C. Parietal Placentation matches with **Mustard** (I).
- D. Basal Placentation matches with **Marigold** (III).

The correct combination derived is A-II, B-IV, C-I, D-III.

---

136. **Answer: c**

Explanation:

Frog Anatomy Statements Analysis

This question requires identifying correct statements about the anatomy of a frog.

Statement A: Hepatic Portal System

The **hepatic portal system** is a specialized network of veins that carries blood from the digestive organs (like the intestine) directly to the liver. This system allows the liver to process nutrients absorbed from the intestine before they enter the general circulation. Therefore, statement A is **correct**.

Statement B: Cranial Nerves

Frogs possess **ten pairs** of cranial nerves that originate from the brain. The statement claims there are twelve pairs, which is incorrect for frogs (though correct for humans). Therefore, statement B is **incorrect**.

Statement C: Ureters, Oviducts, and Cloaca

In female frogs, the **ureters** (carrying urine from kidneys) and the **oviducts** (carrying eggs from ovaries) both terminate and open into the **cloaca**. The cloaca is a common chamber that serves the digestive, urinary, and reproductive systems. Therefore, statement C is **correct**.

Statement D: Hind-brain Composition

The frog's **hind-brain** is primarily composed of the **cerebellum** and the **medulla oblongata**. The **optic lobes** are part of the mid-brain, not the hind-brain. Therefore, statement D is **incorrect**.

Statement E: Sinus Venosus and Heart

The **sinus venosus** is a chamber that receives deoxygenated blood from the body and empties it into the right atrium of the frog's heart. This is a standard part of the amphibian circulatory system. Therefore, statement E is **correct**.

Conclusion on Correct Statements

Based on the analysis, the correct statements regarding frog anatomy are A, C, and E.

- Statement A: Correct
- Statement B: Incorrect
- Statement C: Correct
- Statement D: Incorrect
- Statement E: Correct

The combination of correct statements is A, C, and E only.

---

137. **Answer: a**

Explanation:

Flightless Bird Adaptation for Swimming

The question asks to identify a **flightless bird** characterized by **forelimbs modified as paddle-like structures**, specifically adapted for **swimming**.

Key Characteristics Analysis

- **Flightless:** The bird cannot fly.
- **Modified Forelimbs:** Wings have evolved into paddle shapes.
- **Swimming Adaptation:** These paddles are efficient for movement in water.

Evaluating the Options

- **A. *Aptenodytes*:** This genus includes penguins. Penguins are well-known **flightless birds**. Their wings are modified into stiff, flat flippers or **paddle-like structures**, which are highly effective for **swimming** and diving.

- **B. *Neophron*:** This genus represents the Egyptian vulture. Vultures are birds of prey capable of flight, and their forelimbs are wings used for aerial locomotion, not paddles for swimming.
- **C. *Psittacula*:** This genus includes various species of parrots. Parrots are known for their ability to fly, and their forelimbs function as wings.
- **D. *Struthio*:** This genus represents the ostrich. Ostriches are **flightless birds**, but their forelimbs are relatively small and primarily used for balance and display, not for swimming. Their legs are adapted for running.

Conclusion

Based on the analysis, *Aptenodytes* (penguins) perfectly matches the description of a flightless bird with forelimbs modified into paddle-like structures for swimming.

---

138. **Answer: d**

Explanation:

The statement **"Bulging eyes"** is incorrect.

Male frogs are distinguished from female frogs by the presence of **vocal sacs and nuptial pads** (in many species), which help in mating and calling.

Bulging eyes are **not a sex-determining feature** in frogs.

---

139. **Answer: a**

Explanation:

Analyzing Animal Characteristics

The question describes three key characteristics (A, B, C) observed in a group of fish-like animals and asks for the best fit among the options.

- **Characteristic A:** Endoskeleton is made of cartilage.

- **Characteristic B:** Ectoparasites with a circular sucking mouth.
- **Characteristic C:** Absence of paired fins and scales, presence of 7 pairs of gill slits.

Evaluating Animal Options

Let's examine each option based on the given characteristics:

- **Petromyzon sp. (Lamprey):**
  - Endoskeleton: Cartilaginous (Matches A).
  - Lifestyle: Many species are ectoparasites with a distinct sucking mouth (Matches B).
  - Fins/Scales/Gills: Lacks paired fins and scales. Possesses 7 pairs of external gill openings (Matches C).

  Conclusion: Petromyzon fits all described characteristics.
- **Branchiostoma sp. (Lancelet):**
  - Endoskeleton: Possesses a notochord, not a typically cartilaginous endoskeleton like vertebrates.
  - Lifestyle: Free-living filter feeders, not ectoparasites.
  - Fins/Scales/Gills: Lacks paired fins and scales, but has numerous gill slits, not specifically 7 pairs externally.

  Conclusion: Does not fit well.
- **Scoliodon sp. (Dogfish Shark):**
  - Endoskeleton: Cartilaginous (Matches A).
  - Lifestyle: Predatory fish, not ectoparasites attached with sucking mouths.
  - Fins/Scales/Gills: Possesses paired fins and scales (placoid), contradicting C.

  Conclusion: Does not fit well.
- **Exocoetus sp. (Flying Fish):**
  - Endoskeleton: Bony skeleton, not cartilaginous.
  - Lifestyle: Free-swimming fish, not ectoparasites.
  - Fins/Scales/Gills: Possesses paired fins and scales, contradicting C.

  Conclusion: Does not fit well.

Identifying the Best Fit

Based on the analysis, **Petromyzon sp.** is the only species that exhibits all the observed characters: a cartilaginous skeleton, ectoparasitic nature with a sucking mouth, absence of paired fins and scales, and 7 pairs of gill slits.

140. **Answer: c**

Explanation:

Respiration Steps Order

Understanding the sequence of events in human respiration is key. These steps involve breathing, gas exchange, transport, and cellular metabolism.

Pulmonary Ventilation Explained

This is the first stage, involving the intake of atmospheric air into the lungs and the expulsion of carbon dioxide-rich alveolar air. This corresponds to step C.

Gas Exchange Across Alveoli

Next, oxygen (O_2) moves from the lung alveoli into the blood, while carbon dioxide (CO_2) moves from the blood into the alveoli. This is step B.

Gas Transport by Blood

The blood then circulates, carrying oxygen (O_2) from the lungs to the body tissues and returning carbon dioxide (CO_2) from the tissues to the lungs. This is step E.

Gas Exchange Between Blood and Tissues

At the body tissues, oxygen (O_2) diffuses from the blood into the cells, and carbon dioxide (CO_2) diffuses from the cells into the blood. This is step A.

Cellular Respiration's Role

The final step is cellular respiration, where cells use oxygen (O_2) to break down nutrients and produce energy (ATP), releasing carbon dioxide (CO_2) as a byproduct. This is step D.

The correct order integrating these physiological processes is C, B, E, A, D.

141. **Answer: d**

Explanation:

Identifying Non-Membrane Bound Organelles in Prokaryotes and Eukaryotes

The question asks to identify cell organelles that lack a membrane and are present in both prokaryotic and eukaryotic cells. Understanding the structure and presence of various organelles is key to answering this.

Organelle Characteristics Analysis

Let's analyze the options based on whether they are membrane-bound and their presence in prokaryotic and eukaryotic cells:

- **Mitochondria**: These are double-membrane bound organelles found only in eukaryotic cells. They are absent in prokaryotes.
- **Lysosomes**: These are single-membrane bound organelles containing digestive enzymes, found in eukaryotic cells. They are absent in prokaryotes.
- **Centrosomes**: These are structures involved in cell division in eukaryotic cells, typically lacking a membrane but not found in prokaryotes.
- **Ribosomes**: These are granular structures responsible for protein synthesis. They are non-membrane bound and are found in both prokaryotic and eukaryotic cells (though differing slightly in size, 70S in prokaryotes and 80S in eukaryotes).

Conclusion on Organelle Presence

Based on the analysis, only ribosomes fit the criteria of being non-membrane bound and present in both prokaryotic and eukaryotic cells.

---

142. **Answer: c**

Explanation:

GIFT Infertility Procedure Explained

GIFT (Gamete Intrafallopian Transfer) is an assisted reproductive technology used to treat infertility by enabling fertilization within the woman's body.

GIFT Definition Overview

The GIFT procedure involves retrieving eggs (ova) from the female and sperm from the male partner. These gametes (egg and sperm) are then mixed and transferred directly into the woman's **fallopian tube**. The primary goal is for fertilization to occur naturally within the fallopian tube (*in vivo*), similar to natural conception.

Analyzing Correctness of GIFT Statements

- **Option 1: Incorrect.** This describes an IVF procedure using donor eggs where ova are transferred to the **uterus**, not the fallopian tube as required for GIFT.
- **Option 2: Incorrect.** Transferring **early embryos** (up to 8 blastomeres) into the fallopian tube is known as Tubal Embryo Transfer (TET). GIFT specifically involves the transfer of **gametes** (egg and sperm), not embryos.
- **Option 3: Correct.** This statement accurately describes GIFT. It details the transfer of an **ovum** (potentially from a donor) into the **fallopian tube**. It correctly mentions the recipient female can provide a suitable environment for fertilization and development, implying *in vivo* fertilization, which is the hallmark of GIFT.
- **Option 4: Incorrect.** Transferring early **embryos** into the **uterus** is a common technique in IVF, distinct from GIFT's gamete transfer into the fallopian tube.

Conclusion on GIFT

Option 3 is the only statement that accurately reflects the core principle of GIFT: the transfer of gametes into the fallopian tube to allow fertilization to occur naturally within the woman's body.

---

143. **Answer: b**

Explanation:

Muscle Contraction: Statement Analysis

This section analyzes the statements related to the mechanism of muscle contraction to identify the correct ones.

Analyzing Statement A: Motor Neuron Signal

- Statement: "A motor neuron carries a signal sent by the Central Nervous System (CNS) to the sarcolemma of the muscle fibre."
- **Analysis:** This statement is correct. Motor neurons are responsible for transmitting nerve impulses from the CNS to muscle fibres, initiating muscle contraction at the neuromuscular junction on the sarcolemma.

Analyzing Statement B: Action Potential and Calcium Release

- Statement: "The neural signal generates an action potential which causes the release of Ca^{++} into sarcoplasm."
- **Analysis:** This statement is correct. The arrival of a nerve impulse triggers an action potential in the muscle fibre membrane (sarcolemma) and T-tubules, leading to the release of calcium ions (Ca^{++}) from the sarcoplasmic reticulum into the sarcoplasm.

Analyzing Statement C: Calcium and Actin Interaction

- Statement: "Increase in Ca^{++} inactivates the actin for breaking cross bridges."
- **Analysis:** This statement is incorrect. Calcium ions (Ca^{++}) bind to troponin, causing tropomyosin to shift away from the myosin-binding sites on actin. This process *activates* actin, allowing myosin heads to bind and form cross bridges, rather than inactivating it.

Analyzing Statement D: Actin-Myosin Binding

- Statement: "Actin binds to the myosin head to form a cross bridge."
- **Analysis:** This statement is correct. The interaction between the myosin heads and the actin filaments forms the cross bridges, which are essential for generating force during muscle contraction.

Analyzing Statement E: Sarcomere Shortening

- Statement: "Shortening of sarcomere takes place, by pulling actin filaments towards the centre of 'A' band."
- **Analysis:** This statement is correct. According to the sliding filament theory, during contraction, the actin filaments slide past the myosin filaments towards the M-line (center of the A band), causing the sarcomere to shorten.

Conclusion: Correct Statements

Based on the analysis, the correct statements are A, B, D, and E.

Therefore, the option that includes all these correct statements is the right choice.

---

144. **Answer: b**

Explanation:

pBR322 BamHI Site and Antibiotic Resistance

The cloning vector *pBR322* is a common plasmid used in genetic engineering, particularly in *E. coli*.

Antibiotic Resistance Genes in pBR322

pBR322 contains genes that provide resistance to two antibiotics:

- Ampicillin (resistance conferred by the *amp* gene)
- Tetracycline (resistance conferred by the *tet* gene)

BamHI Restriction Site Location

The recognition site for the restriction enzyme *BamHI* is located within the **tetracycline resistance gene** (*tet* gene) of the *pBR322* vector.

Consequence of Foreign DNA Insertion

Inserting foreign DNA into a restriction site within a functional gene inactivates that gene.

Therefore, inserting foreign DNA at the *BamHI* site disrupts the *tetracycline resistance gene*, rendering it non-functional.

Impact on Antibiotic Resistance

As a result of the inactivation of the *tetracycline resistance gene*, bacteria transformed with this recombinant plasmid will lose their resistance to **Tetracycline**. The ampicillin resistance gene is typically unaffected by insertion at the *BamHI* site, so resistance to ampicillin is usually retained.

---

145. **Answer: a**

Explanation:

Synaptic Neurotransmitter Receptors Location

In a synapse, the junction between two neurons, communication occurs via chemical messengers called neurotransmitters. These neurotransmitters are released from the pre-synaptic neuron and travel across the synaptic cleft to bind with specific receptors on the next neuron.

Receptor Location in Synapses

The specific binding sites, or receptors, for neurotransmitters are primarily located on the membrane of the neuron that receives the signal. This receiving neuron's membrane is known as the post-synaptic membrane.

- **Neurotransmitter Release:** Occurs from the pre-synaptic terminal.
- **Signal Transmission:** Neurotransmitters diffuse across the synaptic cleft.
- **Receptor Binding:** Specific receptors on the **post-synaptic membrane** bind to these neurotransmitters, initiating a response in the receiving neuron.

The pre-synaptic membrane is involved in releasing neurotransmitters, while the myelin sheath and Schwann cells are related to nerve impulse insulation and speed, not direct neurotransmitter reception.

Therefore, the specific receptors for neurotransmitters are present on the **post-synaptic membrane**.

---

146. **Answer: b**

Explanation:

Let's check each statement:

A. Human skull is monocondylic → **Incorrect**
Human skull is **dicondylic** (two occipital condyles).

B. The joint between any two adjoining vertebrae is a cartilaginous joint → **Correct**
Intervertebral joints are **cartilaginous (slightly movable)** joints.

C. Number of cervical vertebrae is seven → **Correct**
Humans have **7 cervical vertebrae**.

D. All ribs except last 2 pairs are bicephalic → **Incorrect**
Ribs typically have a single head; classification here is incorrect.

E. Occipital bone articulates with atlas vertebra → **Correct**
The **occipital condyles** articulate with the **atlas (C1)** forming the atlanto-occipital joint.

Correct answer:

B, C and E

---

147. **Answer: b**

Explanation:

Alpha-1 Antitrypsin Use in Emphysema Treatment

The protein α-1-antitrypsin (AAT) is crucial for protecting the lungs from damage caused by certain enzymes, like neutrophil elastase.

- **Function:** AAT acts as a protease inhibitor, preventing the uncontrolled breakdown of lung tissue.
- **Deficiency:** A genetic deficiency in AAT leads to its reduced levels or ineffective function.
- **Disease Link:** Without sufficient AAT, neutrophil elastase can damage lung alveoli, leading to **emphysema**, a severe lung disease characterized by shortness of breath.
- **Therapeutic Use:** Human AAT, produced using transgenic animal technology, can be administered as a therapy (enzyme replacement therapy) to individuals with severe AAT deficiency to help manage or slow the progression of **emphysema**.

Therefore, human α-1-antitrypsin obtained from transgenic animals is primarily used for the treatment of **Emphysema**.

---

148. **Answer: c**

Explanation:

Rh Grouping Statements Analysis

This analysis identifies incorrect statements regarding Rh grouping, focusing on Erythroblastosis foetalis and blood transfusion compatibility.

Analyzing Rh Incompatibility Statement A

Statement A: Erythroblastosis foetalis is a condition observed having fetus with Rh^{-ve} blood and mother with Rh^{+ve} blood.

Reasoning: This is **incorrect**. The condition, Hemolytic Disease of the Newborn/Fetus, specifically occurs when an Rh^{-ve} mother carries an Rh^{+ve} fetus. Maternal Rh^{+ve} blood and fetal Rh^{-ve} blood do not cause this condition.

Analyzing Rh Antigen Statement B

Statement B: Rh antigen is observed on RBCs in the majority of human beings.

Reasoning: This is **correct**. Approximately 85% of the human population is Rh^{+ve}, meaning they possess the Rh antigen on their red blood cells.

Analyzing Rh Transfusion Statement C

Statement C: Before blood transfusion, Rh group should also be matched.

Reasoning: This is **correct**. Rh factor compatibility is crucial during blood transfusions to prevent immune responses and potential transfusion reactions, especially for individuals who may become sensitized.

Analyzing Rh Incompatibility Statement D

Statement D: Rh incompatibility is observed when a pregnant mother is Rh^{-ve} and the fetus is Rh^{+ve}.

Reasoning: This is **correct**. This specific scenario defines Rh incompatibility during pregnancy, which can lead to complications like Erythroblastosis foetalis.

Analyzing Rh Prevention Statement E

Statement E: Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the second child.

Reasoning: This is **incorrect**. Effective prevention involves administering anti-Rh antibodies (e.g., RhoGAM) to Rh^{-ve} mothers during pregnancy (around 28 weeks) and within 72 hours after delivering an Rh^{+ve} baby. This prevents the mother from developing her own antibodies (sensitization). The statement's timing ("after the second child") is inaccurate and incomplete for preventing initial sensitization.

Conclusion on Incorrect Rh Statements

The analysis confirms that statements A and E are factually incorrect regarding Rh grouping principles and clinical applications.

---

149. **Answer: a**

Explanation:

Drug Effects Matching Explanation

This question requires matching specific drugs with their primary physiological effects. We will analyze each drug individually based on its known pharmacological properties.

Matching Drug A: Nicotine

- **Nicotine** is primarily known as a stimulant.
- It acts rapidly, stimulating the central nervous system and adrenal glands.
- This stimulation leads to the release of catecholamines (like adrenaline) into the bloodstream.
- Therefore, Nicotine matches with **II. Stimulates adrenal gland to release catecholamines into blood circulation**.

Matching Drug B: Morphine

- **Morphine** is a potent opioid analgesic.
- Its main therapeutic uses are pain relief and sedation.
- Therefore, Morphine matches with **III. Effective sedative and painkiller**.

Matching Drug C: Heroin

- **Heroin** is a highly addictive opioid.
- While it can produce euphoria, it is fundamentally a central nervous system depressant.
- Opioid depressants slow down vital bodily functions.
- Therefore, Heroin matches with **IV. A depressant; slows down body function**.

Matching Drug D: Cocaine

- **Cocaine** is a powerful central nervous system stimulant.
- It significantly increases energy levels and alertness.
- Users often experience intense feelings of euphoria shortly after use.
- Therefore, Cocaine matches with **I. Causes sense of euphoria and increased energy**.

Final Match and Answer

Based on the analysis:

- A matches with II
- B matches with III
- C matches with IV
- D matches with I

This corresponds to the pairing **A-II, B-III, C-IV, D-I**.

The correct option is the one that lists this specific matching.

---

150. **Answer: b**

Explanation:

Muscular and Skeletal System Conditions Matching

This question requires matching specific conditions related to the muscular and skeletal systems (List I) with their correct descriptions (List II).

Matching Terms and Descriptions

- **A. Tetany** matches with **III. Wild contraction in muscle due to low Ca^{++} in body fluid**. Tetany is characterized by muscle spasms caused by abnormal calcium levels.
- **B. Arthritis** matches with **I. Inflammation of joints**. Arthritis is a condition defined by inflammation of one or more joints.
- **C. Myasthenia gravis** matches with **II. Autoimmune disorder affecting neuromuscular junction**. This condition involves the immune system attacking the communication between nerves and muscles.
- **D. Muscular dystrophy** matches with **IV. Progressive degeneration of skeletal muscle**. This genetic disorder leads to the gradual weakening and breakdown of skeletal muscles.

Correct Option Identification

Based on the matches:

- A - III
- B - I
- C - II
- D - IV

This corresponds to option 2 (A-III, B-I, C-II, D-IV).

---

151. **Answer: c**

Explanation:

Matching Contraceptive Methods

This question requires matching specific contraceptive methods from List I with their descriptions or types in List II. Let's analyze each item:

Detailed Matching

- **A. Progestasert**: This is an intrauterine device (IUD) that releases progesterone. Therefore, it matches with **III. Hormone releasing IUD**.
- **B. Multiload 375**: This is a type of IUD that contains copper, acting as a copper-releasing device. Thus, it matches with **IV. Copper releasing IUD**.
- **C. Diaphragm**: This is a flexible, dome-shaped barrier, typically made of silicone or rubber, inserted into the vagina to prevent pregnancy. It matches with **I. Barrier made of rubber used by females**.
- **D. Saheli**: This is a non-steroidal, single-pill-a-week oral contraceptive. It matches with **II. Oral contraceptive**.

Correct Matching Combination

Based on the individual matches, the correct combination is:

- A - III
- B - IV
- C - I

- D - II

This corresponds to option 3.

---

152. **Answer: d**

Explanation:

Cell Membrane Statements Correctness

Let's evaluate each statement about the cell membrane in eukaryotic cells:

- **A. Membrane of human RBCs has approximately 52% protein.**

  This is generally accepted. Human red blood cell membranes contain a high proportion of protein.

- **B. The phospholipids are arranged in a bilayer.**

  This is correct. The plasma membrane's core structure is a phospholipid bilayer.

- **C. Extensions of the plasma membrane into the cell form mesosomes.**

  This is incorrect. Mesosomes are characteristic infoldings in prokaryotic cells, not eukaryotic ones.

- **D. Tails towards the inner part of lipids are hydrophobic and thus protected from aqueous medium.**

  This is correct. The hydrophobic lipid tails face inward, shielded from the aqueous environment.

- **E. Glycocalyx is present on the outer surface of the plasma membrane.**

  This is considered incorrect in this context. While found on animal cells, glycocalyx is not a universal feature of all eukaryotic cells.

The statements identified as correct are A, B, and D.

---

153. **Answer: d**

Explanation:

Cell Organelle Statements Analysis

We need to evaluate each statement about cell organelles and inclusions to determine which are correct.

Statement A: Endomembrane System Components

- The endomembrane system is a group of membranes and organelles in eukaryotic cells that works together to modify, package, and transport lipids and proteins.
- It includes the nuclear envelope, lysosomes, vesicles, the endoplasmic reticulum, and the Golgi apparatus.
- **Mitochondria are generally not considered part of the endomembrane system** because they have a different evolutionary origin and operate relatively independently.
- Therefore, statement A is incorrect.

Statement B: Rough Endoplasmic Reticulum (RER)

- The **Rough endoplasmic reticulum (RER)** is characterized by the presence of ribosomes attached to its outer surface.
- These attached ribosomes synthesize proteins that are often destined for secretion, insertion into membranes, or delivery to other organelles.
- Therefore, statement B is correct.

Statement C: DNA in Mitochondria and Plastids

- **Mitochondria** and **plastids** (like chloroplasts) are unique organelles because they contain their own genetic material.
- This DNA is typically in a **circular** form, similar to the DNA found in bacteria.
- This feature supports the endosymbiotic theory regarding their origin.
- Therefore, statement C is correct.

Statement D: Cytoskeleton Components

- The **cytoskeleton** is an intricate network within the cytoplasm of eukaryotic cells.
- It is composed of protein filaments and tubules, including **microtubules**, **microfilaments** (actin filaments), and **intermediate filaments**.
- It provides mechanical support, maintains cell shape, and enables cell movement.
- Therefore, statement D is correct.

Statement E: Mitochondrion Membrane Structure

- **Mitochondria** are characterized by having **two** membranes: an outer membrane and a highly folded inner membrane (forming cristae).
- This double-membrane structure is crucial for its function in cellular respiration.
- Therefore, statement E is incorrect.

Conclusion on Correct Statements

Based on the analysis:

- Statement A is incorrect.
- Statement B is correct.
- Statement C is correct.
- Statement D is correct.
- Statement E is incorrect.

The correct statements are B, C, and D.

---

154. **Answer: a**

Explanation:

Embryonic Development Matching

This question involves matching key developmental milestones of the foetus (List I) with specific weeks of pregnancy (List II). The solution identifies the correct pairing for each stage based on typical prenatal development timelines.

Developmental Milestones and Gestation Periods

- **A. The foetus movement starts and hair appears on the head**: Foetal movement, often termed quickening, typically becomes noticeable to the mother around **II. 20 weeks of pregnancy**. Hair on the head also starts to develop significantly by this stage.
- **B. The foetus develops limbs and digits**: While limb and digit formation commences earlier in gestation, by **IV. 12 weeks of pregnancy**, the limbs and digits are well-formed and clearly distinct.
- **C. The foetus develops external genital organs**: The initial development and differentiation of external genital organs begin around **III. 8 weeks of pregnancy**.
- **D. The foetus body is covered with fine hair; eyelids separate and eyelashes are formed**: The foetus's body typically becomes covered with fine hair (lanugo) around 20 weeks. Eyelids then separate, and eyelashes are formed later, around **I. 24 weeks of pregnancy**.

Correct Matching Summary

The accurate pairings derived from analyzing the developmental stages are:

- A corresponds to II (20 weeks)
- B corresponds to IV (12 weeks)
- C corresponds to III (8 weeks)
- D corresponds to I (24 weeks)

This set of matches is represented by the combination **A-II, B-IV, C-III, D-I**.

---

155. **Answer: c**

Explanation:

Sexual Deceit Example Explained

Sexual deceit refers to a deceptive strategy used by one organism to trick another into performing an action related to reproduction, often involving mimicry.

Analysis of Options

- **Option 1: Sea anemone and clown fish** This is an example of mutualism, where both species benefit. The clownfish gets protection, and the anemone is cleaned. It does not involve sexual deceit.
- **Option 2: Female wasp and fig** This involves a specialized pollination relationship where the fig provides shelter and resources for the wasp, and the wasp pollinates the fig. It's mutualistic, not typically sexual deceit.
- **Option 3: Ophrys and bumblebee** This is a classic example of **sexual deceit**. The *Ophrys* orchid mimics the shape and scent of a female bumblebee. Male bumblebees attempt to mate with the orchid (pseudocopulation), transferring pollen in the process.
- **Option 4: Cuckoo and crow** This is brood parasitism, where the cuckoo lays eggs in the crow's nest, deceiving the crow into raising its young. While deceptive, it's not sexual deceit involving mating mimicry.

Therefore, the *Ophrys* orchid deceiving the male bumblebee into attempting mating is the appropriate example of **sexual deceit**.

---

156. **Answer: c**

Explanation:

Identifying Osteichthyes Fishes

The class **Osteichthyes** comprises the bony fishes, characterized by a skeleton made primarily of bone and gills covered by a protective bony flap called an operculum.

Classifying Fish in Options

Let's examine the fish listed in each option to determine their class:

- **Option 1:** Saw fish (Class Chondrichthyes - cartilaginous fish), Fighting fish (Class Osteichthyes - bony fish), Dog fish (Class Chondrichthyes - cartilaginous fish). This set contains non-Osteichthyes.
- **Option 2:** Devil fish (Class Chondrichthyes - cartilaginous fish), Cuttlefish (Phylum Mollusca - not a fish), Hagfish (Class Hyperoartia/Agnatha - jawless fish). This set contains non-Osteichthyes and a mollusc.

- **Option 3:** Flying fish (Class Osteichthyes - bony fish), Angel fish (Class Osteichthyes - bony fish), Fighting fish (Class Osteichthyes - bony fish). All listed organisms are bony fishes.
- **Option 4:** Starfish (Phylum Echinodermata - not a fish), Hagfish (Class Hyperoartia/Agnatha - jawless fish), Cuttlefish (Phylum Mollusca - not a fish). This set contains non-fishes and non-Osteichthyes.

Osteichthyes Set Confirmation

Based on the classification:

- Flying fish are bony fish.
- Angel fish are bony fish.
- Fighting fish (e.g., Betta splendens) are bony fish.

Therefore, Option 3 is the correct set containing only fishes from the class Osteichthyes.

---

157. **Answer: a**

Explanation:

Evolution of Life Forms Chronology Matching

This question requires matching key events in the evolution of life with their approximate time periods in million years ago (mya).

Matching Events with Time Periods

Let's analyze the correct matches provided:

- **A. About 65 mya** corresponds to **II. The dinosaurs suddenly disappeared from the earth**. This marks the end of the Cretaceous period and the K-Pg extinction event.
- **B. About 500 mya** corresponds to **IV. Invertebrates were formed and became active**. This period aligns with the Cambrian Explosion, a time of rapid diversification of invertebrate life.
- **C. About 350 mya** corresponds to **I. Jawless fish probably evolved**. While jawless fish originated earlier, this period (Carboniferous) might represent significant

diversification or evolution of specific groups.

- **D. About 320 mya** corresponds to **III. Seaweeds and few plants probably existed**. By this time (Carboniferous), land plants were well-established and diverse, but this match may refer to simpler forms or specific ecological stages.

Correct Option Identification

Based on the established matches:

- A - II
- B - IV
- C - I
- D - III

This corresponds to Option A.

---

158. **Answer: b**

Explanation:

Haploid Gamete Production in Male Honeybees

The question concerns the unique process where **haploid cells** divide mitotically to produce gametes. This biological phenomenon is characteristic of the haplodiploid sex-determination system found in certain insects, like honeybees.

Male Honeybee Reproduction

- Male honeybees, known as drones, develop parthenogenetically from unfertilized eggs. Consequently, they are haploid, having a chromosome number represented as 'n'.
- The reproductive cells (germline cells) within the male honeybee are also haploid ('n').
- To produce sperm, which are the male gametes and must also be haploid ('n'), these haploid germline cells undergo **mitotic division**. Mitosis creates genetically identical copies, thus maintaining the haploid state necessary for gametes in this system.

Contrast with Diploid Reproduction

In contrast, most animals like male frogs, grasshoppers, and earthworms follow a diploid system ('$2n$'). In these organisms:

- Diploid germline cells ('$2n$') proliferate via mitosis.
- The production of haploid gametes ('n') requires **meiosis**, a process that reduces the chromosome number by half.

Therefore, the specific condition of haploid cells undergoing mitosis to form gametes is observed in male honeybees.

---

159. **Answer: d**

Explanation:

Calculating Eosinophil and Lymphocyte Counts from WBC

This solution estimates the approximate number of eosinophils and lymphocytes based on a given total White Blood Cell (WBC) count of 8000 cells per cubic millimeter (cu.mm).

Differential WBC Count Basics

The total WBC count includes various types of white blood cells. Knowing their typical percentage distribution (differential count) is essential for estimation:

- Eosinophils: Constitute approximately 1% to 6% of total WBCs.
- Lymphocytes: Constitute approximately 20% to 45% of total WBCs.

WBC Count Calculation Steps

Total WBC count provided = 8000 cells/cu.mm.

Eosinophil Count Calculation

Applying the standard percentage range (1% - 6%) to the total WBC count:

- Minimum Eosinophils: $1\% \times 8000 = \frac{1}{100} \times 8000 = 80$ cells/cu.mm
- Maximum Eosinophils: $6\% \times 8000 = \frac{6}{100} \times 8000 = 480$ cells/cu.mm

This gives an approximate expected range of 80 to 480 eosinophils per cu.mm.

Lymphocyte Count Calculation

Applying the standard percentage range (20% - 45%) to the total WBC count:

- Minimum Lymphocytes: $20\% \times 8000 = \frac{20}{100} \times 8000 = 1600$ cells/cu.mm
- Maximum Lymphocytes: $45\% \times 8000 = \frac{45}{100} \times 8000 = 3600$ cells/cu.mm

This gives an approximate expected range of 1600 to 3600 lymphocytes per cu.mm.

Result Verification with Options

We need to find an option that fits these calculated ranges. Option D suggests:

- Eosinophils: 160 - 240/cu.mm
- Lymphocytes: 1600 - 2000/cu.mm

Comparing these values:

- The suggested eosinophil count (160-240) falls within the calculated standard range (80-480). It corresponds to about 2% to 3% of the total WBC count.
- The suggested lymphocyte count (1600-2000) falls within the calculated standard range (1600-3600). It corresponds to 20% to 25% of the total WBC count.

Option D provides values consistent with the lower to mid-range of typical lymphocyte percentages and a plausible mid-range estimate for eosinophils, aligning with the question's request for approximate values.

---

160. **Answer: d**

Explanation:

Blood Group Probability Calculation

This question involves basic Mendelian genetics, specifically the ABO blood group system, which has three alleles: I^A, I^B, and i (often represented as I^O). Alleles I^A and I^B are codominant, and both are dominant over i. Blood group 'O' has the genotype ii $(I^O I^O)$.

Determining Parental Genotypes

The mother is heterozygous for 'A' blood group. This means her genotype is $I^A i$ $(I^A I^O)$. She has the 'A' antigen but also carries the recessive 'O' allele.

The father is heterozygous for 'B' blood group. This means his genotype is $I^B i$ $(I^B I^O)$. He has the 'B' antigen but also carries the recessive 'O' allele.

Predicting Offspring Genotypes

We can use a Punnett square to determine the possible genotypes of the offspring. Each parent can produce two types of gametes:

- Mother: I^A and i (I^O)
- Father: I^B and i (I^O)

	I^B	i (I^O)
I^A	$I^A I^B$	$I^A i$ $(I^A I^O)$
i (I^O)	$I^B i$ $(I^B I^O)$	ii $(I^O I^O)$

Identifying the 'O' Blood Group Probability

The Punnett square shows four possible genotypes for the offspring:

- $I^A I^B$ (Blood Group AB)
- $I^A i$ $(I^A I^O)$ (Blood Group A)
- $I^B i$ $(I^B I^O)$ (Blood Group B)
- ii $(I^O I^O)$ (Blood Group O)

Each genotype has an equal probability of occurring, which is 1 out of 4, or 25%.

The genotype required for blood group 'O' is ii $(I^O I^O)$. This occurs in 1 out of the 4 possible combinations.

Therefore, the probability of having a child with blood group 'O' is 25%.

---

161. **Answer: a**

Explanation:

Ordering Renin-Angiotensin Mechanism Events

The Renin-Angiotensin mechanism regulates blood pressure and fluid balance. Here's the correct sequence of events:

- **Step 1: Trigger (C)**: A decrease in blood pressure or glomerular filtration rate (GFR) occurs.
- **Step 2: Hormone Cascade (E)**: In response, renin is released. Renin converts angiotensinogen into angiotensin I, which is then converted to angiotensin II.
- **Step 3: Angiotensin II Effects (D)**: Angiotensin II causes vasoconstriction (narrowing of blood vessels) and stimulates the release of aldosterone.
- **Step 4: Aldosterone Action (B)**: Aldosterone acts on the distal tubules of the nephron, promoting the reabsorption of sodium ions (Na^+) and water.
- **Step 5: Result (A)**: Increased sodium and water reabsorption leads to increased blood volume, consequently raising blood pressure and GFR.

Therefore, the correct order is C, E, D, B, A.

Correct Sequence Summary

The events in the Renin-Angiotensin mechanism occur in the following order:

C (Fall in Glomerular filtration rate) → E (Renin converts Angiotensinogen into Angiotensin I, followed by Angiotensin II) → D (Vasoconstriction by Angiotensin II and release of Aldosterone) → B (Reabsorption of Na^+ and water from distal parts of tubule due to Aldosterone) → A (Increase in blood pressure and Glomerular filtration rate).

---

162. **Answer: a**

Explanation:

Here is the clear explanation for each statement:

B. Both species benefit in mutualism → Correct
Mutualism is an interaction where both organisms gain benefits. Example: bees get nectar and flowers get pollinated.

D. In parasitism, only one species is harmed and the other species is unaffected → Given as correct in option, but biologically incorrect
Actually, in parasitism, one species (parasite) benefits while the other (host) is harmed. The host is not "unaffected."
So scientifically, this statement is wrong, but it is included in the given answer key.

E. In amensalism, one species is harmed and the other is unaffected → Correct
In amensalism, one organism is negatively affected while the other remains completely unaffected. Example: Penicillium inhibiting bacterial growth.

---

163. **Answer: a**

Explanation:

Spermatogonia Cell Division Analysis

The question asks to identify the correct statements regarding the process of sperm production (spermatogenesis).

Statement Evaluation

- **A. Spermatogonia always undergo meiotic cell division.** This is incorrect. Spermatogonia first undergo mitotic divisions to increase their numbers and then differentiate into primary spermatocytes, which then undergo meiosis.

- **B. Primary spermatocytes divide mitotically to produce secondary spermatocytes.** This is incorrect. Primary spermatocytes undergo the *first meiotic division* (Meiosis I) to produce secondary spermatocytes.
- **C. Secondary spermatocytes, through their second meiotic division, produce haploid spermatids.** This is correct. Secondary spermatocytes are haploid cells that complete Meiosis II to form haploid spermatids.
- **D. Spermatids produce spermatozoa through mitosis.** This is incorrect. Spermatids undergo differentiation, not mitosis, to become spermatozoa.
- **E. Spermatids transform into spermatozoa by spermiogenesis.** This is correct. Spermiogenesis is the specific process where spermatids mature into functional spermatozoa (sperm).

Conclusion on Correct Statements

Based on the analysis, statements C and E are correct.

Identifying the Correct Option

The option that includes only the correct statements (C and E) is Option 1.

---

164. **Answer: a**

Explanation:

Plasmodium Life Cycle Stages Explained

The question asks to arrange the stages of the Plasmodium life cycle in the correct chronological order within the human host. This involves understanding the progression from initial infection to the development of stages transmissible back to the mosquito.

Sequence of Plasmodium Stages in Humans

1. **Stage E: Initial Infection**
   The cycle begins when a female Anopheles mosquito, acting as a vector, injects infectious sporozoites into the human bloodstream during a blood meal.

2. **Stage D: Sporozoites Travel to the Liver**
   Following injection, the sporozoites circulate in the blood and rapidly migrate to the liver.
3. **Stage B: Asexual Reproduction in Liver Cells**
   In the liver, sporozoites invade hepatocytes (liver cells) and undergo asexual multiplication (exo-erythrocytic schizogony), producing numerous merozoites. These liver cells eventually rupture, releasing merozoites into the bloodstream.
4. **Stage A: Asexual Reproduction in Red Blood Cells (RBCs)**
   Merozoites infect red blood cells and undergo further asexual multiplication (erythrocytic schizogony). This process involves the rupture of RBCs, releasing more merozoites that infect other RBCs, leading to the characteristic symptoms of malaria.
5. **Stage C: Gametocyte Development**
   During the erythrocytic cycle, some merozoites differentiate into male and female gametocytes within the RBCs. These are the forms that can be picked up by a mosquito during a subsequent blood meal, enabling the parasite's transmission.

Following this sequence, the correct order of the Plasmodium life cycle stages in humans is E, D, B, A, C.

---

165. **Answer: c**

Explanation:

Bioactive Molecules Matching Solution

This question requires matching bioactive molecules from List I with their specific importance or function described in List II.

Matching List I with List II:

- **Streptokinase (A):** This enzyme is known for its ability to dissolve blood clots. Therefore, it matches with "Removal of clots from blood vessels" (II). So, **A-II**.
- **Statins (B):** Statins are a class of drugs primarily used to reduce cholesterol levels in the blood. This corresponds to "Blood cholesterol-lowering agent" (III). So, **B-III**.

- **Lipases (C):** Lipases are enzymes that break down fats (lipids). They are commonly used in cleaning products like detergents to help remove greasy stains. Thus, it matches with "Detergent formulations" (IV). So, **C-IV**.
- **Cyclosporin A (D):** Cyclosporin A is a crucial medication used to suppress the immune system, particularly important in preventing rejection after organ transplants. This aligns with "Immunosuppressive agent" (I). So, **D-I**.

Correct Matching Combination:

Combining the matches:

- A - II
- B - III
- C - IV
- D - I

This combination corresponds to option 3 (A-II, B-III, C-IV, D-I), which is presented as the correct answer.

---

166. **Answer: c**

Explanation:

Identifying Divergent Evolution: Whales and Bats Forelimbs

The question asks to identify which option is **not** an example of convergent evolution. Convergent evolution occurs when unrelated species independently evolve similar traits due to similar environmental pressures or ecological niches. These similar traits are called analogous structures.

Conversely, divergent evolution occurs when related species evolve different traits from a common ancestor, often due to different environmental pressures. Structures derived from a common ancestor, even if modified for different functions, are called homologous structures.

Analysis of Options:

- **Wings of butterflies and birds:** Butterflies (insects) and birds (vertebrates) are distantly related. Their wings serve the same function (flight) but evolved independently. This is an example of **convergent evolution** (analogous structures).
- **Flippers of penguins and dolphins:** Penguins (birds) and dolphins (mammals) are unrelated. Their flippers are adapted for swimming, demonstrating independent evolution for aquatic life. This is an example of **convergent evolution** (analogous structures).
- **Fore limbs of whales and bats:** Whales and bats are both mammals and share a common mammalian ancestor. Their forelimbs have a similar underlying bone structure (homologous) but are adapted for different functions (swimming flippers vs. flying wings). This modification from a common ancestral structure into different forms is an example of **divergent evolution**. Therefore, this is **not** an example of convergent evolution.
- **Eyes of octopuses and mammals:** Octopuses (mollusks) and mammals (vertebrates) are very distantly related. Their complex eyes function similarly but evolved independently. This is an example of **convergent evolution** (analogous structures).

Conclusion:

The fore limbs of whales and bats represent homologous structures modified through divergent evolution, making it the correct answer as it is not an example of convergent evolution.

---

167. **Answer: d**

Explanation:

Swiss Cheese Holes Production Explained

The characteristic large holes, often called 'eyes', in Swiss cheese are a direct result of specific bacterial activity during the ripening process.

Key Bacterium and Gas Production

- The primary reason for hole formation is the metabolic activity of the bacterium *Propionibacterium sharmanii.*
- This bacterium consumes lactic acid, a byproduct of earlier fermentation by other bacteria (like lactic acid bacteria), and produces carbon dioxide gas (CO_2) as a waste product.
- The chemical reaction can be simplified as: Lactic Acid $\rightarrow$ Propionic Acid + Acetic Acid + CO_2 + Biomass.

Mechanism of Hole Formation

- As *Propionibacterium sharmanii* produces CO_2 gas within the cheese curd, the gas accumulates.
- Because the cheese curd is relatively solid but still pliable, the accumulating CO_2 gas forms bubbles.
- These gas bubbles stretch the curd and, upon escaping or remaining trapped, leave behind the distinctive holes or 'eyes' characteristic of Swiss cheese.

Evaluation of Options

- Option 1 is incorrect because *Trichoderma polysporum* is a fungus and not involved in Swiss cheese production, nor does it produce H_2 in this context.
- Option 2 is incorrect. While *Clostridium* species can produce gases, they are typically associated with different types of cheese spoilage or specific fermentations, not the characteristic holes of Swiss cheese.
- Option 3 is incorrect. Lactic acid bacteria like *Lactobacillus* are crucial for initial acid production in cheese making but do not produce the large amounts of CO_2 needed for Swiss cheese eyes. They primarily produce lactic acid.
- Option 4 correctly identifies *Propionibacterium sharmanii* as the bacterium responsible for producing the significant amount of CO_2 gas that creates the holes in Swiss cheese.

---

168. **Answer: d**

Explanation:

Hormone Functions Matching Explained

This question requires matching hormones from List I to their primary functions in List II.

Matching List I with List II

- **A. Cortisol** is correctly matched with **II. Produces anti-inflammatory reactions**. Cortisol is a steroid hormone known for its role in regulating the immune response and suppressing inflammation.
- **B. Aldosterone** is correctly matched with **III. Stimulates reabsorption of Na^+ and water from renal tubule**. Aldosterone is a mineralocorticoid hormone that regulates electrolyte balance and blood pressure by acting on the kidneys.
- **C. Cholecystokinin (CCK)** is correctly matched with **IV. Stimulates secretion of pancreatic enzymes and bile juice**. CCK is a gastrointestinal hormone that triggers the pancreas to release digestive enzymes and the gallbladder to release bile.
- **D. Progesterone** is correctly matched with **I. Stimulates the formation of alveoli in mammary glands**. Progesterone is a crucial hormone for the reproductive system, particularly in preparing the uterus for pregnancy and development of mammary glands.

Final Correct Matching

Based on the analysis, the correct matching is:

- A - II
- B - III
- C - IV
- D - I

This corresponds to option A-II, B-III, C-IV, D-I.

---

169. **Answer: b**

Explanation:

The question asks to arrange the structures surrounding the female gamete (ovum) from the outermost to the innermost layer.

Ovum Structure Layers: Order

The correct arrangement of the layers around the female gamete, from outer to inner, is based on their position during fertilization:

- **Corona Radiata (C):** This is the outermost layer, composed of follicular cells that surround the ovum after ovulation.
- **Zona Pellucida (A):** Located just inside the corona radiata, this is a thick, glycoprotein layer essential for sperm binding and the acrosome reaction.
- **Perivitelline Space (B):** This is a narrow space situated between the zona pellucida and the ovum's plasma membrane.
- **Plasma Membrane of Ovum (D):** Also known as the oolemma, this is the innermost membrane, directly enclosing the ovum's cytoplasm.

Therefore, the sequence from outer to inner is C, A, B, D.

Correct Option Identification

Based on the established order:

- C - Corona radiata (Outermost)
- A - Zona pellucida
- B - Perivitelline space
- D - Plasma membrane of ovum (Innermost)

The sequence **C, A, B, D** matches option 2.

---

170. **Answer: b**

Explanation:

Verhulst-Pearl Logistic Growth Equation Identification

The Verhulst-Pearl logistic growth model describes population growth that is limited by environmental resources. It introduces a carrying capacity (K) that restricts exponential growth.

Understanding the Logistic Growth Formula

The general differential equation for population growth is:

$\frac{dN}{dt} = rN$

where:

- N is the population size.
- t is time.
- r is the intrinsic rate of natural increase.

The logistic model modifies this by adding a term that reduces the growth rate as the population approaches the carrying capacity (K). The standard form is:

$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$

Analyzing the Options

Let's examine the provided options to see which one matches the standard logistic growth equation:

- **Option 1:** $\frac{dN}{dt} = rN\left(\frac{K-N}{N}\right)$. Simplifying this gives $rN\left(\frac{K}{N} - 1\right) = rK - rN$, which is not the logistic equation.
- **Option 2:** $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$. This can be rewritten as $rN\left(\frac{K}{K} - \frac{N}{K}\right) = rN\left(1 - \frac{N}{K}\right)$. This exactly matches the standard Verhulst-Pearl logistic growth equation.
- **Option 3:** $\frac{dN}{dt} = rN\left(\frac{K}{K-N}\right)$. This form does not represent logistic growth, as the growth rate would increase disproportionately as N approaches K.
- **Option 4:** $\frac{dN}{dt} = rN\left(\frac{K+N}{K}\right)$. Simplifying gives $rN\left(1 + \frac{N}{K}\right)$, which represents growth that accelerates beyond exponential growth, not logistic growth.

Conclusion

Option 2 correctly represents the Verhulst-Pearl logistic population growth equation.

---

171. **Answer: a**

Explanation:

Bt Toxin Genes Targeting Specific Pests

The bacterium *Bacillus thuringiensis* (Bt) produces crystal (Cry) proteins that act as insecticides. Different Cry proteins are effective against different insect pests.

Cry Genes and Pest Specificity

Specific Cry genes within *Bacillus thuringiensis* are responsible for producing toxins that target particular types of insects. Understanding these specificities is crucial for effective pest management.

- **Cotton Bollworms:** The toxin protein coded by the ***crylAc*** gene is known to be highly effective against Lepidopteran pests, including the cotton bollworm (*Helicoverpa spp.*).
- **Corn Borers:** The toxin protein coded by the ***crylAb*** gene is primarily effective against certain species of borers, such as the European corn borer (*Ostrinia nubilalis*).

Therefore, the genes coding for toxin proteins that control cotton bollworms and corn borers, respectively, are ***crylAc*** and ***crylAb***.

---

172. **Answer: d**

Explanation:

JGA Components Identification

The Juxta Glomerular Apparatus (JGA) is a critical structure within the kidney responsible for regulating blood pressure and filtration rate.

This specialized region is formed by the close association of specific cellular modifications from two distinct parts related to the same nephron:

- The **Distal Convoluted Tubule (DCT)**: Specifically, the specialized cells called the macula densa, which are located where the DCT passes near the glomerulus.
- The **Afferent Renal Arteriole**: The wall of this arteriole contains specialized smooth muscle cells known as juxtaglomerular cells.

Therefore, the JGA is formed by the distal convoluted tubule and the afferent renal arteriole.

This corresponds to Option D.

---

173. **Answer: b**

Explanation:

Let's match them correctly:

A. Molluscs → II. Branchial respiration
Most aquatic molluscs (like bivalves) respire through gills (branchial respiration).

B. Reptiles → I. Pulmonary respiration only
Reptiles breathe only through lungs.

C. Adult amphibians → IV. Pulmonary and cutaneous respiration
Adult frogs and similar amphibians use both lungs and skin.

D. Amoeba → III. Cellular respiration
Amoeba respires through simple diffusion at the cellular level.

Final correct matching:

A → II
B → I
C → IV
D → III

So the correct answer is:
A-II, B-I, C-IV, D-III

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174. **Answer: b**

Explanation:

Mutant Codon in Beta Globin Gene Explained

The question concerns a specific mutation in the **beta globin gene** responsible for altering red blood cell (RBC) shape and causing haemoglobin polymerization. This condition is characteristic of sickle cell anemia.

Identifying the Sickle Cell Mutation

- The mutation occurs at the **sixth position** of the beta globin protein.
- The normal codon is GAG, which codes for the amino acid Glutamic acid.
- The specific mutant codon causing the disease is GUG.
- This change replaces Glutamic acid with the amino acid Valine at the sixth position.
- This single amino acid substitution (Glu → Val) alters the haemoglobin molecule (HbS), causing it to polymerize under low oxygen conditions.
- The polymerization distorts the RBCs into a sickle shape, leading to various complications.

Therefore, the sixth mutant codon responsible for these effects is GUG.

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175. **Answer: c**

Explanation:

Grasshopper Chromosome Numbers Explained

The difference in chromosome numbers between male and female grasshoppers is due to their specific sex determination system.

- **Sex Determination System:** Grasshoppers typically exhibit the X0 sex determination system.
- **Female Chromosomes:** Females have two sex chromosomes (XX). If the autosome number is 'n', the total chromosome number is $2n + 2$.
- **Male Chromosomes:** Males have only one sex chromosome (X0, meaning just one X chromosome and no Y chromosome). If the autosome number is 'n', the total chromosome number is $2n + 1$.

- **Chromosome Count Analysis:**
    - A count of 23 chromosomes (an odd number) indicates the male configuration $(2n + 1)$.
    - A count of 24 chromosomes (an even number) indicates the female configuration $(2n + 2)$.

Therefore, the members with 23 chromosomes are males, and the members with 24 chromosomes are females.

Conclusion

The 23 and 24 chromosome-bearing members in this grasshopper species are **males and females, respectively**.

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176. **Answer: b**

Explanation:

Statement Analysis for Incorrect Statements

The question asks to identify the **incorrect** statements among the given options related to animal characteristics.

- **Statement A:** Digestive system in Platyhelminthes is incomplete. This is **correct**. Platyhelminthes (flatworms) typically have a single opening for both ingestion and egestion, characteristic of an incomplete digestive system.
- **Statement B:** Bilateral symmetry is a characteristic feature of adult Echinoderms. This is **incorrect**. Adult Echinoderms (like starfish) exhibit radial symmetry, while their larval forms show bilateral symmetry.
- **Statement C:** Pseudocoelom is possessed by Aschelminthes. This is **correct**. Aschelminthes (roundworms) are known as pseudocoelomates, possessing a body cavity that is not completely lined by mesoderm.
- **Statement D:** Notochord is persistent throughout life in the class Chondrichthyes. This is considered **correct** in the context of this question. While the notochord is largely replaced by cartilaginous vertebrae in many adult chondrichthyans, its

persistence in certain forms or regions is acknowledged, aligning with the provided answer's premise.
- **Statement E:** Members of class Reptilia maintain a constant body temperature. This is **incorrect**. Reptiles are ectothermic (cold-blooded), meaning their body temperature varies with the environment, unlike endothermic (warm-blooded) animals like mammals and birds.

Conclusion on Incorrect Statements

Based on the analysis, the incorrect statements are B and E.

Therefore, the correct option is the one that includes only statements B and E.

Final Answer: The final answer is Option B

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177. **Answer: c**

Explanation:

Enzyme Class Identification: Lyase Activity

The reaction shows enzyme 'E' acting on a substrate containing a $\mathrm{C-C}$ single bond.

The products include a $\mathrm{C=C}$ double bond and the combined $\mathrm{X-Y}$ molecule.

Key Characteristics of Lyases:

- Lyases catalyze the breaking of chemical bonds via elimination reactions.
- This often results in the formation of a double bond (e.g., $\mathrm{C=C}$) or a ring structure.
- These reactions occur *without* hydrolysis or oxidation.

The observed formation of the $\mathrm{C=C}$ double bond directly matches the defining activity of Lyases.

Other enzyme classes do not fit this reaction pattern:

- Ligases join molecules.
- Transferases move functional groups.

- Isomerases rearrange atoms within a molecule.

Therefore, enzyme 'E' belongs to the class **Lyases**.

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178. **Answer: b**

Explanation:

Ecological Pyramids Basics

Ecological pyramids illustrate the relationship between different trophic levels in an ecosystem. They can represent biomass, number of organisms, or energy. Typically, these pyramids are upright, showing a decrease at each successive trophic level.

Inverted Pyramids Explained

However, certain ecosystems exhibit inverted pyramids under specific conditions. An inverted pyramid occurs when an organism or group at a higher trophic level has a larger biomass, number, or energy than the level below it. This is unusual but occurs in specific ecological contexts.

Analyzing Ecosystem Biomass Pyramids

Let's examine the options concerning inverted pyramids:

- **Pyramid of biomass in grassland:** Generally upright. Producers (grass) have large biomass compared to primary consumers (herbivores).
- **Pyramid of biomass in sea:** Often inverted. The primary producers are microscopic phytoplankton. They have a very high reproduction rate but a short lifespan, resulting in a smaller standing biomass at any given moment compared to the larger biomass of zooplankton (primary consumers) feeding on them.
- **Pyramid of number in grassland:** Typically upright. Many grass plants support fewer herbivores, which support even fewer carnivores.
- **Pyramid of energy in pond ecosystem:** Always upright. Energy transfer between trophic levels is inefficient (around 10%), so energy content decreases significantly at each higher level.

Reasoning for Inverted Biomass in Sea

The pyramid of biomass in the sea is a classic example of an inverted pyramid. Phytoplankton, despite being the producers, have a very rapid turnover rate. Their total biomass at a specific time might be less than the biomass of the zooplankton consuming them. This inversion is characteristic of aquatic ecosystems dominated by phytoplankton.

Therefore, the ecological pyramid that is generally inverted is the pyramid of biomass in the sea.

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179. **Answer: a**

Explanation:

Human Evolution: Species Sequence

The evolution of humans is closely linked to the development of brain capacity and language skills. Identifying the correct sequence of ancestral species represents a key aspect of understanding human origins.

Correct Evolutionary Timeline

The established sequence for the appearance of major hominin species is as follows:

- Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens

This order reflects the progressive changes leading to modern humans.

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180. **Answer: b**

Explanation:

The question requires matching respiratory volumes from List I with their corresponding capacities in milliliters (mL) from List II.

Respiratory Volumes Matching Solution

Matching Respiratory Volumes and Capacities

To solve this, we need to identify the typical values for each respiratory volume:

- **Tidal Volume (TV):** The volume of air inhaled or exhaled during a normal breath. It is typically around **500 mL**. (Matches with II)
- **Expiratory Reserve Volume (ERV):** The additional volume of air that can be forcefully exhaled after a normal exhalation. It is typically around **1000 - 1100 mL**. (Matches with III)
- **Inspiratory Reserve Volume (IRV):** The additional volume of air that can be forcefully inhaled after a normal inhalation. It is typically around **2500 - 3000 mL**. (Matches with I)
- **Residual Volume (RV):** The volume of air remaining in the lungs even after maximal exhalation. It is typically around **1100 - 1200 mL**. (Matches with IV)

Correct Matches Found

Based on the typical values:

- A. ERV corresponds to 1000 - 1100 mL (III)
- B. RV corresponds to 1100 - 1200 mL (IV)
- C. IRV corresponds to 2500 - 3000 mL (I)
- D. TV corresponds to 500 mL (II)

Therefore, the correct matching is A-III, B-IV, C-I, D-II.

List I (Respiratory Volume)	List II (Capacity in mL)
A. ERV	III. 1000 - 1100 mL
B. RV	IV. 1100 - 1200 mL
C. IRV	I. 2500 - 3000 mL
D. TV	II. 500 mL

This corresponds to Option 2 in the choices provided.